Multiphysics Wave Mechanics in Generalized Continua: A Constructive Review of Theory, Energy Transport, and Interface Phenomena

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22 October 2026

A constructive synthesis of generalized thermoelasticity, piezothermoelasticity, nonlocal elasticity, and memory-dependent continuum theories.

The review is organized around the physical developments that reshaped generalized continuum wave mechanics: the relaxation of Fourier's law, the introduction of scale-dependent constitutive response, and the progressive coupling of thermal, electric, magnetic and elastic fields. Each theory is presented with its governing equations, the assumptions that license them, the solution approaches used to analyse it, and the wave phenomena that follow.

This review examines how extensions of classical continuum mechanics change thermomechanical wave propagation, energy transport and interface phenomena. It organizes generalized thermoelastic, multiphysics, memory-dependent, fractional, gradient and nonlocal models by constitutive mechanism, governing structure, predicted modes and validation. With total energy per unit volume and outward-positive flux, the classical mechanical contribution is minus stress power, giving $J_i = q_i - \sigma_{ij} v_j$. Lord–Shulman and Green–Lindsay are distinct from two-temperature thermoelasticity: the former relaxes heat flux, whereas the latter introduces temperature-rate dependence in a single-temperature formulation. The standard Eringen-type nonlocal dispersion relation considered here has bounded frequency at high wavenumber, and quasi-electrostatic piezoelectric models do not contain a propagating electromagnetic branch. The review compares model-specific assumptions and identifies questions for further analytical, numerical and experimental study.

Notation

Acronyms

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Acronyms **1**

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Table 1: Master symbol table with units and physical meaning.

Symbol	Meaning	Units	First appearance
u_i	Displacement component	m	§3.1
v_i	Velocity component, $\dot{u}_i$	m/s	§3.1
ε_{ij}	Infinitesimal strain tensor	dimensionless	§3.1
ω_{ij}	Rotation tensor	dimensionless	§3.1
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$	§3.6; §8.1
ω_M	MDD finite-history window (distinct from angular frequency)	s	§6.4
t'	Dummy integration time in hereditary kernels	s	§6.3–6.4
σ_{ij}	Cauchy stress tensor	Pa	§3.2
ρ	Mass density	$\mathrm{kg/m^3}$	§3.2
f_i	Body force component	N/kg	§3.2
q_i	Heat flux component	$\mathrm{W/m^2}$	§3.3
T	Temperature increment, $T - T_0$	K	§3.3
T_0	Reference temperature	K	§3.3
ψ	Specific free energy	J/kg	§3.5
s	Specific entropy	J/(kg·K)	§3.5
J_i	Total energy flux component	$\mathrm{W/m^2}$	§3.8
$\mathcal{E}$	Total energy density	$\mathrm{J/m^3}$	§10.1
$\mathcal{P}$	Mechanical power input	$\mathrm{W/m^3}$	§10.1
$\mathcal{D}$	Dissipation rate	$\mathrm{W/m^3}$	§10.1
τ_0	Lord–Shulman relaxation time	s	§4.3
τ_q, τ_T, τ_ν	Dual/triple phase-lag times	s	§4.7–4.8
c_T	Thermal wave speed	m/s	§4.3
c_p, c_s	Longitudinal and shear wave speeds	m/s	§3.6
e_{ijk}	Piezoelectric stress coefficient	$\mathrm{C/m^2}$	§5.1
D_i	Electric displacement component	$\mathrm{C/m^2}$	§5.1
ϕ	Electric potential	V	§5.1
κ_{ij}	Dielectric permittivity tensor	F/m	§5.1
κ, κ^*	GN-III scalar coefficients; distinct from κ_{ij}	model-dependent	§4.6
E_i	Electric field component, $-\phi_{,i}$	V/m	§5.1
β_{ij}	Thermal stress coefficient	Pa/K	§4.2
c_v	Specific heat at constant strain	J/(kg·K)	§4.2
k	Thermal conductivity	W/(m·K)	§4.2
k_{w}	One-dimensional wave number in the displayed dispersion law (distinct from conductivity k)	$\mathrm{m^{-1}}$	§7.3
λ, μ	Lamé elastic constants	Pa	§3.6
α	Thermal expansion coefficient (context-qualified; distinct from fractional order)	1/K	§4.2
α (fractional)	Fractional derivative order	dimensionless	§4.11; §6.2
α_{GN}	GN-II thermal displacement	K s	§4.5
ξ	Nonlocal length scale parameter	m	§7.1
K_{MDD}	MDD kernel, evaluated at the time lag $t - t'$; units depend on its definition	model-dependent	§6.4

Table 2: Acronym table.

Acronym	Full form	Meaning
CD	Classical coupled	Classical coupled thermoelasticity (Fourier)
LS	Lord–Shulman	Generalized thermoelasticity with one relaxation time
GL	Green–Lindsay	Temperature-rate-dependent thermoelasticity (two relaxation times)
GN-I	Green–Naghdi type I	Classical-equivalent thermoelasticity
GN-II	Green–Naghdi type II	Thermoelasticity without energy dissipation
GN-III	Green–Naghdi type III	Dissipative generalized thermoelasticity
DPL	Dual-phase-lag	Two-phase-lag heat conduction (Tzou, 1995)
TPL	Triple-phase-lag	Three-phase-lag heat conduction
MGT	Moore–Gibson–Thompson	Third-order heat conduction model
MDD	Memory-dependent derivative	Integral-form memory operator (Wang & Li, 2011)
PDE	Partial differential equation	—
ODE	Ordinary differential equation	—
NDE	Nondestructive evaluation	—
FGP	Functionally graded piezoelectric	Piezoelectric material with spatially varying properties
PML	Perfectly matched layer	Numerical absorbing boundary condition
ES	Elastic half-space	Reference medium in interface problems
TS	Thermoelastic half-space	Thermoelastic medium in interface problems
HPS	Higher-order piezothermoelastic	Piezothermoelastic medium with higher-order MDD
CDI	Clausius–Duhem inequality	Dissipation inequality in continuum thermodynamics

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1 Review Scope and Purpose

1.1 Motivation and scientific context

Wave propagation in generalized continua lies at the intersection of continuum mechanics, thermodynamics, and coupled-field physics. In the classical Fourier model, combining Fourier heat conduction with energy balance gives a parabolic heat equation whose fundamental solution has instantaneous mathematical support. This property is often called the infinite-speed paradox; it is a consequence of the idealized continuum model, not evidence that the model is simply wrong in every regime. Biot's coupled linear thermoelastic formulation remains useful within its assumptions, while later theories replace or augment selected constitutive structures to represent finite-rate transport, memory, additional fields, or characteristic lengths. These alternatives are not interchangeable and do not all remove instantaneous support or guarantee finite-speed propagation.

Common modifications include relaxation of the heat flux (Lord–Shulman), temperature-rate dependence with two relaxation parameters (Green–Lindsay, a single-temperature formulation), thermal-displacement theories (Green–Naghdi), phase-lag laws, hereditary kernels, fractional operators, and spatially nonlocal or gradient constitutive laws. Each class has variants, assumptions, and limiting regimes; the exact governing equations, not a family label, determine propagation, thermodynamic admissibility, and well-posedness. The historical and constitutive comparisons below are therefore presented as a map of distinct models rather than a single sequence of increasingly accurate replacements.

These modifications change the mathematical character of the model and may introduce additional parameters, state variables, memory, or spatial scales. Their consequences for support, stability, thermodynamic admissibility, and initial data are model-specific. For example, claims about relaxation-time orderings, entropy production, or the function spaces required by fractional or nonlocal operators must be tied to the precise equations and boundary conditions; they are not asserted uniformly across each named family in this review.

A separate modelling axis concerns additional coupled fields: electrical polarization, porosity, micro-rotation, or charge carriers. Piezothermoelastic, porous, micropolar, semiconductor, and magneto-thermoelastic models introduce field equations and coupling terms whose effects on modal structure and interface conditions depend on the constitutive model and geometry. Simultaneous temporal memory and multiphysics coupling can increase analytical and computational complexity, but it does not imply a fixed number of additional modes or boundary conditions.

These coupled formulations underpin smart structures, energy harvesters, nondestructive evaluation, geophysical prospecting, biomedical ultrasound and nanodevice design, and they have produced a substantial literature in their own right. Yet the field lacks the kind of synthesis that a practitioner needs: a single account that states the constitutive structure of each family, derives the wave problem it produces, establishes the conditions under which each is thermodynamically admissible, shows how the resulting quantities are computed, and identifies what remains open. Existing reviews tend to cover one family or one application Shakeriaski et al. [2021], Chandrasekharaiah [1998], Joseph and Preziosi [1989], Askes and Aifantis [2006]; this review is organised so that the same question can be followed across all of them.

1.2 Objectives and contributions

This review has the following objectives:

1. To present the first-principles foundations of continuum wave mechanics: kinematics, bal-

ance laws, thermodynamic restrictions, and constitutive principles (Section 3).

2. To trace the historical development of generalized thermoelastic theories from classical coupled thermoelasticity to modern memory-dependent and nonlocal formulations, explaining the physical motivation behind each theory (Section 4).

3. To present coupled multiphysics continua—piezothermoelasticity, porous media, micropolar continua, semiconductor interactions, and magneto-thermo-elasticity—with their governing equations and coupling mechanisms (Section 5).

4. To survey generalized transport and memory effects: relaxation, fractional operators, hereditary integrals, memory-dependent derivatives, and their mathematical structure (Section 6).

5. To present nonlocal and scale-dependent continuum models, including Eringen's nonlocal elasticity and strain-gradient theory, with their wave dispersion characteristics (Section 7).

6. To review wave-propagation methods: plane waves, Christoffel and secular equations, transform methods, and numerical approaches (Section 8).

7. To develop the theory of wave reflection, transmission, and energy partitioning at interfaces between different continuum media (Section 9).

8. To derive the classical local energy flux under explicit assumptions and identify, without overclaiming, the model-specific energy-flux derivations that remain open (Section 10).

9. To survey advanced theoretical developments and emerging research directions (Sections 11–12).

10. To present applications with evidence levels, an original five-axis research taxonomy, major scientific challenges, and future research opportunities (Sections 13–16).

1.3 Scope and organization

The review covers the following theory families:

- **Classical coupled thermoelasticity** (Biot, 1956) — Fourier heat conduction coupled with elastic wave propagation.
- **Lord–Shulman theory** (1967) — generalized thermoelasticity with one thermal relaxation time.
- **Green–Lindsay theory** (1972) — temperature-rate-dependent thermoelasticity with two relaxation times and one temperature field; distinct from two-temperature theories.
- **Green–Naghdi types I–III** — the type framework dates to 1991; the type-III equations treated here are from 1992 and the type-II primary formulation from 1993.
- **Dual-phase-lag and triple-phase-lag theories** (DPL: Tzou, 1995; TPL: Roy Choudhuri, 2007) — phase-lag heat conduction models.
- **Moore–Gibson–Thompson-type thermoelastic models** — later thermal analogues of Moore and Gibson's 1960 third-order acoustic equation for relaxing gases; the 1960 work is not a heat-conduction law.

- **Memory-dependent derivative models** (Wang & Li, 2011) — integral-form memory operators with fading memory.
- **Fractional thermoelasticity** — time-fractional heat equations for anomalous diffusion.
- **Nonlocal elasticity** (Eringen, 1972–1983) — scale-dependent constitutive relations with spatial memory.
- **Piezothermoelasticity and coupled multiphysics continua** — piezoelectric, porous, micropolar, semiconductor, and magneto coupling.
- **Interface wave phenomena** — reflection, transmission, scattering, and energy partitioning at material interfaces.

The manuscript is organized into seventeen sections:

- **Section 1** (this section): scope, purpose, and organization.
- **Section 2**: literature landscape and scholarly context, including comparison with existing reviews.
- **Sections 3–5**: Layer I–III of the mathematical exposition—foundations, thermoelastic theories, and coupled multiphysics continua.
- **Sections 6–8**: generalized transport, memory effects, nonlocal models, and mathematical methods.
- **Sections 9–10**: interface phenomena and energy transport.
- **Sections 11–12**: advanced developments and comparative synthesis.
- **Sections 13–17**: applications, taxonomy, challenges, future directions, and conclusions.

Appendix B derives selected balance relations, and Appendix C defines benchmark problems for future computational comparison.

1.4 Intended readership

This review is written for multiple audiences:

- **Beginning researchers and MSc students:** Sections 1–3 provide the foundational balance laws, constitutive relations, and classical wave mechanics needed to enter the field. The notation tables and Section 17 provide orientation.
- **Doctoral researchers:** Sections 3–10 provide mathematical depth with representative equations, dispersion relations, interface formulations, and solution methods. Section 16 identifies dissertation topics.
- **Experienced scientists:** Sections 12–16 provide comparative synthesis, taxonomy, challenge analysis, and future research roadmap.
- **Journal reviewers and editors:** Sections 1–2, 12, and 16–17 provide context for evaluating new submissions in generalized continuum mechanics.

1.5 Scholarly principles

This review adheres to the following principles:

1. **Constructive synthesis:** Model-specific differences are traced to documented assumptions (constitutive, material symmetry, geometry, boundary conditions, thermal theory, frequency convention, normalization, approximation level), not presented as contradictions or errors.

2. **Mathematical accuracy:** Important equations should be stated with their assumptions, conventions, and applicable model.

3. **Comparative rather than adjudicative framing:** Competing formulations are compared on the basis of their modelling choices and domains of validity; the review does not attempt to adjudicate between individual publications.

4. **Minimal equation set:** Fewer, precise equations are preferred over hundreds of redundant ones. Representative equations are selected to illustrate each theory family.

5. **Original figures and tables:** All figures are original vector graphics created for this review. All tables synthesize knowledge across theory families rather than listing individual papers.

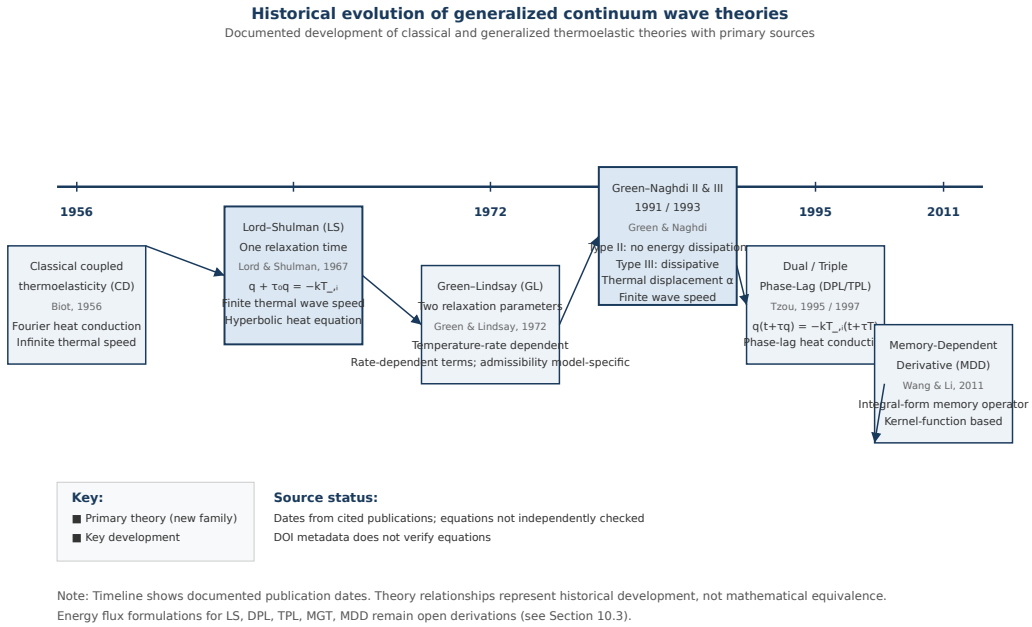


Figure 1: Selected historical milestones in classical and generalized thermoelasticity. Publication dates and model descriptions are taken from the inherited bibliography; the figure does not assert that every generalized theory has finite propagation speed.

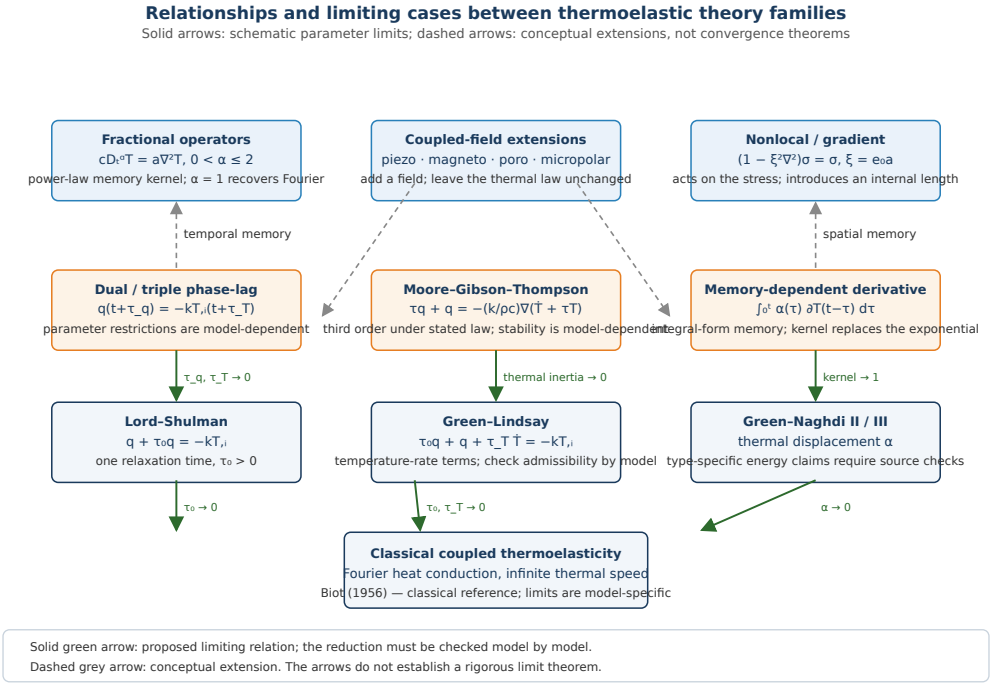


Figure 2: Schematic relationships between thermoelastic model families. Parameter limits are model-specific and are not asserted to exist for every family; the arrows are conceptual rather than rigorous convergence results.

2 Literature Landscape and Scholarly Context

2.1 The organising problem and the shape of the field

The literature of generalized continuum wave mechanics addresses the infinite propagation speed associated with Fourier heat conduction. Joseph and Preziosi's retrospective traces early finite-speed heat-flux formulations to Cattaneo and describes Vernotte's later presentation of the relaxation law as a simple generalization Joseph and Preziosi [1989].

That pattern — identification, modification, cost — is the organising theme of this review, and it is why the literature is best read by theory family rather than by paper. Each family below answers the same question in a different way, and each pays for the answer in a different currency: a second temperature, a second wave, a loss of ellipticity, a loss of uniqueness, a loss of local well-posedness. Sections 4 through 11 take the families in turn; this section establishes how they arose, where the field's centre of gravity has moved, and what the geographic distribution of the work reveals about the structure of the problem.

2.2 Four phases in the development of the field

Cattaneo and Vernotte proposed relaxation-type modifications of Fourier heat conduction; Lord and Shulman incorporated a relaxation time into coupled thermoelasticity. Green and Lindsay introduced a temperature-rate-dependent, single-temperature formulation with two relaxation parameters. These models have different constitutive structures and limiting assumptions.

Phase II — second sound and thermal-displacement theories (1966–1993). Chester's 1963 model predates this phase: Eq. (5) adds a relaxation term to Fourier flux, Eqs. (6)–(7) give

the associated damped thermal-wave equation and model speed, and Eq. (8)'s phonon-speed relation is restricted to an idealized dispersionless phonon gas (printed pp. 2014–2015) Chester [1963]. Guyer and Krumhansl's p. 766 article derives a nonlocal coefficient from the linearized phonon Boltzmann equation; Eq. (59) supports $\lambda^2 = c^2\tau_N\tau_z/5$ only in the stated scattering regime (printed p. 773) Guyer and Krumhansl [1966a]. Their separate p. 778 companion article treats thermal conductivity, second sound, and phonon-hydrodynamic phenomena in nonmetallic crystals; its findings are not transferred to the p. 766 equation Guyer and Krumhansl [1966b]. Green–Naghdi type-II and type-III formulations are treated as separate systems; no cross-type dissipation or energy claim is inferred Green and Naghdi [1993, 1992]. Chandrasekharaiah's 1998 review surveys recent hyperbolic-thermoelasticity literature and is used here for historical context, not as primary equation evidence Chandrasekharaiah [1998].

Phase III — the microscale turn (1995–2011). Phase II assumed that the Fourier paradox was a matter of taking a limit too early. Phase III questioned the premise, and asked instead what a thermal wave should look like when the material has microstructure. Tzou's dual-phase-lag model Tzou [1995] and Roy Choudhuri's triple-phase-lag formulation Choudhuri [2007] distinguish the flux and temperature-gradient lags and include a thermal-displacement-gradient contribution in the latter; the exact truncation and admissibility conditions remain model-specific. In parallel, Eringen's nonlocal elasticity Eringen [1972, 1974, 1983] and the strain-gradient formulations Askes and Aifantis [2006, 2011], Polizzotto [2001] transferred the same logic to the mechanical side, and the size-dependent wave speeds measured and computed for nanostructures Maranganti et al. [2006], Maranganti and Sharma [2007] supplied the corresponding experimental evidence.

Phase IV — memory, fractional operators and multiphysics (2011–present). Recent models use memory kernels and coupled fields. Gorenflo and Mainardi provide mathematical background on fractional calculus Gorenflo and Mainardi [1997]. Memory-dependent derivatives use time-history weights Wang and Li [2011]; fractional models replace integer-order derivatives with fractional orders in the governing equations Caputo [1967], Povstenko [2004, 2007, 2009, 2011a, 2015]. Couplings to electrical, porous, micro-rotational, and carrier fields produce multiphysics formulations Ieşan [1986], Ieşan [2013], Cowin and Nunziato [1983], Nunziato and Cowin [1979], Chandrasekharaiah [1996], Marotti de Sciarra [2013], Sherief and Hussein [2022], Abouelregal et al. [2023]. These citations identify model families, not shared equations; kernels, fractional orders, couplings, admissibility conditions, and physical consequences remain model-dependent.

2.3 International structure of the field

The development of generalized thermoelasticity involved several research communities. Foundational heat-conduction and thermoelasticity works, micropolar extensions, anisotropic models, fractional and memory-dependent formulations, and more recent multiphysics studies are represented in the cited literature Joseph and Preziosi [1989], Biot [1956], Lord and Shulman [1967], Green and Lindsay [1972], Eringen [1966], Tauchert et al. [1968], Eringen [1972], Dhaliwal and Sherief [1980], Sherief and Hussein [2022], Povstenko [2004, 2007, 2009, 2011a, 2015], Wang and Li [2011], Ieşan [1986], Ieşan [2013], Cowin and Nunziato [1983], Nunziato and Cowin [1979], Chandrasekharaiah [1996], Marotti de Sciarra [2013], Abouelregal et al. [2023], Quintanilla [2019], Barak et al. [2023a,b,c,d], Gupta et al. [2023, 2022], Kumar et al. [2023a], Gupta et al. [2026].

One interpretive reading is that research communities approached generalized thermoelasticity from different interests — heat conduction in metals, micropolar fluid mechanics, anomalous diffusion in geophysics, and piezoelectricity in smart materials — and brought different constitutive tools. In the cited literature, questions about propagating thermal modes are addressed through different devices: relaxation times, second temperatures, thermal displacements, phase lags, memory kernels, and fractional orders. Organizing by these devices can make cross-family recurrences visible, though the proposed links between communities and modelling choices are interpretive rather than a complete causal history. The field can be viewed as negotiating finite propagation speed, thermodynamic admissibility, and well-posedness, which may compete under particular model assumptions.

2.4 The literature corpus

The studies discussed here range from foundational analyses and surveys to applied work on coupled, nonlocal, fractional, and memory-dependent models. Comparisons are organized by constitutive structure and stated assumptions rather than by bibliographic metadata alone. This is a qualitative map of cited examples, not an exhaustive census or an evidence-strength ranking.

2.5 Development axes and the gaps they expose

The corpus can be classified along five axes — physical domain, mathematical structure, modal content, coupling topology, and validation regime — and this classification is developed formally in Section 14. In the cited examples, work appears more concentrated on single-field, one-dimensional, interface-focused classical, relaxation-time, and phase-lag problems than on experimental discrimination of constitutive novelties. The cited nonlocal and fractional studies include theoretical dispersion analyses, while fewer of these examples fit predictions to measured dispersion data. Head-to-head comparisons of competing mechanisms — nonlocal versus gradient, exponential versus power-law memory, phase lag versus relaxation time — also appear less represented in the cited set. Interfaces with memory or nonlocality require attention to both constitutive history and geometric matching; energy partition depends on the full model, not the local constitutive law alone. These density comparisons are qualitative observations from the cited set, not an exhaustive count or a universal claim about model-family maturity.

These apparent gaps need not indicate defects in individual contributions. One possible explanation is that many studies extend a constitutive law and solve the resulting modal problem, while direct experimental comparisons between competing extensions are less common in the cited set. Section 16 takes this diagnosis as the basis for its research directions.

2.6 Existing reviews and the contribution of this article

Reviews already exist for several individual theory families: nonlocal elasticity and its application to nanostructures Askes and Aifantis [2006, 2011], Maranganti and Sharma [2007], fractional thermoelasticity Chandrasekharaiah [1998], Povstenko [2015], hyperbolic thermoelasticity and second sound Chandrasekharaiah [1996], Joseph and Preziosi [1989], and generalized thermoelasticity with modified models Shakeriaski et al. [2021]. Their common structure is the one this review sets aside: an enumeration of theory families, each with its governing equations, its application domains and a list of its principal results, with the families treated in isolation and the transitions between them left implicit.

The present review differs in three respects. It is organised by the problems — finite propagation speed, thermodynamic admissibility, well-posedness, scale dependence, interface partition

of energy — rather than by the theories, so that a reader interested in one problem can follow it across all the theories that address it. It states the cost of each modification alongside the benefit, which the standard format does not do, and which the literature itself rarely does: the second temperature, the loss of ellipticity, the loss of uniqueness of the initial-value problem, and the non-locality of the energy flux are each the price of something that was bought. It separates a mechanism's plausibility from the empirical support for it, treating the evidence base as an explicit axis of classification rather than a matter of emphasis.

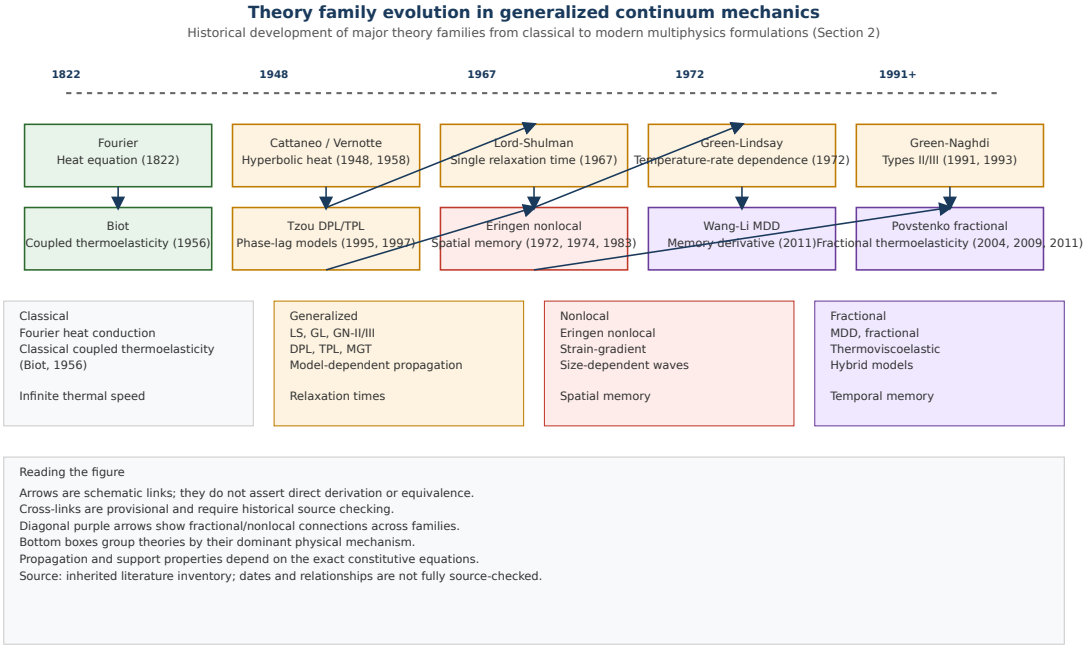


Figure 3: Selected theory families and relationships in generalized continuum mechanics. This schematic is a conceptual taxonomy of the model families discussed in the text.

2.7 Citation and assumptions

A useful comparison states the validity domain and conventions of each equation and distinguishes a derivation from an attributed result. The assumptions used in each cited formulation are therefore identified alongside the relevant discussion.

Where material errors are conclusively established in the literature, they are recorded transparently and without personal characterisation. Where uncertainty exists, it is acknowledged as an open question. No error allegations are made against any publication, and model-specific differences are never presented as mathematical contradictions without sufficient evidence.

3 Foundations of Continuum Wave Mechanics

This section develops the first-principles foundation (Layer I of the layered mathematical exposition) for all subsequent theory families. We begin with kinematics, proceed through the conservation laws of continuum mechanics, establish the thermodynamic restrictions on constitutive relations, derive classical elasticity and wave propagation, and conclude with the classical

energy flux Biot [1956] and interface conditions. Every equation in this section is derived from first principles or drawn from established standard results with documented assumptions.

3.1 Kinematics: displacement, strain, rotation

Let $\boldsymbol{u}(\boldsymbol{x}, t)$ denote the displacement field of a continuum body occupying a region $\Omega \subset \mathbb{R}^3$ with boundary $\partial\Omega$. The displacement gradient tensor decomposes additively into symmetric and antisymmetric parts:

$$u_{i,j} = \varepsilon_{ij} + \omega_{ij}, \tag{1}$$

where the infinitesimal strain tensor and rotation tensor are defined by

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}), \qquad \omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i}). \tag{2}$$

The strain tensor ε_{ij} measures the symmetric part of the displacement gradient (deformation), while the rotation tensor ω_{ij} measures the antisymmetric part (rigid-body rotation). The velocity field is the material time derivative of displacement:

$$v_i = \dot{u}_i = \frac{\partial u_i}{\partial t}. \tag{3}$$

3.1.1 Physical interpretation

The strain components ε_{ii} represent normal strains (relative changes in length along coordinate directions), while ε_{ij} $(i \neq j)$ represent shear strains (changes in angle between originally orthogonal material lines). The infinitesimal strain assumption requires $\|\varepsilon_{ij}\| \ll 1$, which is valid for small deformations typical of linearized wave propagation problems.

3.2 Conservation laws: linear momentum and angular momentum

3.2.1 Linear momentum balance

Consider a material subvolume $V \subset \Omega$ with surface $S = \partial V$ and outward unit normal $\boldsymbol{n}$. The linear momentum balance states that the time rate of change of linear momentum equals the sum of surface tractions and body forces:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho v_i \,\mathrm{d}V = \int_S \sigma_{ji} n_j \,\mathrm{d}S + \int_V \rho f_i \,\mathrm{d}V, \tag{4}$$

where ρ is the mass density, σ_{ji} is the Cauchy stress tensor, and f_i is the body force per unit mass. Applying the divergence theorem to the surface integral and localizing (since V is arbitrary) yields the pointwise form:

$$\rho \dot{v}_i = \sigma_{ji,j} + \rho f_i. \tag{5}$$

Since $v_i = \dot{u}_i$, this can also be written in terms of displacement:

$$\rho \ddot{u}_i = \sigma_{ji,j} + \rho f_i. \tag{6}$$

Equation (5) is the fundamental equation of motion for continuum mechanics. It is sometimes called Cauchy's first law or the equation of linear momentum.

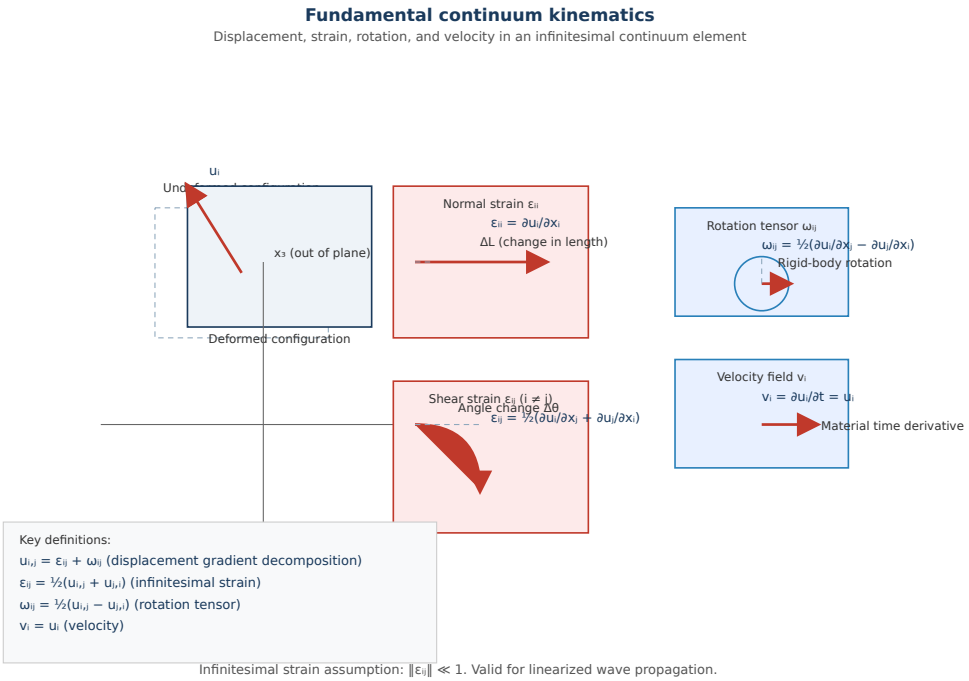


Figure 4: Kinematics of a continuum element. The displacement field $\boldsymbol{u}(\boldsymbol{x}, t)$ defines the deformation gradient, which in the infinitesimal regime splits additively into a symmetric part (strain, the extensional part of the deformation) and an antisymmetric part (rotation, the rigid part). The rigid part does no work and is removed from the balance laws by the objectivity requirement, so that only the strain enters the stress; this is the first of the two places where irreversibility enters the formulation, the second being the second law of Section 3.4.

3.2.2 Angular momentum balance

The angular momentum balance for an arbitrary subvolume V states:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{x}\times\boldsymbol{v})_k\,\mathrm{d}V = \int_S \epsilon_{kij}x_i\sigma_{jm}n_m\,\mathrm{d}S + \int_V \rho\epsilon_{kij}x_i f_j\,\mathrm{d}V, \tag{7}$$

where ϵ_{kij} is the Levi-Civita permutation symbol. Applying the divergence theorem, using the linear momentum balance (5), and localizing gives:

$$\sigma_{ij} = \sigma_{ji}. \tag{8}$$

Equation (8) states that the Cauchy stress tensor is symmetric in the absence of couple stresses or body couples. This symmetry is essential for the energy flux derivations in Sections 3.8 and 10.2.

3.2.3 Physical interpretation

Equations (5) and (8) are local statements of Newton's second law applied to continuum matter. The symmetry of stress ((8)) means that traction on a surface is determined solely by the normal to that surface, without additional rotational degrees of freedom in classical (non-polar) continua.

3.3 Mechanical power and internal energy: first law of thermodynamics

3.3.1 Mechanical power balance

Multiply the linear momentum equation (5) by v_i and integrate over V:

$$\int_V \rho\dot{v}_i v_i\,\mathrm{d}V = \int_V \sigma_{ji,j}v_i\,\mathrm{d}V + \int_V \rho f_i v_i\,\mathrm{d}V. \tag{9}$$

The left-hand side is the rate of change of kinetic energy:

$$\int_V \rho\dot{v}_i v_i\,\mathrm{d}V = \frac{\mathrm{d}}{\mathrm{d}t}\int_V \frac{1}{2}\rho v_i v_i\,\mathrm{d}V. \tag{10}$$

For the stress term, use the identity $\sigma_{ji,j}v_i = (\sigma_{ji}v_i)_{,j} - \sigma_{ji}v_{i,j}$. Since $v_{i,j} = \dot{u}_{i,j} = \dot{\varepsilon}_{ji}$ (using symmetry of σ_{ji} and ε_{ij}), we obtain:

$$\int_V \sigma_{ji,j}v_i\,\mathrm{d}V = \int_S \sigma_{ji}v_i n_j\,\mathrm{d}S - \int_V \sigma_{ji}\dot{\varepsilon}_{ij}\,\mathrm{d}V. \tag{11}$$

Combining (9) through (11) gives the mechanical power balance:

$$\rho\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}v_i v_i\right) = \sigma_{ji,j}v_i + \rho f_i v_i = -\sigma_{ij}\dot{\varepsilon}_{ij} + (\sigma_{ij}v_j)_{,i} + \rho f_i v_i. \tag{12}$$

The term $\sigma_{ij}v_j n_i$ on the surface represents mechanical power entering through the boundary per unit area; its density $(\sigma_{ij}v_j)_{,i}$ is the mechanical power flux divergence.

3.3.2 Internal energy balance

The first law of thermodynamics for a continuum states that the rate of change of internal energy equals the mechanical power input minus the heat flux out of the body plus heat supply:

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i} + \rho r, \tag{13}$$

where e is the specific internal energy (per unit mass), q_i is the heat flux vector, and r is the heat supply per unit mass. In the absence of external heat supply ($r = 0$):

$$\rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i}. \tag{14}$$

3.3.3 Physical interpretation

Equation (13) expresses energy conservation: the internal energy of a material element changes due to (i) mechanical work done by stresses ($\sigma_{ij}\dot{\varepsilon}_{ij}$), (ii) heat conduction out of the element ($-q_{i,i}$), and (iii) external heat supply (ρr). The term $\sigma_{ij}\dot{\varepsilon}_{ij}$ is the stress power density — the rate of mechanical work converted into internal energy or stored as elastic energy.

3.4 Entropy and dissipation inequality: Clausius–Duhem inequality

The second law of thermodynamics, in the Clausius–Duhem form, states that the entropy production within a continuum is non-negative:

$$\rho \dot{s} \geq -\left(\frac{q_i}{\Theta}\right)_{,i} + \frac{\rho r}{\Theta}, \tag{15}$$

where s is specific entropy and Θ is absolute temperature. In the linear constitutive equations below, $T = \Theta - T_0$ denotes the temperature increment. Using the internal-energy balance (13) and defining the Helmholtz free energy by $\psi = e - \Theta s$, we obtain after standard manipulation:

$$\sigma_{ij}\dot{\varepsilon}_{ij} - \rho\left(\dot{\psi} + s\dot{\Theta}\right) - \frac{q_i \Theta_{,i}}{\Theta} \geq 0. \tag{16}$$

Equation (16) is the Clausius–Duhem dissipation inequality. The left-hand side represents the local entropy production rate (multiplied by temperature) and must be non-negative for all admissible thermodynamic processes.

3.4.1 Physical interpretation

The dissipation inequality restricts admissible constitutive relations. Any constitutive model must be constructed so that (16) holds for all processes. This principle is used in Section 3.5 to derive Coleman–Noll restrictions and in Section 10.7 to assess thermodynamic admissibility of generalized theories.

3.5 Constitutive principles: free energy and linear relations

3.5.1 Coleman–Noll procedure

Assuming the free energy depends on strain and temperature, $\psi = \psi(\varepsilon_{ij}, T)$, we have:

$$\dot{\psi} = \frac{\partial \psi}{\partial \varepsilon_{ij}} \dot{\varepsilon}_{ij} + \frac{\partial \psi}{\partial T} \dot{T}. \tag{17}$$

Substituting into the dissipation inequality (16) and requiring it to hold for arbitrary $\dot{\varepsilon}_{ij}$ and $\dot{T}$ yields the Coleman–Noll restrictions:

$$\sigma_{ij} = \rho \frac{\partial \psi}{\partial \varepsilon_{ij}}, \qquad s = -\frac{\partial \psi}{\partial T}. \tag{18}$$

With these restrictions, the dissipation inequality reduces to the thermal conduction restriction:

$$-\frac{q_i T_{,i}}{T} \geq 0. \tag{19}$$

For isotropic linear conduction with Fourier's law $q_i = -kT_{,i}$ ($k > 0$), we have $-q_i T_{,i}/T = k(T_{,i})^2/T \geq 0$, which is satisfied.

3.5.2 Linear isotropic constitutive relations

For a linear isotropic thermoelastic material, with temperature increment $T = \Theta - T_0$ and thermal-stress coefficient β in Pa K^{-1}, take the specific Helmholtz free energy:

$$\psi = \frac{1}{2\rho}(\lambda \varepsilon_{kk}\varepsilon_{ll} + 2\mu \varepsilon_{ij}\varepsilon_{ij}) - \frac{\beta}{\rho}\varepsilon_{kk} T - s_0 T - \frac{c_v}{2T_0} T^2 + \text{const.}, \tag{20}$$

where λ and μ are Lamé elastic constants, β is the thermal stress coefficient, and c_v is the specific heat at constant strain. Applying (18) gives:

$$\sigma_{ij} = \lambda \varepsilon_{kk}\delta_{ij} + 2\mu \varepsilon_{ij} - \beta T \delta_{ij}, \tag{21}$$

$$s = s_0 + \frac{\beta}{\rho}\varepsilon_{kk} + \frac{c_v}{T_0} T. \tag{22}$$

Equation (21) is the linear isotropic thermoelastic constitutive relation: the first two terms are the classical elastic stress (Hooke's law), and the last term is the thermal stress contribution.

3.5.3 Physical interpretation

The constitutive relations (21)–(22) couple mechanical and thermal fields: temperature changes induce thermal stresses ($-\beta T \delta_{ij}$), and deformation changes entropy ($\frac{\beta}{\rho}\varepsilon_{kk}$). This coupling is the physical origin of thermoelastic wave propagation.

3.6 Classical elasticity: Navier equations and wave speeds

Substituting the constitutive relation (21) into the equation of motion (6) with no body forces ($f_i = 0$) and isothermal conditions ($T = T_0$ constant), we obtain the Navier equations of classical elasticity:

$$\rho \ddot{u}_i = (\lambda + \mu) u_{j,ji} + \mu u_{i,jj}. \tag{23}$$

3.6.1 Plane wave solutions

For harmonic plane waves of the form $u_i = \hat{u}_i e^{i(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}$, substitution into (23) gives the Christoffel equation. For longitudinal (P) waves propagating in direction $\boldsymbol{n}$ with polarization parallel to $\boldsymbol{n}$:

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \tag{24}$$

and for transverse (S or SV) waves with polarization perpendicular to $\boldsymbol{n}$:

$$c_s = \sqrt{\frac{\mu}{\rho}}. \tag{25}$$

These are the classical elastic wave speeds. The P-wave speed exceeds the S-wave speed since $\lambda + 2\mu > \mu$ for stable materials ($\lambda + 2\mu/3 > 0$, $\mu > 0$).

3.6.2 Physical interpretation

Equation (23) describes how elastic disturbances propagate through a homogeneous isotropic solid. The P-wave involves volume change (compressional), while the S-wave involves shear (distortional) deformation. Both propagate without dispersion in classical elasticity — all frequencies travel at the same speed.

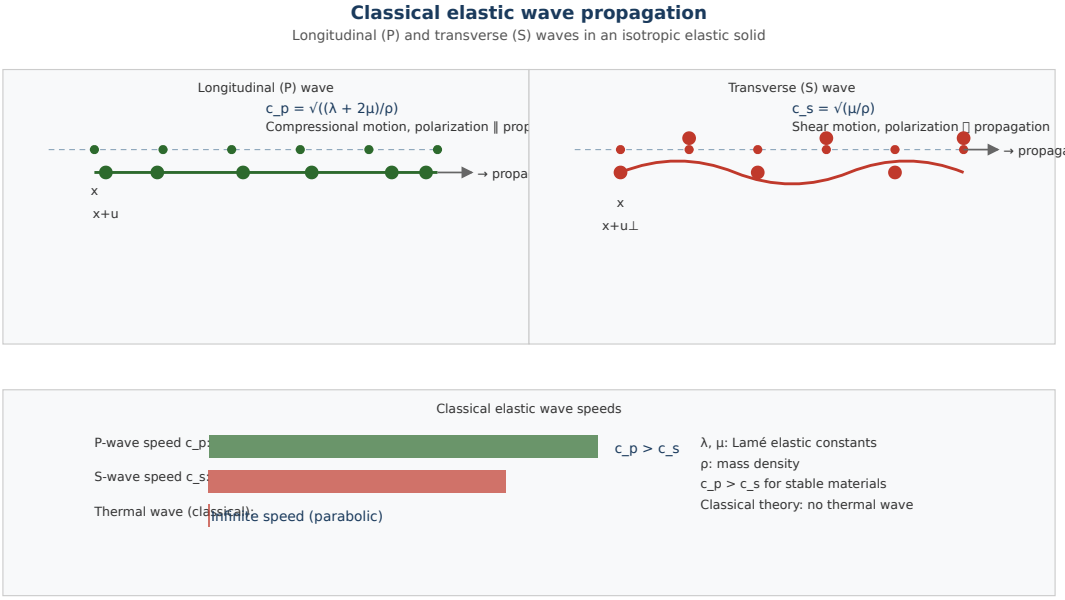


Figure 5: Classical elastic wave propagation in an isotropic solid. A longitudinal (P) wave, whose polarization is parallel to the propagation direction, travels at $c_p = \sqrt{(\lambda + 2\mu)/\rho}$; a transverse (S) wave, whose polarization is perpendicular to it, travels at $c_s = \sqrt{\mu/\rho}$. Because the elastic moduli are independent of frequency, both speeds are dispersion-free: this constancy is the reference against which the dispersive branches of Sections 4–7 are measured, and it is the property that a generalized theory must forfeit when it introduces a length scale or a memory kernel.

3.7 Plane wave assumptions and harmonic wave form

For time-harmonic problems, all field variables take the form:

$$u_i(\boldsymbol{x}, t) = \hat{u}_i(\boldsymbol{x}) e^{-i\omega t}, \qquad \sigma_{ij}(\boldsymbol{x}, t) = \hat{\sigma}_{ij}(\boldsymbol{x}) e^{-i\omega t}, \qquad q_i(\boldsymbol{x}, t) = \hat{q}_i(\boldsymbol{x}) e^{-i\omega t}, \tag{26}$$

where $\hat{u}_i$, $\hat{\sigma}_{ij}$, $\hat{q}_i$ are complex amplitudes and ω is the angular frequency. The time convention $e^{-i\omega t}$ implies that positive imaginary parts of complex wave numbers correspond to attenuation

(decay in space) and negative imaginary parts of complex frequencies correspond to temporal decay.

3.7.1 Time-average identity

For two harmonic quantities $a(t) = \Re[\hat{a}e^{-i\omega t}]$ and $b(t) = \Re[\hat{b}e^{-i\omega t}]$, the time average over one period is:

$$\langle ab \rangle_t = \frac{1}{2}\Re[\hat{a}\hat{b}^*], \tag{27}$$

where * denotes complex conjugation. This identity is used throughout Sections 9 and 10 to compute average energy fluxes and energy ratios from complex wave amplitudes.

3.8 Energy flux in classical elasticity

Adding the mechanical power balance (12) and the internal-energy balance (14) (with $r = 0$), and defining total energy per unit volume as $\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$, gives an outward-positive flux $J_j = q_j - \sigma_{ji}v_i$:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial}{\partial x_j}\left(q_j - \sigma_{ji}v_i\right) = \rho f_i v_i. \tag{28}$$

With no body forces ($f_i = 0$), this becomes the local energy conservation law:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial J_i}{\partial x_i} = 0, \qquad J_i = q_i - \sigma_{ij}v_j. \tag{29}$$

Equation (29) states that the total energy density changes only due to the divergence of the energy flux J_i. The energy flux has two contributions:

- $-\sigma_{ij}v_j$: mechanical contribution to the outward-positive flux, and
- q_i: thermal energy flux (heat conduction across a surface per unit area).

Integrating (29) over a volume V and applying the divergence theorem:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \mathcal{E}\,\mathrm{d}V + \int_S J_i n_i\,\mathrm{d}S = 0. \tag{30}$$

For an isolated system ($J_i n_i = 0$ on S), the total energy is conserved: $\frac{\mathrm{d}}{\mathrm{d}t}\int_V \mathcal{E}\,\mathrm{d}V = 0$.

3.8.1 Physical interpretation

Equation (29) is the classical energy flux Biot [1956] for elastic and thermoelastic media. With the outward-positive convention, the mechanical contribution is $-\sigma_{ij}v_j$ and the thermal contribution is q_i. Reversing the normal or defining inward-positive flux reverses the signs consistently. This local balance is derived for the classical thermoelastic model only. Whether an additional stored-energy or boundary-power term is required in a generalized theory depends on its constitutive equations and must be derived model by model.

3.9 Interface conditions: perfect and imperfect bonding

Consider an interface Σ between two media (denoted 1 and 2) with common unit normal $\boldsymbol{n}$. For perfectly bonded interfaces:

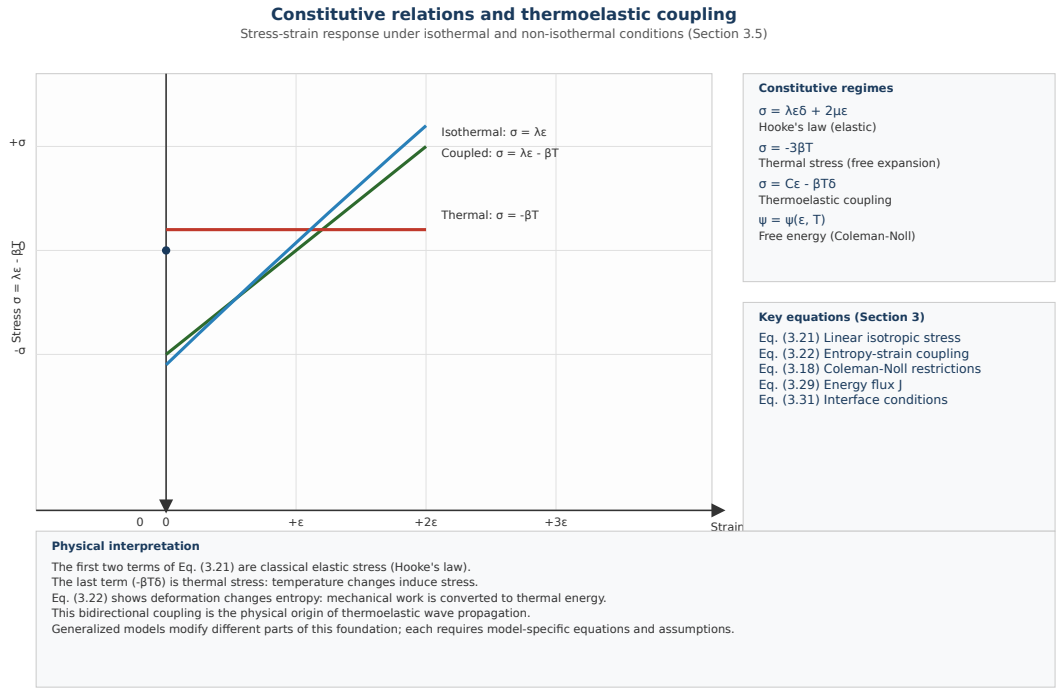


Figure 6: Constitutive relations and thermoelastic coupling: stress-strain response under isothermal and non-isothermal conditions (Section 3.5).

3.9.1 Perfect bonding

$$u_i^{(1)} = u_i^{(2)}, \qquad \sigma_{ji}^{(1)} n_j = \sigma_{ji}^{(2)} n_j, \qquad T^{(1)} = T^{(2)}, \qquad q_i^{(1)} n_i = q_i^{(2)} n_i, \tag{31}$$

on Σ. The first condition enforces displacement continuity (no slip or separation), the second enforces traction equilibrium (Newton's third law at the interface), and the last two enforce thermal contact (temperature continuity and heat flux continuity).

3.9.2 Imperfect bonding: spring-type conditions

For imperfectly bonded interfaces, displacement and traction are related by interfacial stiffness parameters:

$$[u_i] = (C^u)^{-1}_{ij} t_j, \qquad \sigma_{ki}^{(1)} n_k = \sigma_{ki}^{(2)} n_k = t_i, \tag{32}$$

where $[u_i] = u_i^{(2)} - u_i^{(1)}$, t_i is the common traction on the shared normal, and C^u_{ij} is a positive-definite interfacial stiffness tensor (units Pa/m). The compliance form is used so that, when all stiffness eigenvalues tend to infinity, the displacement jump vanishes; when they tend to zero through positive values, the traction vanishes for a finite jump. More general imperfect-interface laws require their own constitutive and dissipation assumptions.

3.9.3 Electrical interface conditions (piezoelectric media)

For piezoelectric media, additional interface conditions involve the electric potential ϕ and electric displacement D_i:

$$\phi^{(1)} = \phi^{(2)} \qquad \text{(perfect electrical contact)}, \tag{33}$$

$$D_i^{(1)} n_i = D_i^{(2)} n_i \qquad \text{(electric displacement continuity)}, \tag{34}$$

or for an insulated interface:

$$D_i n_i = 0. \tag{35}$$

3.9.4 Physical interpretation

Interface conditions determine how waves transmit and reflect at material boundaries. Perfect bonding ((31)) is the standard assumption for welded interfaces. Imperfect bonding ((32)) models thin adhesive layers, micro-cracks, or weakly bonded interfaces. Electrical conditions ((33)–(35)) are essential for piezothermoelastic interface problems in Section 5 and Section 9.

Table 3: Wave types and representative characteristics across continuum models.

Wave type	**Polarization or field pattern**	**Classical reference feature**	**Model-specific interpretation**
Longitudinal (P)	Parallel to propagation	$c_p = \sqrt{(\lambda + 2\mu)/\rho}$ for isotropic elasticity	Generalized mode labels and speeds follow the selected constitutive equations
Transverse (S)	Perpendicular to propagation	$c_s = \sqrt{\mu/\rho}$ for isotropic elasticity	Coupling can alter the modal structure
Thermal	Temperature and heat-flux fields	Diffusion in the Fourier model	Relaxation, phase-lag, and memory formulations have model-dependent propagation behavior
Coupled modes	Mixed mechanical, thermal, or electrical fields	Eigenfrequencies of the coupled system	Branches follow the coupled field equations and boundary conditions
Surface waves	Amplitude decays with depth from the surface	Rayleigh-type branch in an elastic half-space	Constitutive laws and boundary conditions determine the surface-wave spectrum
Evanescent modes	Spatially decaying amplitude	Imaginary part of the wave number	Near-field contributions can enter interface and scattering solutions

Wave labels are representative; mode count, speed, and attenuation depend on the full constitutive system and boundary conditions.

3.10 Notation summary for Section 3

Table 4 summarizes the notation introduced in this section.

3.11 Open questions for this section

OQ3.1 What is the most general form of energy flux that remains valid across all generalized theories (LS, GL, GN, DPL, TPL, MGT, MDD, nonlocal)?

OQ3.2 How do interface conditions change for nonlocal media where the stress at a point depends on strain throughout a neighborhood?

Table 4: Notation introduced in Section 3.

Symbol	Meaning	Units	Equation
u_i	Displacement	m	(1)
ε_{ij}	Infinitesimal strain	dimensionless	(2)
ω_{ij}	Rotation tensor	dimensionless	(2)
v_i	Velocity	m/s	(3)
σ_{ij}	Cauchy stress	Pa	(5)
ρ	Mass density	kg/m^3	(5)
f_i	Body force	N/kg	(5)
q_i	Heat flux	W/m^2	(13)
T	Temperature increment $\Theta - T_0$	K	(20)
e	Specific internal energy	J/kg	(13)
ψ	Specific free energy	J/kg	(18)
s	Specific entropy	J/(kg·K)	(18)
$\mathcal{E}$	Total energy density	J/m^3	(29)
J_i	Energy flux	W/m^2	(29)
λ, μ	Lamé constants	Pa	(23)
c_p, c_s	P-wave and S-wave speeds	m/s	(24)–(25)

OQ3.3 What is the relationship between the classical energy flux Biot [1956] (29) and the entropy flux in thermoelasticity without energy dissipation (GN-II)?

4 Classical and Generalized Thermoelastic Theories

This section presents the historical development, governing equations, wave propagation characteristics, and dissipation structure of classical and generalized thermoelastic theories. Each theory is introduced with its physical motivation, mathematical formulation, and domain of applicability. Representative equations are selected to illustrate each theory family; derivations from first principles are given where they illuminate the theory's structure. Citations to primary sources are provided throughout to distinguish established results from open questions.

4.1 Historical evolution

The development of generalized thermoelasticity spans nearly two centuries, from Fourier's heat equation to modern memory-dependent and nonlocal formulations. Table 5 summarizes the principal milestones.

Chandrasekharaiah's review provides historical context for generalized thermoelasticity; it is not primary evidence for the equations used here Chandrasekharaiah [1998].

For the historical overview, see Figure 1 in Section 1.

A recurring motivation for some generalized heat-transport models is that Fourier conduction is parabolic and has formally infinite propagation speed. Generalized constitutive choices explore alternatives for selected regimes, but finite-speed behavior and thermodynamic consistency are model-specific properties that require derivation; neither follows from the label "generalized."

Joseph and Preziosi's review (1989) and 1990 addendum summarize the second-sound evidence and the objections raised in the intervening discussion Joseph and Preziosi [1989, 1990]. This debate helped motivate later finite-speed heat-transport models; it does not establish a universal need for them.

Table 5: Selected publication milestones and broad modelling mechanisms.

Model/family	**Year**	**Source**	**Broad modelling feature**
Classical coupled thermoelasticity	1956	Biot [1956]	Coupled mechanical and thermal fields with Fourier conduction
Lord–Shulman	1967	Lord and Shulman [1967]	Heat-flux relaxation model
Green–Lindsay	1972	Green and Lindsay [1972]	Single temperature; rate dependence with two relaxation times
Green–Naghdi II / III	1993 / 1992	Green and Naghdi [1993, 1992]	Thermal-displacement formulations; type III includes a rate term
Dual/triple phase lag	1995/2007	Tzou [1995], Choudhuri [2007]	Lagged constitutive forms, including first-order TPL
Eringen-type nonlocality	1972 onward	Eringen [1972, 1974, 1983]	Spatially nonlocal constitutive response
Memory-dependent operator	2011 onward	Wang and Li [2011]	History-dependent heat-transport formulation
Moore–Gibson–Thompson	1960; thermoelastic analogues later	Moore and Gibson [1960], Quintanilla [2019]	Relaxing-gas acoustics and later thermal analogues

This table is a navigation aid, not a validated chronology or exhaustive taxonomy; no propagation-speed or dissipation comparison is asserted.

4.2 Classical coupled thermoelasticity (Fourier heat conduction)

The classical coupled thermoelasticity theory, developed by Biot (1956) Biot [1956], combines the equations of linear elasticity with Fourier's law of heat conduction:

$$\rho \ddot{u}_i = (\lambda + \mu) u_{j,ji} + \mu u_{i,jj} - \beta T_{,i}, \tag{36}$$

$$\rho c_v \dot{T} = k \nabla^2 T - \beta T_0 \dot{u}_{i,i}, \tag{37}$$

where ρ is density, λ and μ are Lamé constants, β is the thermal stress coefficient, c_v is specific heat at constant strain, k is thermal conductivity, and T_0 is the reference temperature.

4.2.1 Wave speeds

The classical theory predicts two elastic wave types:

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \qquad c_s = \sqrt{\frac{\mu}{\rho}}. \tag{38}$$

The thermal field satisfies a parabolic diffusion equation, predicting infinite thermal propagation speed — the classical paradox that motivated all subsequent generalized theories.

4.2.2 Physical interpretation

Equations (36)–(37) are two-way coupled: volumetric strain rate sources temperature, while the temperature gradient contributes to mechanical stress. In the fully dynamic problem neither field can in general be solved independently; sequential treatments are approximations requiring explicit scale assumptions.

4.3 Lord–Shulman (LS) theory (1967)

Lord and Shulman (1967) Lord and Shulman [1967] introduced a single thermal relaxation time τ_0 into Fourier's law:

$$q_i + \tau_0 \dot{q}_i = -k T_{,i}, \tag{39}$$

where $\tau_0 > 0$ is the relaxation time and q_i is the heat flux vector.

4.3.1 Modified heat equation

Combining (39) with the energy balance yields the LS heat equation:

$$\rho c_v \left(\dot{T} + \tau_0 \ddot{T} \right) = k \nabla^2 T - \beta T_0 (\dot{u}_{i,i} + \tau_0 \ddot{u}_{i,i}). \tag{40}$$

4.3.2 Finite thermal wave speed

The thermal wave speed in LS theory is:

$$c_T = \sqrt{\frac{k}{\rho c_v \tau_0}}. \tag{41}$$

For the displayed LS/Cattaneo law, the formal thermal characteristic speed is finite when $\tau_0 > 0$. The limiting substitution $\tau_0 \to 0$ recovers Fourier conduction at the level of the

constitutive equation; convergence of solutions is a separate question. These statements are conditional on the displayed model.

4.3.3 Well-posedness

For the displayed Cattaneo-type heat-flux law, $\tau_0 > 0$ gives a finite characteristic speed for the thermal branch. This parameter condition alone is not a proof that the full coupled initial-boundary-value problem is well-posed; the latter depends on coefficients, domain, and boundary/initial data. A separate fractional Lord–Shulman formulation couples a Klein–Gordon nonlocal operator to the heat law; its printed $\alpha \to 0$ branch is not used here as a limiting identity Das et al. [2026].

4.4 Green–Lindsay (GL) theory (1972)

Green–Lindsay denotes a temperature-rate-dependent thermoelastic model with one temperature field. Related formulations may introduce additional temperature-like fields; the constitutive equations and admissible parameter ranges depend on the selected model.

4.5 Green–Naghdi type II theory (1993)

The type-II system treated here is Green and Naghdi's 1993 thermoelasticity without energy dissipation. Let α_{GN} be the thermal displacement, with $\dot{\alpha}_{\mathrm{GN}} = T$; it is distinct from the fractional order α.

4.5.1 Thermal displacement gradient formulation

For the source-free form, time integration with zero integration constant gives the relation Green and Naghdi [1993]:

$$\kappa^* \nabla^2 \alpha_{\mathrm{GN}} = \rho c_e \ddot{\alpha}_{\mathrm{GN}} + \beta T_0 \dot{u}_{i,i}. \tag{42}$$

The strain-rate coupling carries one fewer time derivative than the inherited form. Here κ^* is the thermal-displacement coefficient; below, κ denotes the coefficient on the temperature-rate term in GN-III. In the displayed isotropic convention, the candidate entropy flux uses the same scalar coefficient:

$$p_i = \kappa^* \alpha_{\mathrm{GN},i}, \tag{43}$$

where p_i is the entropy-flux vector. No anisotropic tensor generalization is asserted here.

4.5.2 Energy conservation

The 1993 type-II paper describes harmonic-wave propagation without damping for the displayed system and stated coefficients. The schematic balance below is not an energy identity.

$$\frac{\partial \mathcal{E}}{\partial t} + \partial_i J_i = \mathcal{P}. \tag{44}$$

Wave attenuation and energy conservation for type II must be obtained from the specific model equations and boundary-value problem; no application suitability claim is made here. A separate rheological network derivation relates CV/GN/MCV/MGN flux laws and includes second-order coefficients proportional to τ^2; those coefficients do not establish energy ratios or admissibility for the displayed system Huang et al. [2025].

4.6 Green–Naghdi type III theory (1992)

Green and Naghdi's 1992 type-III system contains both the temperature-rate and thermal-displacement contributions. The displayed source-free isotropic equation uses κ for the rate term and κ^* for the non-dissipative term, consistently with Eq. (42) Green and Naghdi [1992].

4.6.1 Governing equations

$$\rho c_e \ddot{T} + \beta T_0 \ddot{u}_{i,i} = \kappa \nabla^2 \dot{T} + \kappa^* \nabla^2 T. \tag{45}$$

A distinct nonlocal GN-III formulation prints a flux law containing both the temperature-gradient and thermal-displacement-gradient contributions; it is not substituted into the local source-free form shown here Biswas [2020b]. This source-free form is conditional on the displayed linear isotropic system; the source term and material normalization must be restored for other conventions. The transcribed restrictions include $\kappa \geq 0$, $c \geq 0$, and $\kappa^{**} = 0$; no additional sign restriction on κ^* is inferred here.

Any proposed type-III-to-type-II or classical limit must be verified by substitution into the exact system and by a convergence analysis. No general limit theorem is asserted here.

4.7 Dual-phase-lag (DPL) theory (Tzou, 1995)

Tzou (1995) introduced two phase lags into Fourier's law:

$$q_i(t + \tau_q) = -kT_{,i}(t + \tau_T), \tag{46}$$

where τ_q is the heat flux phase lag and τ_T is the temperature gradient phase lag.

4.7.1 Taylor expansion

First-order Taylor expansion about t:

$$q_i + \tau_q \dot{q}_i = -k(T_{,i} + \tau_T \dot{T}_{,i}). \tag{47}$$

Some later DPL variants retain a $\tau_q^2/2$ term in a second-order flux expansion (Yang et al.; Gupta et al.); these are distinct model-specific truncations, not part of Eq. (47). No universal lag-order admissibility follows Yang et al. [2021], Gupta and Barak [2023]. Formal reductions can sometimes be obtained by setting one or both lag parameters to zero in a specified DPL equation, but the resulting equations and solution limits must be checked for that model. The ordering $\tau_q \geq \tau_T$ is not asserted here as a universal thermodynamic-admissibility condition; a model-specific free-energy/entropy derivation is required.

Guyer–Krumhansl (GK) phonon transport. Guyer and Krumhansl's 1966 kinetic treatment derives the nonlocal phonon-flux structure from the phonon Boltzmann equation, not from a ready-made continuum thermoelastic law. In the rapid-normal-scattering regime ($\Omega\tau_N \ll 1$), its characteristic length is $\lambda^2 = c^2\tau_N\tau_z/5$, where τ_z is the Ziman-averaged resistive-process relaxation time Guyer and Krumhansl [1966a]. Separately, Tzou (1995, Eq. 15, printed p. 10) gives $\dot{\boldsymbol{q}} + (c^2 C_p/3)\nabla T + \boldsymbol{q}/\tau_R = (\tau_N c^2/5)[\nabla^2 \boldsymbol{q} + 2\nabla(\nabla \cdot \boldsymbol{q})]$ for the pure phonon field; this particular continuum reduction is not attributed to the 1966 paper Tzou [1995]. Tzou's Eq. 17 (printed p. 10) maps this phonon case to DPL parameters $\alpha = \tau_R c^2/3$, $\tau_T = 9\tau_N/5$, and $\tau_q = \tau_R$ Tzou [1995]. These continuum and mapping statements are specific to Tzou's stated phonon approximation; they are not universal DPL identities.

4.8 Triple-phase-lag (TPL) theory (Roy Choudhuri, 2007)

Roy Choudhuri (2007) Choudhuri [2007] proposed the three-lag TPL law:

$$\begin{aligned} q_i(t+\tau_q) &= -kT_{,i}(t+\tau_T) \\ &\quad - k^* v_{,i}(t+\tau_v), \qquad \dot{v} = T. \end{aligned} \tag{48}$$

The cited source gives a first-order expansion in the three lags with $0 \le \tau_v < \tau_T < \tau_q$. We use no $\tau_\nu/2$ term. Second-order DPL variants retain a $\tau_q^2/2$ term, but they are distinct from this first-order Roy Choudhuri law and establish no shared admissibility condition Yang et al. [2021], Gupta and Barak [2023]. Other published variants include second-order DPL/TPL truncations with a free coefficient or tensor conductivity Kumar et al. [2023b], Singh et al. [2019], Biswas [2019]; a higher-order two-temperature TPL model uses a temperature lag rather than τ_v Abouelregal et al. [2020]. Further formulations combine memory-kernel bounds Singh et al. [2020] or nonlocal operators on both stress and inertia Bayones et al. [2021]; related DPL/TPL model studies are also reported by Tibullo et al. Chiriţă et al. [2017]. These are distinct constitutive variants, not one universal TPL law; no common stability or admissibility condition is inferred.

4.9 Moore–Gibson–Thompson (MGT) theory

Moore and Gibson (1960) derived a third-order acoustic wave equation for gas dynamics with relaxation; it is not a heat-conduction law for thermoelastic solids. Later Moore–Gibson–Thompson thermoelastic models are analogues that borrow this higher-order structure Moore and Gibson [1960], Quintanilla [2019]. A candidate constitutive system used in some later thermal formulations is

$$\tau \dot{q}_i + q_i = -kT_{,i} - k^* \vartheta_{m,i}, \qquad \dot{\vartheta}_m = T. \tag{49}$$

Here ϑ_m is a thermal-displacement variable and k^* must have units consistent with the selected model. This constitutive variant is discussed below with its source-specific assumptions and stability scope.

Thermoelastic MGT evidence. The displayed Eq. (49) is the thermoelastic MGT variant with relaxation in the heat-flux law and no relaxation term in the stress. Quintanilla's Eqs. (2.4), (2.5), and (3.5) give the source-specific coercivity condition $K_{ij} = k_{ij} - \tau k^*_{ij}$ positive definite, with the scalar condition $k - \tau k^* > 0$; in one dimension, $k > k^*\tau$ yields exponential stability, while $k < k^*\tau$ gives instability Quintanilla [2019]. The source uses $\alpha_{,j}$ in (2.1) but $\theta_{,j}$ in (2.5) and (3.5), without reconciling that notation; the conclusion is kept model-specific.

Bazarra et al. document a distinct radial thermoelastic MGT variant in which τ also appears in the stress coupling, $-\beta(\theta_{,i} + \tau\dot{\theta}_{,i})$; their radial decay result assumes $\bar{k} = k - \tau k^* > 0$ Bazarra et al. [2021]. Thus the relaxation placement in the stress law is variant-specific.

Acoustic MGT evidence. Acoustic MGT-with-memory PDE decay uses its own acoustic parameter set and is separate from thermoelastic admissibility. The gas-dynamics parameter $\gamma_{\text{ac}} = b - \tau c^2$ is acoustic only; it is not the thermoelastic condition $k - \tau k^* > 0$.

The available abstract for Lasiecka and Wang's acoustic MGT equation with memory reports exponential decay in the subcritical regime $\alpha\beta > \gamma$; this is an acoustic result, not a thermoelastic claim Lasiecka and Wang [2015].

The available abstract of Dell'Oro, Lasiecka, and Pata reports that at $\alpha\beta = \gamma$ exponential stability holds if and only if A is bounded; otherwise the energy tends to zero without an exponential rate Dell'Oro et al. [2016].

These results apply to the specified MGT variants and do not establish a family-wide classification.

4.10 Memory-dependent derivative (MDD) models (Wang and Li, 2011)

Memory-dependent derivative formulations use a kernel to weight the time history of a field. The kernel and normalization are part of the constitutive model, so comparisons must state the chosen form.

4.11 Fractional thermoelasticity

Fractional thermoelasticity generalizes the heat equation using fractional derivatives:

$$ {}^{C}D_t^{\alpha} T = a_\alpha \nabla^2 T, \qquad 0 < \alpha \leq 2, \tag{50} $$

where ${}^{C}D_t^{\alpha}$ is the Caputo fractional derivative of order α and a_α has dimensions $\mathrm{m}^2\,\mathrm{s}^{-\alpha}$ in this scalar equation. Thus a_1 is a thermal diffusivity, while $\sqrt{a_2}$ has dimensions of speed in the special case $\alpha = 2$. For $1 < \alpha \leq 2$ this is a fractional diffusion–wave equation; that terminology alone does not establish a thermodynamically admissible, causal, or well-posed coupled thermoelastic model. Initial data and a constitutive embedding must be specified for a physical application.

4.12 Comparative summary

Table 6 summarizes selected assumptions, state variables, flux closures, finite-speed statements, stability scope, and dissipation evidence. Cells marked not established are deliberate evidence limits, not claims that the property is absent.

Common field-equation scaffold. For the local classical Cauchy continuum, the momentum and internal-energy balances provide the comparison scaffold:

$$ \rho \ddot{u}_i = \sigma_{ji,j} + \rho f_i, \qquad \rho \dot{e} = \sigma_{ij} \dot{\varepsilon}_{ij} - q_{i,i} + \rho r. \tag{51} $$

The constitutive closures may be recorded abstractly as

$$ \sigma_{ij} = \mathcal{S}_{ij}[\varepsilon, T, \dot{T}, \alpha_{\mathrm{GN}}, \boldsymbol{z}; \boldsymbol{\theta}], \qquad \mathcal{H}_i[\boldsymbol{q}, \nabla T, \nabla \dot{T}, \mathcal{D}_t^{\alpha} T, \boldsymbol{z}; \boldsymbol{\theta}] = 0, \tag{52} $$

where $\mathcal{S}_{ij}$ and $\mathcal{H}_i$ are placeholders for the selected stress and heat-transport laws, $\boldsymbol{z}$ denotes any additional internal, thermal-displacement, or history variables, and $\boldsymbol{\theta}$ denotes the model's parameters. They are not common constitutive functions. The balances in Eq. (51) do not cover higher-gradient, micropolar, or nonlocal boundary-power terms; those require augmented balances. This scaffold therefore aligns notation without asserting one universal generalized thermoelastic model.

Table 6: Evidence-bounded comparison of selected thermoelastic and heat-transport formulations. Model-specific cells do not define a family-wide theorem.

Formulation	**Assumptions / scope**	**State variables**	**Heat-flux law / closure**	**Finite-speed property**	**Stability / well-posedness**	**Dissipation**
Classical coupled	Local coupled baseline with Fourier conduction in the displayed formulation Biot [1956].	u_i, T; q_i is constitutive.	$q_i = -kT_{,i}$.	No finite-speed property is asserted in this comparison.	Not classified here beyond the stated local balance and closure.	Fourier entropy production is nonnegative for $k, T > 0$; the full rate depends on the free energy.
Lord–Shulman (LS)	One displayed flux-relaxation formulation; source-equation convention remains under review Lord and Shulman [1967].	u_i, T, q_i.	Displayed candidate: $q_i + \tau_0 \dot{q}_i = -kT_{,i}$.	Not established here; do not infer a speed from the family label alone.	Coupled stability/well-posedness conditions are not established here.	No dissipation identity is asserted for this comparison.
Green–Lindsay (GL)	Primary abstract: linearized thermoelastic generalization with symmetric heat-conduction tensor; detailed equation assumptions are not transcribed Green and Lindsay [1972].	Exact equation-level state set not established here; do not equate GL with a two-temperature model.	Exact GL flux/stress closure not transcribed from a readable equation page.	Abstract reports second-sound effects; branch and speed are not established here.	Abstract states a uniqueness theorem; its detailed hypotheses were not recovered.	The source framework restricts entropy production; the exact rate identity remains open.
Green–Naghdi I (GN-I)	Requested comparison slot; exact source identity and equation-level assumptions remain open.	Not established.	Not established; no flux law asserted.	Not established.	Not established.	Not established.
Green–Naghdi II (GN-II)	Selected 1993 thermal-displacement formulation; source-free isotropic display uses a stated zero integration constant Green and Naghdi [1993].	$u_i, \alpha_{\mathrm{GN}}$ with $T = \dot{\alpha}_{\mathrm{GN}}$.	No heat-flux law is asserted here; p_i in Eq. (43) is entropy flux, not q_i.	The 1993 source abstract reports finite-speed thermal waves; no family-wide speed is assigned.	A uniqueness theorem is stated in the abstract; its hypotheses are not summarized here.	The cited nondissipation result is limited to this type-II formulation.

Continued on next page

Table 6: Evidence-bounded comparison (continued).

Formulation	Assumptions / scope	State variables	Heat-flux law / closure	Finite-speed property	Stability / well-posedness	Dissipation
Green–Naghdi III (GN-III)	Displayed source-free linear isotropic temperature equation; other source terms and normalizations may differ Green and Naghdi [1992].	u_i, T for the displayed equation; no internal-variable inventory is inferred.	Eq. (45) is a temperature equation, not a constitutive q_i law.	Finite-speed waves are reported for the selected source formulation; no universal speed is asserted.	No family-wide stability condition is listed; check the exact coefficients and initial-boundary problem.	Not classified here; no dissipation identity is asserted.
Dual-phase-lag (DPL)	Two shifted lags; Eq. (47) is the displayed first-order truncation; other variants are not identified with it Tzou [1995].	u_i, T, q_i in the displayed coupled form.	Displayed candidate: $q_i + \tau_q \dot{q}_i = -k(T_{,i} + \tau_T \dot{T}_{,i})$.	Not classified here for the displayed coupled problem.	In one 1D Chandrasekharaiah–Tzou problem, $\tau_\theta > \tau_q/2$ gives exponential stability and $\tau_\theta < \tau_q/2$ admits unstable solutions; equality is not classified here Quintanilla [2003].	No dissipation identity is asserted for this comparison.
Triple-phase-lag (TPL)	Source-specific first-order three-lag formulation; later truncations are distinct Choudhuri [2007].	u_i, T, q_i, v in the displayed formulation.	Eq. (48); the cited first-order expansion and lag ordering are not generalized to other TPL variants.	Not classified here for the selected closure.	Stability is not classified for this selected first-order closure.	No dissipation identity is asserted for this comparison.
MGT thermoelastic variant	One selected candidate thermoelastic formulation; Eq. (49) is not asserted as a family-wide law Quintanilla [2019].	u_i, T, q_i, ϑ_m in the displayed candidate, with $\dot{\vartheta}_m = T$.	Candidate closure: $\tau \dot{q}_i + q_i = -kT_{,i} - k^* \vartheta_{m,i}$.	Not classified here for the selected coupled problem.	A source-specific 1D coercivity note remains under review; no family-wide condition is asserted.	No dissipation identity is established here.
Memory-dependent derivative (MDD)	Finite-window operator shown as a candidate; kernel normalization, regularity, and source convention remain open Wang and Li [2011].	Field history; a finite-dimensional realization is not implied.	Candidate $D_{\omega_M} q_i$ closure; see Eq. (77); no specific kernel is selected.	Finite propagation speed is not established here.	Well-posedness depends on the exact kernel, normalization, data, and boundary problem; not classified here.	No kernel sign condition or dissipation rate is established.
Fractional scalar example	Selected scalar Caputo diffusion–wave equation, not a coupled thermoelastic model by itself Povstenko [2004, 2009].	Scalar temperature T only; no coupled state set is asserted.	${}^C D_t^\alpha T = a_\alpha \nabla^2 T$; this scalar PDE supplies no vector heat-flux law.	The $\alpha = 2$ endpoint has wave-equation form; intermediate-order propagation is not classified here.	Not classified here; depends on the specified operator, data, and any coupled embedding.	No coupled dissipation result is asserted.

Continued on next page

Table 6: Evidence-bounded comparison (continued).

Formulation	Assumptions / scope	State variables	Heat-flux law / closure	Finite-speed property	Stability / well-posedness	Dissipation
Spatially nonlocal elasticity	The Section 7 dispersion example is an elastic-only differential Helmholtz display; kernel and boundary formulations are distinct Eringen [1972].	u_i and stress for that example; no thermal state is included.	No heat-flux law is asserted for the elastic-only example.	Mechanical dispersion only; thermal propagation is not addressed.	Not classified here; kernel, domain, and boundary formulation are model-specific.	No dissipation property is asserted for a coupled nonlocal model.

Reading the table. "Not established" means that the reviewed equations or source anchors do not support a comparison in this audit; it does not mean the property is absent. The Green–Lindsay and GN-I equation-level cells remain open. DPL, TPL, MGT, MDD, fractional, and nonlocal entries are specific to the displayed formulation or explicitly unresolved.

Published examples include a hyperbolic two-temperature closure Prasad and Kumar [2020], fractional-order heat conduction with two temperatures Ezzat and El Karamany [2011], semiconductor applications of a hyperbolic two-temperature model Alzahrani and Abbas [2020], and a fractional two-temperature hybrid with relaxation times Bassiouny and R [2020]. These examples illustrate distinct constructions only; they do not establish shared admissibility or stability results.

4.13 State variables and constitutive distinctions

The model classes surveyed in this section differ in their chosen state variables and constitutive operators. Relaxation models may promote heat flux to an evolving variable; two-temperature formulations distinguish temperature-like fields; thermal-displacement formulations introduce a variable whose time derivative is related to temperature; phase-lag models shift the arguments of constitutive quantities; hereditary and fractional models use history operators. These descriptions are schematic. Their thermodynamic interpretation, initial data, modal consequences, and limiting behavior must be derived from each model's exact equations. In particular, Green–Lindsay is not a two-temperature construction.

4.14 Thermodynamic admissibility as a selection principle

Thermodynamic admissibility depends on the free energy, entropy, state variables, evolution laws, and admissible processes of each constitutive model.

Thermodynamic restrictions depend on the free energy, entropy, heat-flux law, and internal variables of the chosen model. A single dissipation expression or parameter-ordering condition does not apply to every relaxation, two-temperature, phase-lag, memory-dependent, or fractional formulation.

Parameter restrictions and limiting behavior are model-specific and should be stated with the constitutive equations and assumptions.

4.15 Modal structure is model-specific

The modal structure follows from substituting the chosen constitutive equations into the field equations and solving the resulting characteristic problem. A model label alone does not determine the number of propagating, evanescent, or attenuating roots. Root classification and interface matching are problem-specific.

4.16 Notation summary for Section 4

Table 8 summarizes the notation introduced in this section.

Table 7: Reduction map for the displayed formulations. Each arrow is an algebraic substitution or a derivative-level identity; none is a solution-convergence theorem.

Formulation	**Substitution**	**Displayed-equation result**	**Scope / unresolved condition**
Classical coupled	Reference form	Fourier heat flux with coupled momentum and internal-energy balances.	Baseline only; coefficients, domain, and data remain part of each problem.
LS	$\tau_0 = 0$ in Eq. (39)	Fourier heat-flux law.	Constitutive identity; no solution-convergence claim.
GL	None entered	No equation-level reduction asserted.	Original GL equation pages and convention must be transcribed before a reduction is tested.
GN-I	None entered	No equation-level reduction asserted.	Source identity and equation pages remain open; no family form is inferred.
GN-II / GN-III	In the displayed source-free forms, set $\kappa = 0$ and $T = \dot{\alpha}_{\mathrm{GN}}$.	The GN-III residual equals the time derivative of the displayed GN-II residual.	Derivative-level compatibility only; integration constants, initial data, source terms, and solution convergence are not resolved.
DPL	$\tau_T = 0$, $\tau_q = \tau_0$; or $\tau_q = \tau_T = 0$ in Eq. (47)	The displayed first-order residual reduces to LS; setting both lags to zero gives Fourier.	Exact only for this truncation; not an admissibility or solution-limit theorem.
TPL	$k^* = 0$ in Eq. (48)'s first-order expansion	The thermal-displacement-gradient contribution is removed, leaving the displayed DPL residual.	Algebra of the displayed law; no shared stability or thermodynamic condition follows.
MGT	$\tau = 0$, $k^* = 0$ in the candidate Eq. (49)	The displayed heat-flux residual reduces to Fourier.	Candidate constitutive variant only; acoustic MGT and other thermoelastic placements are not included.
Fractional scalar	$\alpha = 1$ or $\alpha = 2$ in Eq. (50)	The scalar equation has heat-equation or wave-equation form, respectively.	Scalar endpoints only; no coupled thermoelastic limit, intermediate-order speed claim, or solution convergence is asserted.
Spatially nonlocal	$\xi = 0$ in the displayed Helmholtz form and 1D dispersion expression	The displayed constitutive and dispersion forms recover their local expressions.	Algebraic local limit only; no convergence theorem for a nonlocal boundary-value problem.
MDD	No substitution entered	No classical-derivative or finite-speed limit asserted.	Kernel normalization, regularity, and the source-specific operator must be fixed first.

Table 8: Notation introduced in Section 4.

Symbol	**Meaning**	**Units**	**Equation**
τ_0	LS relaxation time	s	(39)
τ_q, τ_T	DPL lag parameters in the displayed inherited form	s	(46)–(47)
c_T	Thermal wave speed	m/s	(41)
β	Thermal stress coefficient	Pa/K	(36)
c_v	Specific heat at constant strain	J/(kg·K)	(37)
k	Thermal conductivity	W/(m·K)	(37)
α_{GN}	Thermal displacement (GN-II; distinct from fractional order α)	K·s	(42)
κ, κ^*	GN-III rate and thermal-displacement coefficients	model-dependent	(42), (45)
p_i	Entropy flux (GN-II)	W/(m^2·K)	(43)
${}^{C}D_t^{\alpha}$	Caputo fractional derivative	$t^{-\alpha}$	(50)
α	Fractional derivative order	dimensionless	(50)
a_α	Coefficient in displayed scalar fractional equation	m^2 s$^{-\alpha}$	(50)

4.17 Open questions for Section 4

OQ4.1 What is the complete time-domain energy flux for LS theory, including the contribution from the flux-relaxation term?

OQ4.2 What heat-flux law and energy balance follow from the original single-temperature Green–Lindsay relaxation-time convention?

OQ4.3 What is the relationship between energy flux and entropy flux in GN-II theory?

OQ4.4 What is the mixed energy flux for GN-III theory?

OQ4.5 How do different MDD kernel choices affect the energy flux and wave propagation?

OQ4.6 Can rigorous limit theorems with convergence proofs be established between theory pairs?

5 Coupled Multiphysics Continua

This section surveys coupled continua: piezo-thermoelastic, porous and double-porous, micropolar, semiconductor photo-thermoelastic, and magneto-thermoelastic media. It outlines their governing equations, constitutive relations, coupling mechanisms, and applications; primary sources are cited where available.

5.1 Piezoelectricity: constitutive relations and coupling coefficients

Piezoelectricity describes the coupling between mechanical and electrical fields in certain crystalline materials (e.g., quartz, barium titanate, lead zirconate titanate). The linear constitutive relations are:

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k, \tag{53}$$

$$D_i = e_{ikl}\varepsilon_{kl} + \kappa_{ij}E_j, \tag{54}$$

$$E_i = -\phi_{,i}, \tag{55}$$

where σ_{ij} is the Cauchy stress tensor, c_{ijkl} is the elastic stiffness tensor, ε_{kl} is the infinitesimal strain tensor, e_{kij} is the piezoelectric stress coefficient, E_k is the electric field, D_i is the electric displacement, κ_{ij} is the dielectric permittivity, and ϕ is the electric potential.

5.1.1 Charge conservation

In the absence of free electric charge:

$$D_{i,i} = 0. \tag{56}$$

5.1.2 Electrical energy density

The electrical energy density is:

$$\mathcal{E}_{\text{elec}} = \frac{1}{2}D_iE_i. \tag{57}$$

5.2 Piezothermoelasticity: three-field coupling

Piezothermoelasticity couples mechanical, thermal, and electrical fields. The constitutive relations are:

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k - \beta_{ij}T, \tag{58}$$

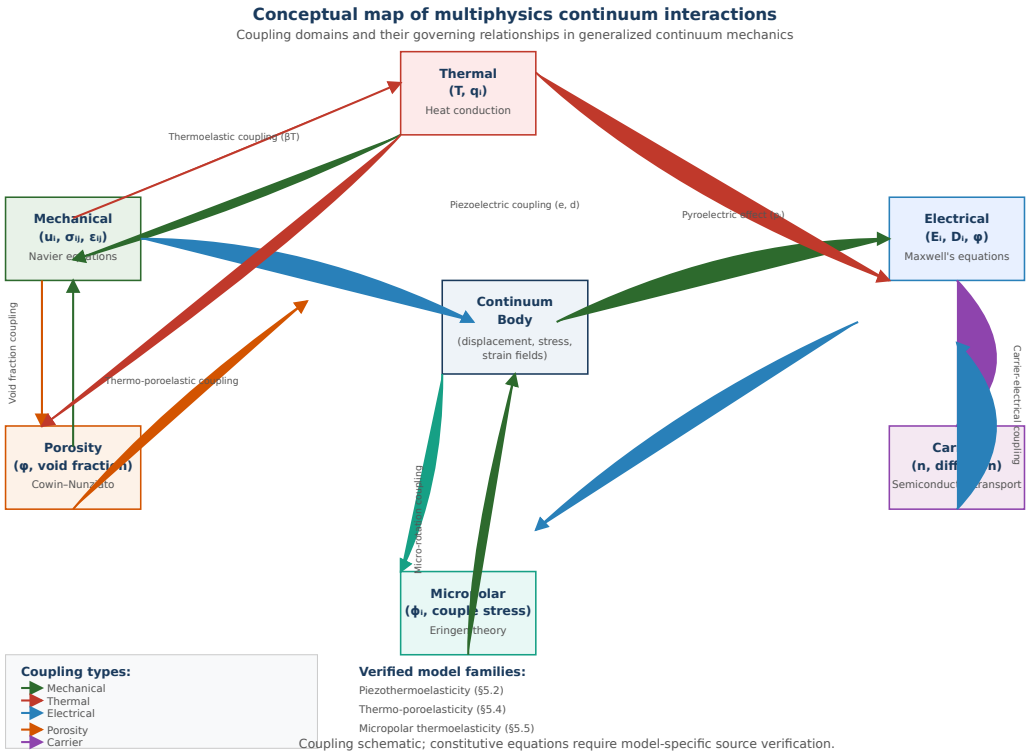


Figure 7: Coupling topology of multiphysics continua. The mechanical field is coupled to a thermal field through the thermoelastic dilatation term, to an electrical field through the piezoelectric tensors, to a porosity field through the void fraction, to a carrier field through the semiconductor transport equations, and to a micro-rotation field through the couple-stress relation. Each coupling is a separate term in a separate equation, and each introduces its own material coefficients and its own additional boundary conditions at an interface; this is the reason that the size of the interface system grows so rapidly as couplings are added (Section 9).

$$\sigma_{ij,j} = \rho \ddot{u}_i, \tag{59}$$

$$E_i = -\phi_{,i}, \tag{60}$$

$$D_i = \kappa_{ij} E_j + e_{ikl} \varepsilon_{kl} + p_i T, \tag{61}$$

$$D_{i,i} = 0, \tag{62}$$

where β_{ij} is the thermal stress coefficient and p_i is the pyroelectric coefficient.

5.2.1 Governing equations for coupled fields

The coupled system consists of the equation of motion (59), the heat conduction equation (theory-dependent: classical, LS, GL, GN, DPL, TPL, MDD, etc.), and the charge conservation equation (62). All three fields are coupled through the constitutive relations (58) and (61): temperature changes induce thermal stresses and pyroelectric effects; mechanical deformation generates piezoelectric polarization; electric fields produce electrostrictive deformation.

Thermopiezoelectric baseline. On printed pp. 625–626, Mindlin (1974) states the classical three-dimensional linear thermopiezoelectric equations (momentum balance, $D_{i,i} = 0$, entropy balance, Fourier heat flux $h_i = -K_{ij}\theta_{,j}$), which are the starting point of his plate reduction, together with linear constitutive equations from an electric Gibbs function Mindlin [1974].

Generalized piezothermoelastic models may use lagged or memory-dependent heat laws. Mindlin's Fourier system provides a classical comparison point; each generalized model must be considered with its own equations and parameters.

Coupled thermopiezoelectric formulations combine thermal, mechanical, and electrical fields; their constitutive details depend on the selected model.

Porosity and void fields are additional modeling ingredients; their governing equations and coupling to thermal and electrical fields depend on the selected formulation.

5.2.2 Modal content of a piezoelectric solid

The coupling of Section 5.2 has an immediate and testable consequence: the number of propagating modes increases. A piezoelectric solid admits a quasi-longitudinal mode qP, in which the mechanical polarization is essentially parallel to the propagation direction but is accompanied by a longitudinal electric field; a quasi-transverse mode qS, whose polarization is essentially transverse but which is accompanied by a transverse electric field whose amplitude is small because the dielectric stiffness dominates; and a quasi-electrostatic potential solution. Under the assumptions used here, $E_i = -\phi_{,i}$ and $D_{i,i} = 0$, Maxwell's displacement-current equation is absent; this solution is not an electromagnetic wave and no speed-of-light mode is implied. A propagating electromagnetic branch requires the full Maxwell system and a stated frequency regime.

5.2.3 The coupling coefficient as a wave-speed diagnostic

Because the electrical field and the mechanical field are both excited by the strain, the effective elastic stiffness of a piezoelectric solid along a given direction is not c_{ijkl} but a stiffness that has been renormalised by the dielectric stiffness, through the electromechanical factor

$$\bar{c}_{ijkl} = c_{ijkl} + e_{aij}(\kappa^{-1})_{ab} e_{bkl}, \tag{63}$$

which is the short-circuit/open-circuit stiffness correction after eliminating the quasistatic field under a specified electrical constraint. Its sign is positive for positive-definite permittivity with the stated tensor contraction. The effective modulus depends on the electrical boundary condition; it is not a universal scalar correction.

5.3 Electrical boundary conditions

5.3.1 Perfect electrical contact

$$\phi^{(1)} = \phi^{(2)} \quad \text{at the interface.} \tag{64}$$

5.3.2 Insulated interface

$$D_i n_i = 0 \quad \text{at the interface.} \tag{65}$$

5.3.3 Electric displacement continuity

$$D_3^{(1)} = D_3^{(2)} \quad \text{at the interface.} \tag{66}$$

5.3.4 Why a void needs its own field of freedom

The kinematics of a porous body require a kinematic variable beyond the displacement, because the material inside the pores can change its volume independently of the solid matrix. This is not a modelling convenience; it is forced by the kinematics of a body whose constituents occupy a variable volume fraction. The void volume fraction ϕ measures the change of that fraction from its reference value, and it enters the balance laws of Section 5.4 as a field in its own right, with its own inertia term, its own equilibrated stress and its own body force. The system is hyperelastic in the sense that the potentials are functions of ε_{ij}, γ_{mn} and ϕ jointly, so that the cross-coupling terms d_{ijk} are constrained by the requirement of a positive-definite quadratic form rather than being free parameters.

5.3.5 Modal content of a void-bearing medium

The presence of the void field adds a further mode to the modal structure: a fast wave associated with the inertia of the solid matrix and the coupled fluid motion, a slow wave associated with the equilibration of the void fraction against the surrounding matrix, and the usual shear waves, which are unaffected by the void field to the lowest order because shear does not change the volume fraction. The slow wave is of practical interest because it is the wave that transports the equilibration of the void pressure, and its speed is controlled by the ratio of the equilibrated inertia to the coupling stiffness. In a thermoporous medium the interaction with the thermal field adds a further degree of freedom, and the resulting modal structure is the one tabulated in Table 9.

5.4 Porous media: voids and double porosity

Porous elastic materials contain vacuous voids or pores distributed throughout the matrix. The Nunziato–Cowin theory (Nunziato and Cowin, 1979 Nunziato and Cowin [1979]; Cowin and Nunziato, 1983 Cowin and Nunziato [1983]) introduces an additional kinematic variable: the void volume fraction η.

5.4.1 Governing equations

The basic equations for thermoelastic materials with voids:

$$\sigma_{ij} = A_{ijmn}\varepsilon_{mn} + B_{ijmn}\gamma_{mn} + d_{ijk}\eta_{,k} - \alpha_{ij}\theta, \tag{67}$$

$$\rho\ddot{u}_i = \sigma_{ji,j} + \rho f_i, \tag{68}$$

$$\rho\ddot{\eta} = h_{i,i} + \rho l, \tag{69}$$

where η is the void volume fraction, h_i is the equilibrated stress vector, l is the equilibrated body force, A_{ijmn} and B_{ijmn} are elastic and equilibrated elasticity tensors, d_{ijk} is the equilibrated stress coupling tensor, and α_{ij} is the thermal coupling tensor. The thermoelastic extension of void theory was developed by Iesan (1986 Ieşan [1986]) and is treated systematically, together with the micropolar and nonsimple variants, in his monograph Ieşan [2013], which also establishes the existence and uniqueness framework for the coupled problems.

5.4.2 Double porosity

Double porosity models introduce two porosity fields to capture both macro- and micro-porosity effects. The governing equations extend the single-porosity theory with two coupled void volume fractions η_1 and η_2, each with its own equilibrated stress and equilibrated body force. Double porosity models are particularly relevant for hydrocarbon reservoirs, fissured rocks, and aquifers where both fracture and matrix porosity contribute to fluid flow and wave propagation.

5.5 Micropolar and micromorphic continua

Micropolar elasticity (Eringen, 1966 Eringen [1966]) extends classical elasticity by incorporating micro-rotational degrees of freedom and couple stresses. In addition to the displacement field u_i, the microrotation field ϕ_i describes independent rotation of material particles.

5.5.1 Governing equations

Micropolar thermoelastic formulations may include independent microrotation fields and couple stresses. Their constitutive equations depend on the selected kinematics, symmetry assumptions, and thermal coupling.

Micropolar models are relevant for granular materials, liquid crystals, composite materials with oriented inclusions, and biological tissues where micro-structural rotation cannot be neglected. A thermoelastic micropolar formulation was published by Tauchert, Claus and Ariman (1968 Tauchert et al. [1968]); earlier related work exists, so no priority claim is made here.

5.5.2 The absorption term and the photothermal source

Illumination does not act on the mechanical field directly; it acts through the heat and the carriers that it generates. The absorbed optical power, attenuated by the Beer–Lambert law, appears as a volumetric source in the thermal equation and as a generation term in the carrier continuity equations. The chain of consequences is then fixed: the thermal source produces a temperature rise; the temperature rise produces a thermoelastic stress through the dilatation term and a pyroelectric polarisation through the pyroelectric coefficient; the stress produces a displacement, and the polarisation produces an electric field, which in turn modifies the carrier distribution. The system is thus closed through the constitutive relations rather than through any additional assumption, and the number of degrees of freedom that must be solved simultaneously is four.

5.5.3 Photoacoustic waves as a diagnostic of the coupling

The photoacoustic wave that follows pulsed absorption is the most direct macroscopic observable of the coupling, and it has a distinctive structure: it contains a thermal contribution that propagates at the sound speed, an electronic contribution that propagates at the speed of light but decays on the absorption length, and a thermoelastic contribution whose amplitude depends on the piezoelectric coupling. The relative weight of these contributions depends on the regime — diffusion-dominated at low frequency, ballistic at high frequency — and it is the crossover between them that makes the technique sensitive to the thermal model. This is a concrete instance of the general point developed in Section 13: a generalized theory earns its place not by being more general, but by being required in a regime where the classical description demonstrably fails.

5.6 Semiconductor photo-thermo-elasticity

Semiconductor photo-thermo-elasticity couples mechanical deformation, thermal fields, electrical fields, and carrier density (electrons and holes). When a semiconductor is illuminated, absorbed photons generate electron–hole pairs, producing carrier diffusion, temperature changes, and thermoelastic stresses.

5.6.1 Governing equations

The coupled system includes:

- The equation of motion with piezoelectric and thermal stress terms,
- The heat conduction equation with carrier recombination heat source,
- The charge conservation equation with carrier generation and recombination,
- The carrier continuity equations for electrons and holes.

Applications include photoacoustic imaging, semiconductor device thermal management, and optoelectronic sensors.

5.7 Magneto-thermo-elasticity

Magneto-thermo-elasticity couples mechanical, thermal, and magnetic fields. In electrically conducting media (e.g., liquid metals, plasmas), magnetic fields induce Lorentz forces that couple with thermal stresses.

5.7.1 Governing equations

The equation of motion includes the Lorentz force term:

$$\rho \ddot{u}_i = \sigma_{ji,j} + \rho f_i + (\boldsymbol{J} \times \boldsymbol{B})_i, \tag{70}$$

where $\boldsymbol{J}$ is the current density and $\boldsymbol{B}$ is the magnetic flux density.

Applications include magnetohydrodynamic (MHD) systems, liquid metal cooling, geophysical dynamo theory, and electromagnetic forming.

5.8 Comparative summary

Table 9 summarizes the coupled multiphysics media: fields, coupling mechanisms, constitutive structure, and applications.

Table 9: Illustrative categories of coupled-field continuum models.

System	Coupled fields	Governing structure	Constitutive feature	Example source
Classical thermoelasticity	Mechanical and thermal	Navier–Fourier system	Thermal-stress coupling; conductivity-based heat equation	Biot [1956]
Piezothermoelasticity	Mechanical, thermal, and electrical	Coupled piezo–thermal equations	Piezothermal moduli and pyroelectric constants	Barak et al. [2023a]
Thermoporoelasticity	Mechanical, thermal, and porosity	Void or porous-medium theory	Void fraction and thermal coupling	Nunziato and Cowin [1979], Cowin and Nunziato [1983]
Thermoelasticity with voids	Mechanical, thermal, and void fields	Coupled thermoelastic equations	Void volume fraction	Ieşan [1986]
Micropolar thermoelasticity	Mechanical, thermal, and microrotation	Linear and angular momentum balances	Couple stress and micro-inertia	Tauchert et al. [1968]
Semiconductor thermoelectricity	Mechanical, thermal, carrier, and electrical	Carrier transport and heat equations	Carrier and electrical–thermal coupling	Model-specific

These categories are illustrative; each coupled system must be specified by its own fields, constitutive equations, and boundary conditions.

5.9 The coupling hierarchy and its computational consequences

The number and character of modes and interface conditions depend on the complete field equations, constitutive parameters, geometry, and boundary conditions. An added field or internal variable need not contribute an independent propagating mode, and matrix size is not fixed by a family label.

The secular equation and interface system follow from each model's field equations and boundary conditions. Their polynomial order and matrix size depend on the selected formulation.

Coupled-field models depend on the free-energy Hessian, constitutive stability, and well-posedness of the initial-boundary-value problem; these properties cannot be inferred from a coupling coefficient alone.

5.10 Notation summary for Section 5

A dual-porous thermoelastic interface study reports that the total sum of energy ratios approximates unity; this is retained as a reported numerical observation, not an exact identity Kumar et al. [2023a].

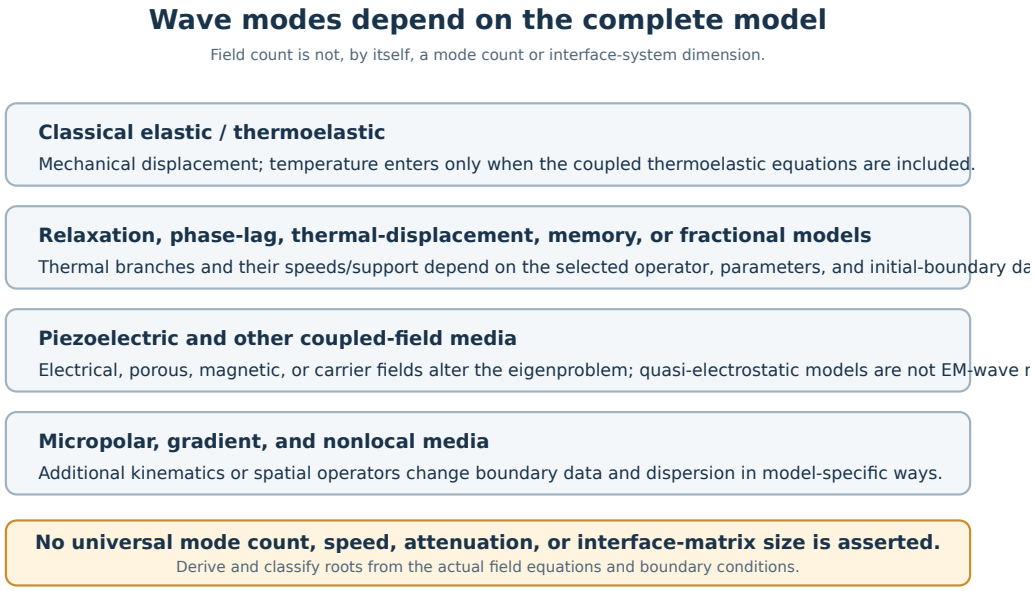


Figure 8: Schematic overview of field types in continuum wave models. Modal content, propagation, and interface-system dimension depend on the selected constitutive equations and boundary conditions; no universal mode count is asserted.

Table 10: Notation introduced in Section 5.

Symbol	Meaning	Units	Equation
c_{ijkl}	Elastic stiffness tensor	Pa	(53)
e_{kij}	Piezoelectric stress coefficient	C/m^2	(53)
κ_{ij}	Dielectric permittivity	F/m	(54)
E_i	Electric field	V/m	(55)
D_i	Electric displacement	C/m^2	(54)
ϕ	Electric potential	V	(55)
β_{ij}	Thermal stress coefficient	Pa/K	(58)
p_i	Pyroelectric coefficient	C/(m^2·K)	(61)
$\bar{c}_{ijkl}$	Piezoelectrically renormalised stiffness	Pa	(63)
η	Void volume fraction	dimensionless	(67)
h_i	Equilibrated stress vector	N/m	(69)

5.11 Open questions for Section 5

OQ5.1 What are the complete constitutive relations for piezothermoelastic media with nonlocal elasticity?

OQ5.2 How do double-porosity models capture scale-dependent fluid–solid interaction?

OQ5.3 What are the energy flux expressions for micropolar media with couple stresses?

OQ5.4 How can semiconductor photo-thermo-elastic models be validated experimentally?

6 Generalized Transport and Memory Effects

Classical Fourier conduction relates heat flux instantaneously to the temperature gradient. Its range of adequacy and any departure from it depend on material, scale, and experiment. This section surveys alternative constitutive structures for heat transport.

6.1 Physical origin of thermal relaxation

Microscopic heat carriers and their scattering can motivate finite-rate transport models. Exponential relaxation is one possible constitutive choice; memory kernels, relaxation parameters, and propagation properties depend on the selected model and material. This section compares alternative transport operators by their stated mathematical structure.

6.2 Three structural routes to memory

Three mathematically distinct ways of making a constitutive law remember have been used, and they are not interchangeable.

- **Differential-rate constitutive laws.** Some models include time derivatives of fluxes, temperatures, or other fields; the order and choice of differentiated variable are model-specific. Whether such a law admits an equivalent finite-dimensional internal-variable representation must be established for the particular formulation.
- **Integral (convolution) memory.** The constitutive relation is written as a convolution of the driving quantity with a memory kernel over a finite or semi-infinite interval. This includes the hereditary-integral formulation of Section 6 and the memory-dependent derivative of Wang and Li Wang and Li [2021]. The memory is then encapsulated in a kernel, and the constitutive relation is nonlocal in time.
- **Spatial nonlocality.** Some constitutive laws average over a material neighbourhood, as in Eringen-type formulations. This is a spatial interaction, not temporal memory, and its operators and boundary conditions require separate treatment in Section 7.

6.3 A transform-domain example, not a universal representation

Some linear flux laws can be written schematically as

$$\mathcal{D}[\boldsymbol{q}] = -k\,\boldsymbol{\nabla} T, \tag{71}$$

but this notation does not make distinct constitutive models equivalent.

For the Lord–Shulman/Cattaneo law, with zero initial heat flux,

$$\boldsymbol{q} + \tau_0 \dot{\boldsymbol{q}} = -k \nabla T.$$

The corresponding Laplace-domain multiplier is

$$\mathcal{D}_{\mathrm{LS}}(s) = 1 + \tau_0 s. \tag{72}$$

Nonzero initial flux contributes an additional initial-data term. Fractional, phase-lag, and hereditary models have their own transform symbols and initial terms, which must be derived from their exact constitutive equations; no universal symbol or shared limiting procedure is asserted here.

A comparison in the transform domain is useful only after each constitutive law, initial condition, and admissible parameter range is stated. Formal limits such as $\tau_0 \to 0$, fractional order tending to an integer, or memory horizon tending to zero require model-specific normalization and convergence arguments. Positivity of a frequency-domain loss modulus, when defined, is not by itself proof of a consistent free energy or a well-posed initial-boundary-value problem. These questions remain model-specific and are not settled uniformly in this review.

6.4 Fractional operators: Caputo and Riemann–Liouville

Fractional derivatives generalise integer-order differentiation to non-integer orders and are used in models with history dependence. Depending on the operator, coupling, kernel, and initial-boundary conditions, they can produce subdiffusive, dispersive, or wave-like responses. A finite propagation speed or a specific support property is not implied by the fractional order alone.

6.4.1 Caputo fractional derivative

The Caputo derivative of order α ($n-1 < \alpha < n$, $n = \lceil \alpha \rceil$):

$$ {}^{C}D_t^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t \frac{f^{(n)}(\tau)}{(t-\tau)^{\alpha+1-n}}\, d\tau. \tag{73} $$

6.4.2 Riemann–Liouville fractional derivative

$$ {}^{RL}D_t^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t \frac{f(\tau)}{(t-\tau)^{\alpha+1-n}}\, d\tau. \tag{74} $$

For suitable orders and regularity, the Caputo definition uses integer-order initial data, while Riemann–Liouville formulations may involve fractional initial data. This difference can matter in model selection, but does not establish a universal preference.

6.4.3 Fractional heat conduction and its limits

$$ {}^{C}D_t^{\alpha} T = a_\alpha \nabla^2 T, \qquad 0 < \alpha \le 2, \tag{75} $$

Here a_α is a model coefficient with dimensions consistent with the selected order (for length and time units, $[a_\alpha] = L^2 T^{-\alpha}$). This scalar equation is illustrative, not a universal fractional thermoelastic law. At integer-order endpoints, the scalar PDE has heat-equation or wave-equation form, with coefficients and initial data specified for each model; intermediate-order behavior depends on the operator definition, coupling, domain, and data.

6.5 Hereditary integrals: convolution kernels

One common representation of linear temporal memory is a hereditary integral:

$$ f(t) = \int_{-\infty}^{t} K(t-t')\, g(t')\, dt', \tag{76} $$

where $K(t-t')$ is a kernel in this schematic convolution. Some models impose fading-memory, positivity, monotonicity, or complete-monotonicity hypotheses, but the required conditions depend on the constitutive law and the chosen thermodynamic framework; none of these properties alone is asserted here as a universal admissibility criterion. For suitable completely monotone kernels, representation results by weighted exponentials may apply, but their hypotheses and implications must be checked for the kernel at hand. No equivalence with a discrete phase-lag hierarchy is claimed.

6.6 Memory-dependent derivative (MDD): operator and kernel

A memory-dependent operator can use a kernel over a finite interval; its definition and normalization are part of the selected constitutive model.

$$ D_{\omega_M} f(t) = \frac{1}{\omega_M} \int_{t-\omega_M}^{t} K_{\mathrm{MDD}}(t-t') f'(t')\, dt', \tag{77} $$

where $\omega_M > 0$ denotes the displayed horizon and K_{MDD} is the kernel. The bounds, normalization, regularity, and admissibility conditions on K_{MDD} must be taken from the exact model definition. A finite integration horizon truncates the history in this formula; it does not by itself imply a finite-dimensional state-space realization, finite computation cost, or that the kernel is a freely chosen rather than material-calibrated quantity. A specific quadratic candidate kernel is omitted because its source definition, dimensions, and admissible parameter range have not been established; no numerical form is asserted here.

6.6.1 Limiting behavior

Short-horizon limits depend on the normalization and regularity assumptions of the chosen finite-horizon operator; no general reduction to a particular relaxation law is asserted here.

6.7 Energy flux in memory-dependent media

For a classical local mechanical balance with a specified heat-flux vector, the outward-positive flux is $J_i = q_i - \sigma_{ij} v_j$. A memory-dependent model may require additional storage or boundary-power terms; these depend on its constitutive operator and must be derived for the selected formulation.

6.8 Well-posedness and propagation in memory-dependent models

Well-posedness and propagation depend on the memory operator, kernel, time horizon, initial data, domain, and boundary conditions. A finite-horizon derivative alone does not specify a propagation speed; the relevant property must be derived for the selected model.

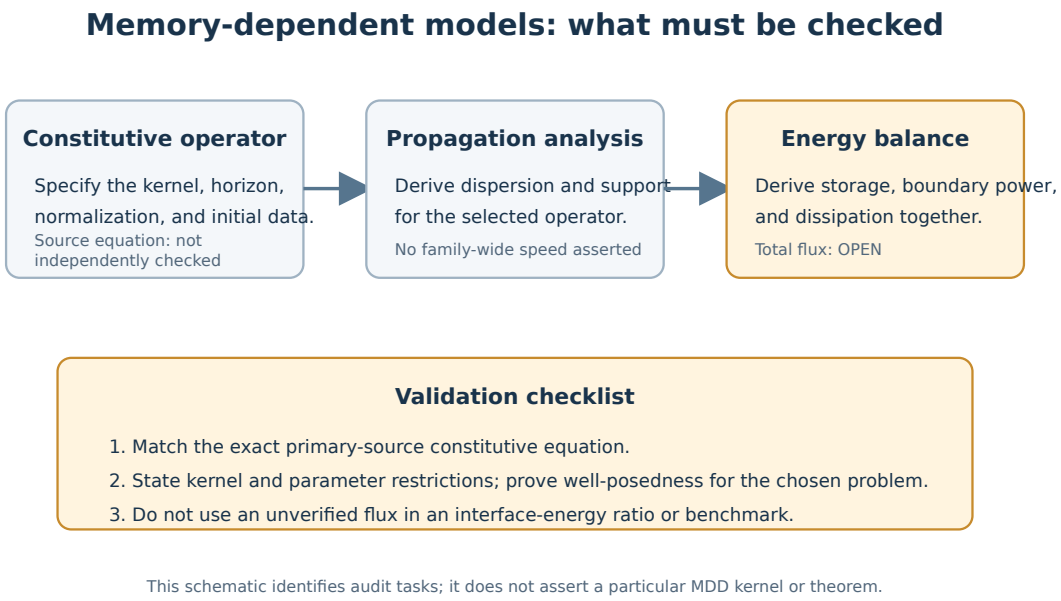


Figure 9: Schematic of a finite-horizon memory-dependent operator and audit questions concerning its kernel, limiting behavior, energy balance, and well-posedness. Model-specific propagation and flux consequences are not asserted; the displayed operator still requires verification against its original source.

6.9 Comparative summary: transport models

Table 11 summarizes transport and memory models by mathematical structure, dissipation, well-posedness, and energy-flux formulation.

Table 11: Transport structures and model-specific features.

Model class	Transport structure	Model-specific feature	Comparison point
Classical Fourier	Local Fourier heat flux	Local balance with Fourier heat conduction	Parabolic heat equation
Lord–Shulman	Relaxing heat flux	Relaxing heat flux	Source-specific constitutive parameters
Green–Lindsay	Temperature-rate-dependent formulation	Single-temperature formulation	Temperature-rate dependence
Phase-lag (dual/triple)	Lagged constitutive laws	Lagged constitutive laws	Truncation and parameter choices
Green–Naghdi II/III	Thermal-displacement formulations	Thermal-displacement formulations	Type-specific equations
Moore–Gibson–Thompson	Higher-order heat equation variants	Selected variant	Relaxation and coupling terms
Memory-dependent	Kernel/time-history operator	Finite- or unbounded-history formulation	Kernel and normalization
Fractional	Fractional time operator	Derivative and initial-data convention	Operator and data space
Nonlocal / gradient	Spatially nonlocal or higher-gradient law	Integral or differential form	Kernel and boundary prescription

6.10 Notation summary for Section 6

Table 12: Notation introduced in Section 6.

Symbol	Meaning	Units	Equation
$\mathcal{D}[\cdot]$	Generic memory operator acting on the flux	—	(71)
$\mathcal{D}_{\mathrm{LS}}(s)$	LS transform multiplier (zero initial flux)	dimensionless	(72)
${}^{C}D_t^{\alpha}$	Caputo fractional derivative	$t^{-\alpha}$	(73)
${}^{RL}D_t^{\alpha}$	Riemann–Liouville derivative	$t^{-\alpha}$	(74)
a	Thermal diffusivity	m^2/s	(75)
$K(t-t')$	Memory kernel	varies	(76)
D_{ω_M}	MDD operator	model-dependent	(77)
ω_M	MDD finite-history window	s	(77)
t'	Dummy integration time in hereditary kernels	s	(76)–(77)
K_{MDD}	MDD kernel	model-dependent	(77)
$\tilde{q}_i(s)$	Laplace transform of the heat flux	—	(72)

A nonlocal Kelvin–Voigt piezo-semiconductor application uses a memory-dependent heat flux; conclusions remain model-specific. Gupta et al. [2026]. A four-lag Taylor expansion is reported, but its printed gradient-field argument is defective and is not copied here. Abouelregal [2020]. A memory-dependent TPL formulation compares candidate kernels and states a model-specific positive-semidefiniteness condition. Sur et al. [2019]. A revised MDD definition normalizes by the kernel integral over the horizon rather than by the horizon alone. Wang and Li [2021]. A model-specific MGT-type flux law is paired with an Nth-order memory expansion using factorial coefficients. Sur et al. [2025]. A nonlocal operator is applied to a fractional Cattaneo law; its printed zero-order branch is not used as a general limiting identity. Bachher and Sarkar [2018]. A tensor positive-semidefiniteness compatibility condition is printed for a particular second-order TPL truncation. Tiwari and Kumar [2021].

6.11 Open questions for Section 6

OQ6.1 What is the complete time-domain energy flux for MDD media with convolution-type memory, expressed solely in terms of the temperature and the stored internal variables?

OQ6.2 How is energy partitioned between storage and dissipation in hereditary media, and what is the correct effective flux that enters a local balance?

OQ6.3 What are the well-posedness conditions for fractional-order thermoelasticity with arbitrary α, in the coupled mechanical setting rather than in isolation?

OQ6.4 How do different MDD kernel choices affect the energy flux and the wave propagation, and can the dispersion of the thermal branch be used to identify the kernel experimentally?

OQ6.5 Can the memory operator be given a finite-dimensional state-space realisation for a general admissible kernel, and what is the numerical cost of that realisation relative to a direct quadrature?

7 Nonlocal and Scale-Dependent Continuum Models

Many classical local constitutive models relate stress to strain at the same material point and contain no explicit internal length. Nonlocal, gradient, couple-stress, and micropolar models introduce spatial interactions or additional kinematics to represent possible size effects. Their use is motivated by particular materials and experiments; evidence of a characteristic microstructural scale does not by itself show that a classical local law is invalid in every small-scale regime. The applicable constitutive model and its length parameter require independent justification.

This section develops that class. The organising question is not which of the competing formulations is correct, but *which physical mechanism each one encodes*, because the mechanisms predict measurably different dispersion and therefore admit different experimental tests.

7.1 Physical motivation and constitutive foundations

The classical assumption fails in a specific and observable way. A beam of a given material, tested in bending, exhibits a stiffness that depends on the specimen thickness: thinner specimens are stiffer than the classical prediction, and the departure becomes significant below a material-dependent critical length. Two distinct physical mechanisms can produce this behaviour, and they are the root of the entire family of models considered here.

The first mechanism is *constitutive*: the internal stress in the material depends on the neighbouring material state, because the mechanism that carries load—bonding, dislocation motion, stress concentration in a heterogeneous microstructure—acts over a finite distance. Eringen's formulation of nonlocal elasticity is the canonical statement of this mechanism Eringen [1972, 1974], in which the stress at a point is obtained by averaging the local stress over a neighbourhood weighted by a kernel:

$$\sigma_{ij}(\boldsymbol{x}) = \int_{\Omega} \alpha(|\boldsymbol{x}' - \boldsymbol{x}|, \tau)\, \sigma_{ij}^{\mathrm{local}}(\boldsymbol{x}')\, d\Omega_{\boldsymbol{x}'}, \tag{78}$$

where α is the attenuation kernel and $\tau = e_0 a/\ell$ is the nonlocal parameter, with e_0 a material constant, a an internal characteristic length such as the lattice spacing, and ℓ an external characteristic length such as the specimen size. The nonlocal parameter is therefore not an arbitrary length: it is a ratio of an internal to an external scale, which means that the nonlocal correction grows in relative importance as the specimen is made smaller relative to the microstructure. This is the scaling property that distinguishes a size-dependent theory from an ad hoc correction.

The second mechanism is *structural*: the material possesses additional internal degrees of freedom—microrotations, relative displacement of the constituents of a porous skeleton, dislocation density—whose motion must be carried by additional stress measures. Micropolar elasticity, formulated by Eringen as a linear theory of a continuum with independent microrotation fields Eringen [1966] and extended to the thermoelastic case by Tauchert, Claus and Ariman Tauchert et al. [1968], is the earliest and most complete statement of this mechanism. Later work extended it to materials with voids, in which the skeleton and the pore fluid carry independent kinematic fields Nunziato and Cowin [1979], Cowin and Nunziato [1983], with a corresponding thermoelastic formulation Ieşan [1986].

7.1.1 Differential form

For special kernel functions—notably the one-parameter exponential and the Helmholtz kernel—the integral equation of Eq. (78) reduces to a differential equation of higher order. For the Helmholtz kernel one obtains

$$(1 - \xi^2 \nabla^2)\sigma_{ij} = \sigma_{ij}^{\text{local}}, \tag{79}$$

where $\xi = e_0 a$ is the nonlocal length scale parameter. The differential form is the one used in wave-propagation applications, because it is a partial differential equation and can therefore be treated with the transform and numerical machinery of Section 8. It is worth recording that the differential and integral forms are not interchangeable in general: the integral relation with a given kernel and the differential operator of Eq. (79) agree only for that kernel, and the difference matters for the boundary conditions, which in the differential form become higher-order Polizzotto [2001].

7.2 Strain-gradient and couple-stress theories

Strain-gradient theory extends classical elasticity by including higher-order gradients of strain in the constitutive relations:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \ell^2 C_{ijkl}\nabla^2 \varepsilon_{kl}, \tag{80}$$

where ℓ is a material length scale parameter. The physical reading of Eq. (80) is that the stress depends not only on the strain but on the curvature of the strain field, so that regions in which the strain varies rapidly—near a notch, a contact edge, or an interface—carry additional stress. The coefficient ℓ is a genuine material constant that must be measured; the identification procedures for it, together with the finite element implementations and the variational formulations that make the theory tractable, have been surveyed by Askes and Aifantis Askes and Aifantis [2006, 2011].

The unifying point of view that connects gradient elasticity to micropolar and couple-stress elasticity is that all three introduce internal degrees of freedom and a length scale, and differ in whether those degrees of freedom are constrained to coincide with the macroscopic deformation or are allowed to evolve independently. Aifantis's formulation, in which the internal length is interpreted as the scale of the microstructural heterogeneity that generates it, is the common root Askes and Aifantis [2006]. It follows that a continuum model with internal length, in whichever of these forms, can be used to represent materials for which no explicit microstructural model is available—which is why the same length-scale parameter appears in treatments of metals, polymers, granular media, and biological tissue with broadly comparable values.

7.3 Size-dependent wave behavior: dispersion from nonlocality

Nonlocality introduces dispersion: the wave speed depends on wavelength. The differential form of nonlocal elasticity was developed by Eringen Eringen [1983], and the size-dependent elastic effects at the nanoscale were quantified by Maranganti and Sharma Maranganti and Sharma [2007], who established the characteristic length below which classical elasticity fails for a range of materials, and by Maranganti, Sharma and Sharma Maranganti et al. [2006], who showed that the nanoscale nonlocal contribution to electromechanical coupling is generically non-zero even for materials that possess no piezoelectric tensor—a result directly relevant to the coupled theories of Section 5, because it means that a purely elastic size effect can generate an apparent electromechanical response in a nominally non-piezoelectric material.

For a nonlocal elastic bar, the dispersion relation is

$$c(k_{\text{w}}) = \frac{c_0}{\sqrt{1 + \xi^2 k_{\text{w}}^2}}, \tag{81}$$

where c_0 is the classical wave speed, k_{w} is the wave number (distinct from the thermal conductivity k), and ξ is the nonlocal length scale. Long waves ($k_{\text{w}} \to 0$) approach the classical speed c_0; short waves are strongly slowed.

7.3.1 The dispersion signature distinguishes the mechanisms

For Eq. (81), the phase speed tends to zero as $k_{\text{w}} \to \infty$, while the angular frequency $\omega(k_{\text{w}}) = c_0 k_{\text{w}}/\sqrt{1 + \xi^2 k_{\text{w}}^2}$ approaches the finite limit c_0/ξ; it is frequency, not phase speed, that saturates. The group velocity is $d\omega/dk_{\text{w}} = c_0(1+\xi^2 k_{\text{w}}^2)^{-3/2}$ and tends to zero. By contrast, the positive strain-gradient law in Eq. (80) gives $c_{\text{ph}}^2 = c_0^2(1+\ell^2 k_{\text{w}}^2)$ for a one-dimensional bar, so it stiffens and phase speed increases. Gradient-inertia variants can behave differently; the model form must accompany any comparison. A dispersion measurement over a sufficiently broad range of k_{w}

can thus distinguish a nonlocal mechanism from a gradient mechanism, and the two are not interchangeable even though both produce the same qualitative signature of a size-dependent stiffness. This is one of the few places in the reviewed literature where two competing formulations of the same effect make quantitatively different and testable predictions, and it deserves more attention than it has received.

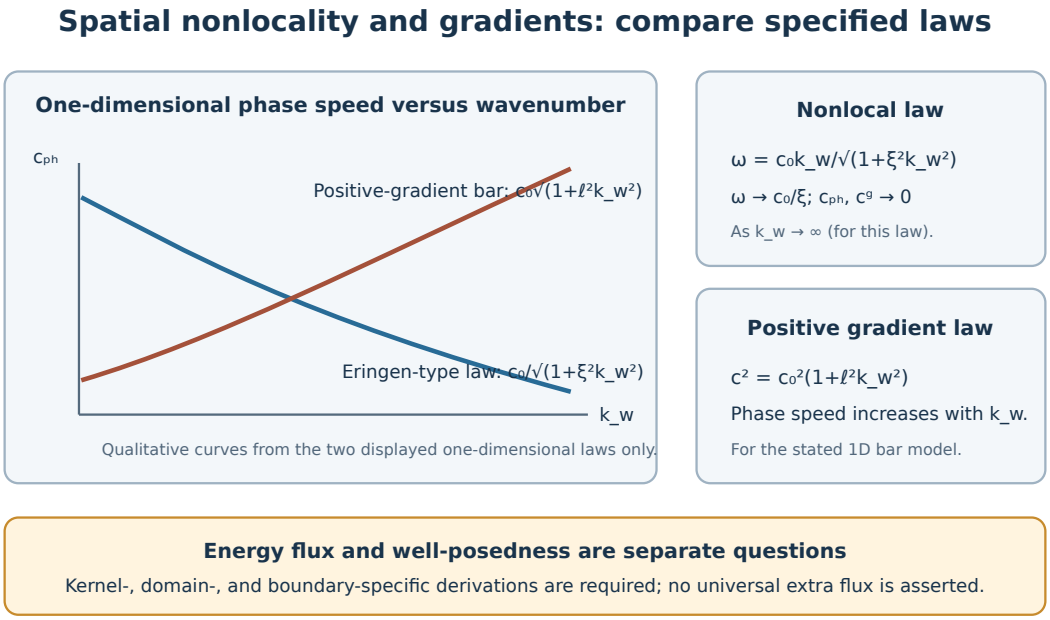


Figure 10: Schematic nonlocal and gradient mechanisms. The plotted dispersion curves correspond only to the particular one-dimensional laws stated in Section 7; other gradient-inertia or kernel choices can differ. No universal nonlocal energy-flux correction is asserted; boundary-power derivations remain open.

7.4 Energy flux in nonlocal media

For nonlocal media, boundary-power terms depend on the kernel, domain, constitutive free energy, and boundary conditions. A flux identity for a given model must be derived from its power balance; no universal correction is assumed.

7.5 Interface conditions for nonlocal media

Interface conditions for nonlocal media differ from the classical ones because the stress at a point depends on the strain in a neighbourhood, and a neighbourhood straddling an interface samples both media. The classical conditions—continuity of displacement and of traction—must be supplemented, and the supplementation is not unique: one may impose conditions on the additional derivatives that appear in the differential form of Eq. (79), or one may impose the conditions in the integral form by requiring the nonlocal stress to be continuous across the interface. The two are not equivalent, and the choice made determines the number and kind of boundary conditions that the resulting boundary-value problem admits. For the differential form, the interface conditions include both the classical conditions and the higher-order conditions arising from the ∇^2 term.

Interface problems in which the nonlocal operator is combined with an additional coupled field have been analysed in the reviewed corpus for a fluid in contact with a nonlocal piezothermoelastic medium Gupta et al. [2022] and for nonlocal Kelvin–Voigt piezo-semiconductors under a memory-dependent heat flux Gupta et al. [2026]. In both cases the analysis proceeds through the modal machinery of Section 9, and the specific difficulty in both is the same: the nonlocal operator acts on the stress while the coupled field acts on the temperature or the electric displacement, and the interface conditions must be formulated consistently for both.

7.6 What nonlocality does not repair

The differential nonlocal model must be distinguished from the integral constitutive law from which it is sometimes obtained. Its boundary-value formulation can exhibit paradoxes under particular geometries and boundary conditions, including the cantilever-bending paradox discussed in the nonlocal-beam literature; that does not establish a bulk-wave group velocity that grows without bound. For Eq. (81), both frequency and group velocity are bounded. Well-posedness claims must specify operator, domain, boundary conditions and function space; no universal Hadamard-instability claim is made here.

Some nonlocal formulations exhibit boundary-value paradoxes for particular kernels, geometries, or boundary conditions. Bounded kernels and variational formulations have been proposed in parts of the literature, but their ability to resolve a given problem must be demonstrated for the specified operator and function space; neither is a universal remedy or proof of well-posedness. The variational route can clarify admissible boundary data, but its conclusions are model-specific.

7.7 Comparative summary: nonlocal models

Table 13: Model-specific comparison of spatially nonlocal and gradient formulations. Dispersion statements are restricted to the one-dimensional laws displayed in this section.

Model/form used here	**Spatial mechanism**	**Dispersion feature**	**Energy/flux terms to specify**
Eringen-type Helmholtz form, Eq. (79)	Kernel-dependent spatial averaging	One-dimensional result of Eq. (81); phase speed decreases and frequency tends to c_0/ξ	Flux depends on the specified kernel and boundary-value problem
Strain-gradient form, Eq. (80)	Strain-curvature term	For the stated positive-gradient 1D bar, phase speed increases with k_{w}	Higher-gradient boundary power depends on additional boundary data
Micropolar / couple-stress	Additional micro-rotation or couple stress	Depends on the selected micropolar model	Model-specific flux terms
Two-phase local/nonlocal	Depends on selected mixture/kernel	Depends on the selected mixture/kernel	Model-specific flux terms

7.8 Notation summary for Section 7

Table 14: Notation introduced in Section 7.

Symbol	**Meaning**	**Units**	**Equation**
$\alpha(\lvert\boldsymbol{x}' - \boldsymbol{x}\rvert, \tau)$	Nonlocal attenuation kernel	varies	(78)
τ	Nonlocal parameter, ratio of internal to external length	dimensionless	(78)
e_0	Material constant of the nonlocal formulation	dimensionless	(78)
a	Internal characteristic length	m	(78)
$\xi = e_0 a$	Nonlocal length scale parameter	m	(79)
ℓ	Strain-gradient or couple-stress length scale	m	(80)
k_{w}	One-dimensional wave number	m^{-1}	(81)
$c(k_{\mathrm{w}})$	Wave speed as a function of wave number	$\mathrm{m\,s}^{-1}$	(81)
c_0	Classical (local) wave speed	$\mathrm{m\,s}^{-1}$	(81)

These examples show that nonlocal operator placement is model-specific. Lata and Singh's two-temperature nonlocal thermoelastic model uses a conductivity-based heat equation, with no phase-lag or Green–Naghdi term in the stated formulation. Lata and Singh [2020]. Gupta et al.'s nonlocal fractional MGT example places the

spatial operator in its constitutive relations, inertial term, and heat-flux relation; the latter includes temperature- and thermal-displacement-gradient terms and a carrier contribution. Gupta et al. [2024]. For one modified-Bessel K_0 attenuation kernel, Li and He give a kernel-specific correspondence with a differential nonlocal relation. Li and He [2018]. A separate cited study treats Klein–Gordon-type nonlocal elasticity in wave propagation; no operator-placement detail is inferred here. Lazar and Agiasofitou [2022]. Tung's nonlocal piezoelastic half-space model applies a spatial operator to the stress and electric-displacement relations, with a corresponding factor in the motion equation; no heat equation is included. Tung [2019]. In Zang et al.'s one-dimensional piezoelectric nanoplate model, scale-dependent terms enter the inertia contributions; the study examines longitudinal-wave dispersion with surface effects. Zang et al. [2014].

7.9 Open questions for Section 7

OQ7.1 What is the correct energy flux at a nonlocal interface, derived from a rigorous modified divergence theorem, and does it reproduce the classical limit as $\xi \to 0$?

OQ7.2 How must the boundary conditions be formulated for the differential nonlocal operator so that the initial-value problem is well posed, and is the resulting formulation variational?

OQ7.3 What is the relationship between the integral and differential nonlocal formulations at the level of the energy flux, given that the two agree only for the specific kernel used to obtain the differential form?

OQ7.4 Can a dispersion measurement over a sufficiently broad range of wave number distinguish a nonlocal mechanism from a gradient mechanism in a specific material?

OQ7.5 Which nonlocal constitutive law is thermodynamically admissible when combined with a memory-dependent thermal law, given that both introduce history into the energy balance?

8 Mathematical Methods for Wave Propagation

The theory families considered here differ in constitutive assumptions and governing equations. Several analytical tools—plane-wave ansätze, transforms, eigenvalue problems, and boundary matching—recur across them, but their applicability and form depend on the specified model, geometry, and boundary conditions. This section summarizes those tools, emphasizing the harmonic sign convention, admissible roots, and consistent normalization when modal energies are compared.

8.1 Plane wave ansatz and harmonic time dependence

For time-harmonic problems, all field variables take the form

$$\text{field}(\boldsymbol{x}, t) = \hat{\text{field}}(\boldsymbol{x}) e^{-i\omega t}, \tag{82}$$

where $\hat{\text{field}}$ is the complex amplitude and ω is the angular frequency. Two consequences of the convention $e^{-i\omega t}$ should be stated, because both are used repeatedly in the sections that follow. First, a wave with complex wave number q whose imaginary part is positive is spatially attenuated, since the spatial factor e^{iqx} acquires a factor $e^{-\operatorname{Im}(q)\, x}$. Second, the sign of the imaginary part of every complex amplitude reverses if the opposite convention $e^{+i\omega t}$ is adopted, while the physical content of the solution does not change. The divergence map of Section 12 records the second point as one of the routine sources of apparent disagreement between studies.

8.2 Christoffel-type equations and wave speeds

For a classical anisotropic elastic medium, substituting a plane-wave ansatz into the elastic equations yields the Christoffel equation

$$[\Gamma_{ik}(\boldsymbol{n}) - \rho c^2 \delta_{ik}]\hat{u}_k = 0, \tag{83}$$

where $\Gamma_{ik}(\boldsymbol{n})$ is the Christoffel tensor, $\boldsymbol{n}$ is the propagation direction, c is the wave speed, and $\hat{u}_k$ is the polarization vector. Non-trivial solutions require

$$\det[\Gamma_{ik}(\boldsymbol{n}) - \rho c^2 \delta_{ik}] = 0. \tag{84}$$

For a classical anisotropic elastic model, the Christoffel tensor depends on the propagation direction $\boldsymbol{n}$; its eigenvalues divided by density give the squared phase speeds for that direction. A generalized formulation that retains the same kinematics may retain this directional dependence, while its constitutive terms alter the resulting characteristic equation. No universal mode count is implied here.

8.3 Secular equations and characteristic polynomials

For a specified model, plane-wave substitution produces a secular equation in the chosen spectral variable. When that equation is polynomial and even in q, it can be written in the generic form

$$\sum_{j=0}^{m} G_j q^{2j} = 0, \tag{85}$$

where the order m and coefficients G_j depend on the governing equations and model parameters. This representation does not prescribe a universal polynomial order or mode count. Admissibility of roots is determined by the relevant boundary and radiation conditions; any effect of relaxation, memory, or nonlocal terms must be established for the selected constitutive model.

8.4 Complex wave numbers: propagating, evanescent, attenuating modes

The roots of the secular equation are classified as

1. **Propagating modes:** real q, waves that propagate without spatial attenuation;
2. **Evanescent modes:** purely imaginary q, waves that decay exponentially with distance;
3. **Attenuating modes:** complex q, waves that both propagate and attenuate.

An early example is Chadwick and Sneddon's (1958) plane-wave study of an elastic solid conducting heat Chadwick and Sneddon [1958].

Only modes satisfying the relevant radiation or decay conditions are retained in the physical solution. In a half-space problem, this selects roots for which the wave decays as $x \to \infty$; in a slab or layer, the transmitted field must satisfy the far-field condition in the adjoining material. Roots that do not satisfy the stated conditions are excluded from the physical solution and should not be included in a flux calculation.

8.5 Modal normalization for energy flux computation

For energy flux computation, each mode must be consistently normalized. For propagating modes a convenient choice is

$$\hat{u}_i^{(n)} \hat{u}_i^{(n)*} = 1, \tag{86}$$

where * denotes complex conjugation. When flux-based normalization is used for evanescent modes, state the convention and evaluate flux using the selected model's energy balance. For superposed fields, report how cross terms are handled; modal amplitudes alone do not determine the total flux.

8.6 Transform methods: Fourier and Laplace

For a time signal for which the bilateral integral converges, define the Fourier transform by:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt, \tag{87}$$

For a causal transient signal for which the transform exists, define the Laplace transform by:

$$\tilde{f}(s) = \int_{0}^{\infty} f(t) e^{-st}\, dt. \tag{88}$$

For a specified classical thermoelastic problem, time-Laplace and spatial Fourier transforms may be applied to the governing equations and boundary data when the transform and inversion assumptions are satisfied. The resulting equations and availability of closed-form solutions depend on the model, geometry, and boundary conditions; no universal closed-form reduction is asserted here.

Transform rules for fractional derivatives depend on the initial data and the chosen derivative. Inversion depends on the spatial operator, geometry, and boundary conditions; no universal inversion method or reference solution is asserted here.

Numerical inverse-transform procedures approximate the inverse using selected sampling and summation rules. Their accuracy depends on the time window, truncation, transform evaluations, and numerical precision; the choice of method is problem-specific.

8.7 Green's functions and fundamental solutions

Where a suitable Green's function is available for the selected coupled operator and geometry, it provides an alternative representation of the response to a prescribed source or incident field. In an interface problem, reflection and transmission quantities must be obtained using the specified boundary conditions and field variables. For multiple coupled fields the representation may be matrix-valued; its construction and computational cost are model-specific. Within the papers reviewed here, modal formulations appear frequently for plane-wave incidence; this is a corpus-level observation limited to the selection described in Section 14.4.

8.8 Spectral methods and numerical approaches

Spectral methods expand a solution in basis functions, such as Fourier series for periodic domains, Chebyshev polynomials on finite intervals, or Legendre polynomials in some axisymmetric formulations. Suitability and convergence depend on regularity, geometry, boundary conditions, and basis choice; convergence should be justified for the selected problem.

Finite element, finite difference, and boundary element methods are possible discretization families for problems without a closed-form solution. The discretization must represent the operators and boundary conditions of the selected nonlocal or memory-dependent model, and its approximation and convergence behavior should be reported.

- **Nonlocal operators.** When the selected constitutive law contains a neighbourhood integral, its discretization must represent that integral. A differential substitute should be used only when its equivalence and the required boundary conditions have been established for the selected model; state which formulation is used.
- **Memory-dependent operators.** For a causal constitutive term defined by a history integral, discretization must approximate the relevant past dependence. Direct history quadrature and validated compressed or transform-based schemes are possible approaches; state the approximation, stability assumptions, and error assessment.
- **Fractional operators.** Fractional operators can be nonlocal in space or history-dependent in time, depending on the definition. Their discretization should match the selected operator and include an appropriate convergence assessment.

8.9 Variational structure and its use as an analytical tool

Alongside the direct transform route, a variational formulation may be considered when a functional is available for the specified coupled model. Marotti de Sciarra's 2013 paper is identified as a study of variational principles for coupled thermoelasticity Marotti de Sciarra [2013]; no specific comparison of admissible spaces, boundary conditions, or solution classes is asserted here. For nonlocal and gradient constitutive laws, the functional and its boundary terms must be derived for the selected model rather than assumed to follow from the local case.

8.10 Classical coupled problem and model-specific approximations

In the classical Fourier thermoelastic model, the momentum and heat equations form a coupled initial-boundary-value problem Biot [1956]. Any sequential or quasi-static treatment should identify the approximation and its model-specific assumptions. For harmonic reductions, report attenuation or dimensionless coupling parameters together with the constitutive convention and geometry that define them.

8.11 Notation summary for Section 8

A layered piezothermoelastic Rayleigh-wave study uses the Green–Naghdi type-III model; its displayed heat equation contains conductivity and additional gradient terms. No energy-partition result is inferred from this example Biswas [2020a].

8.12 Open questions for Section 8

OQ8.1 How are complex modes to be normalized so that the energy ratios computed from them are reproducible across independent implementations, and can a normalization convention be proposed that the field adopts?

Table 15: Mathematical techniques used in continuum wave analysis.

Technique	**Mathematical basis**	**Typical use**	**Model-specific consideration**	**Reference/ example**
Normal mode expansion	Separation of variables; eigenfunction expansion	Layered media and separable geometries	Boundary conditions determine admissible modes	Standard method
Green's functions	Fundamental solution and convolution	Point sources and infinite domains	Anisotropy may complicate integral evaluation	Standard method
Fourier transform	Frequency-domain representation	Time-harmonic problems and dispersion	Initial-value problems require time-domain treatment	Standard method
Laplace transform	Complex-frequency representation	Initial-value problems and transient waves	Inversion may require branch-cut treatment	Standard method
Christoffel equation	$\det[\Gamma_{ik} - \rho c^2 \delta_{ik}] = 0$	Plane waves in anisotropic media	Polynomial order depends on the field equations	Standard method
Scattering matrix	Interface matching formulation	Coupled interface problems	Dimensions and normalization depend on the selected fields	Standard method
Finite element	Weak form and variational principles	Complex geometries and nonlinear problems	Mesh choice and computational cost	Standard method

These are general method roles; convergence and computational behavior depend on the selected problem, discretization, and implementation.

Table 16: Notation introduced in Section 8.

Symbol	**Meaning**	**Units**	**Equation**
Γ_{ik}	Christoffel tensor	varies	(83)
c	Wave speed	$\mathrm{m\,s^{-1}}$	(83)
q	Wave number	$\mathrm{m^{-1}}$	(85)
G_j	Coefficients of the secular equation	varies	(85)
$\hat{f}(\omega)$	Fourier transform	varies	(87)
$\tilde{f}(s)$	Laplace transform	varies	(88)
$\hat{u}_i^{(n)}$	Normalized amplitude of mode n	varies	(86)

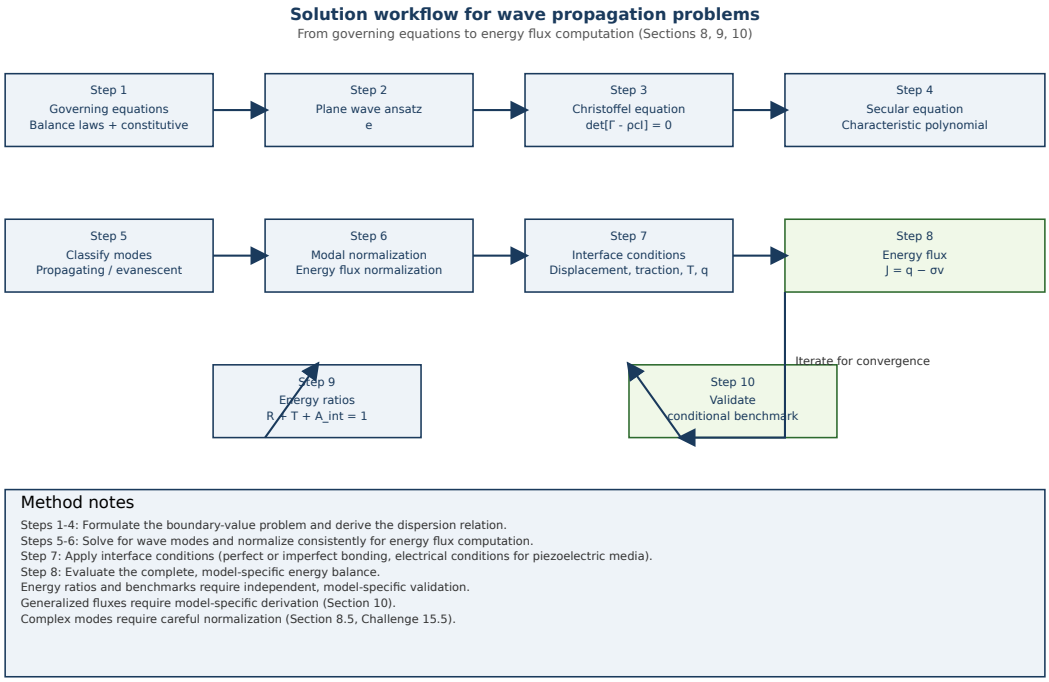


Figure 11: Solution workflow for wave propagation problems: from governing equations to energy flux computation and validation (Sections 8, 9, 10).

OQ8.2 How well do numerical schemes converge for nonlocal and memory-dependent models, and what error results from replacing the integral operator by a differential form?

OQ8.3 Can the two-stage sequential treatment of the classical problem be extended to the generalized families as an approximation, with a bound on the error introduced?

OQ8.4 Which weak formulation is the correct starting point for a variational treatment of a nonlocal and memory-dependent coupled problem?

9 Reflection, Transmission, and Interface Phenomena

This section presents wave reflection, transmission, and energy partitioning at interfaces between different continuum media. The scattering matrix formulation, interface conditions, and energy ratios are developed for elastic, thermoelastic, and piezothermoelastic media.

9.1 Interface geometry

Consider two half-spaces in perfect contact at the interface plane $x_3 = 0$. Medium 1 occupies $x_3 < 0$ and medium 2 occupies $x_3 > 0$. A plane harmonic wave is incident from medium 1 onto the interface, generating reflected waves in medium 1 and transmitted waves in medium 2.

The interface normal is $\boldsymbol{n} = (0, 0, 1)$. All field variables vary harmonically in time with convention $e^{-i\omega t}$ and propagate along the interface with wave number k in the x_1-direction:

$$\text{field}(x_1, x_3, t) = \hat{\text{field}}(x_3) e^{i(kx_1 - \omega t)}. \tag{89}$$

9.2 Wave decomposition: incident, reflected, transmitted modes

In each medium, a finite-mode solution is assembled from partial waves selected for the model and geometry. In the sagittal plane of an isotropic elastic medium, the in-plane polarizations are P and SV; an out-of-plane SH polarization is a separate possibility. Thermoelastic and piezothermoelastic formulations may couple additional

thermal and electrical field variables. The admissible modes and their labels depend on the constitutive equations and electric-field closure.

The displacement field in medium m ($m = 1, 2$) is decomposed as:

$$u_i^{(m)} = \sum_n A_n^{(m)} \hat{u}_i^{(m,n)} e^{i(kx_1 + q_n^{(m)} x_3 - \omega t)}, \tag{90}$$

where $A_n^{(m)}$ are complex amplitude coefficients, $q_n^{(m)}$ are the vertical wave numbers, and $\hat{u}_i^{(m,n)}$ are the polarization vectors.

The incident wave comes from medium 1 with known amplitude. Reflected waves propagate back into medium 1, and transmitted waves propagate into medium 2.

9.3 Interface conditions

For an ideal, perfectly bonded interface with no surface force, the mechanical conditions are displacement and traction continuity. Thermal and electrical conditions below likewise assume the stated perfect-contact and no-surface-source idealizations.

9.3.1 Mechanical conditions

$$u_i^{(1)} = u_i^{(2)}, \qquad \sigma_{i3}^{(1)} = \sigma_{i3}^{(2)} \quad \text{at } x_3 = 0. \tag{91}$$

The first condition enforces displacement continuity; the second expresses traction continuity in the absence of an interfacial force.

9.3.2 Thermal conditions

For thermoelastic media:

$$T^{(1)} = T^{(2)}, \qquad q_3^{(1)} = q_3^{(2)} \quad \text{at } x_3 = 0. \tag{92}$$

These conditions represent perfect thermal contact with no interfacial heat source or thermal resistance; other interface models require different conditions.

9.3.3 Electrical conditions (piezoelectric media)

For piezothermoelastic media:

$$\phi^{(1)} = \phi^{(2)}, \qquad D_3^{(1)} = D_3^{(2)} \quad \text{at } x_3 = 0. \tag{93}$$

For the stated idealization, potential is continuous and normal electric displacement is continuous when no free surface charge is present; a nonzero free surface charge changes the displacement jump. At an insulated interface, the normal displacement condition is instead $D_3 = 0$.

9.3.4 Imperfect bonding

One linear spring-interface model relates traction to the displacement jump:

$$t_i = K_{ij}^u \big(u_j^{(2)} - u_j^{(1)}\big), \qquad t_i = \sigma_{ij}^{(1)} n_j = \sigma_{ij}^{(2)} n_j \quad \text{at } x_3 = 0, \tag{94}$$

where K_{ij}^u is an interfacial stiffness tensor and $\boldsymbol{n}$ is the common interface normal. Under this particular law, the large-stiffness limit enforces displacement continuity, while vanishing stiffness gives zero traction; other imperfect-interface models may use different constitutive relations.

9.4 Scattering matrix formulation

The interface conditions lead to a linear system relating the unknown amplitude coefficients to the known incident wave amplitude. For a thermo-piezoelastic interface problem with N unknown amplitudes and N interface conditions, the system can be written in matrix form:

$$\Delta \boldsymbol{X} = \boldsymbol{Q}, \tag{95}$$

where:

- Δ is the $N \times N$ system matrix whose entries depend on material properties, wave numbers, frequency, and incidence angle,
- $\boldsymbol{X} = (A_1, A_2, \ldots, A_N)^T$ is the vector of unknown amplitude coefficients (reflected and transmitted),
- $\boldsymbol{Q} = (Q_1, Q_2, \ldots, Q_N)^T$ is the known vector determined by the incident wave amplitude.

The entries of Δ are determined by substituting the partial wave expansions into the interface conditions (91)–(93). Each row corresponds to one interface condition; each column corresponds to one unknown amplitude. The solution is obtained by matrix inversion:

$$\boldsymbol{X} = \Delta^{-1}\boldsymbol{Q}. \tag{96}$$

The solution exists and is unique when $\det(\Delta) \neq 0$.

9.5 Secular equation and mode classification

For a selected finite-field model, substituting the modal ansatz into the governing equations gives a homogeneous algebraic system $\boldsymbol{M}(q, \omega, k)\hat{\boldsymbol{a}} = \boldsymbol{0}$. Non-trivial amplitudes require the secular condition

$$\det \boldsymbol{M}(q, \omega, k) = 0, \tag{97}$$

where $\boldsymbol{M}$ is assembled from that model's constitutive equations, orientation, and electric-field closure. Its form, polynomial order when polynomial, and admissible mode count are model-specific; no universal degree or mode set is asserted. The roots are classified as:

- **Propagating modes:** real q — waves that propagate without spatial attenuation,
- **Evanescent modes:** purely imaginary q — waves that decay exponentially with distance,
- **Attenuating modes:** complex q — waves that both propagate and attenuate.

Only outgoing modes (satisfying the radiation condition) are retained in the physical solution.

9.6 Number of boundary conditions vs. unknown amplitudes

A finite-mode interface formulation is often assembled as a square system, but matching the number of independent conditions to the number of unknown amplitudes is not by itself a proof of well-posedness; rank and uniqueness must also be checked. Table 17 summarizes idealized condition configurations for several interface types.

For matrix assembly, Table 18 gives three selected local benchmarks: the Section 10 one-dimensional longitudinal case, an isotropic plane-strain elastic P–SV case, and the displayed classical Fourier thermoelastic plane-strain system. For the last case, the sagittal characteristic determinant factors into one shear pair and two coupled thermo-compressional pairs for generic distinct roots, giving three outgoing modes per side. With one incident mode, the six unknown reflected/transmitted amplitudes are matched by two displacement, two traction, temperature, and normal heat-flux conditions. These counts are model-specific. A square dimension does not by itself prove rank or uniqueness; the determinant and radiation selection still need checking for each parameter set.

The counts are derived for these selected equations and idealized conditions only. A square dimension counts equations and unknowns; it does not establish matrix rank, uniqueness, radiation admissibility, or well-posedness. For LS, GL, GN, phase-lag, piezoelectric, nonlocal, and memory-dependent interfaces, derive the roots and matching conditions from the exact model before assigning a matrix size.

9.7 Energy ratios at interfaces

The interface-plane ratios R_Σ and T_Σ and the interface-local residual A_{int} are defined from total-field normal fluxes as:

$$R_\Sigma = \frac{J_{\text{inc},n} - J^{\text{tot}}_{1,n}}{J_{\text{inc},n}}, \qquad T_\Sigma = \frac{J^{\text{tot}}_{2,n}}{J_{\text{inc},n}}, \qquad A_{\text{int}} = \frac{J^{\text{tot}}_{1,n} - J^{\text{tot}}_{2,n}}{J_{\text{inc},n}}, \tag{98}$$

Here $J_{\text{inc},n} \equiv P_{i,\Sigma} > 0$ is the net, time-averaged normal power flux carried by the incident field alone, evaluated at the interface reference plane used for this normalization; it is not the incident-plus-reflected total-field flux $J^{\text{tot}}_{1,n}$. The latter is the complete incident-side flux, and $J^{\text{tot}}_{2,n}$ is the complete transmitted-side flux at that same plane. All fluxes use one common normal and the total-field fluxes include cross terms. With no interfacial storage or other surface power, $A_{\text{int}} = \mathcal{D}_\Sigma / P_{i,\Sigma}$ for this same-plane normalization. This does not identify $P_{i,\Sigma}$

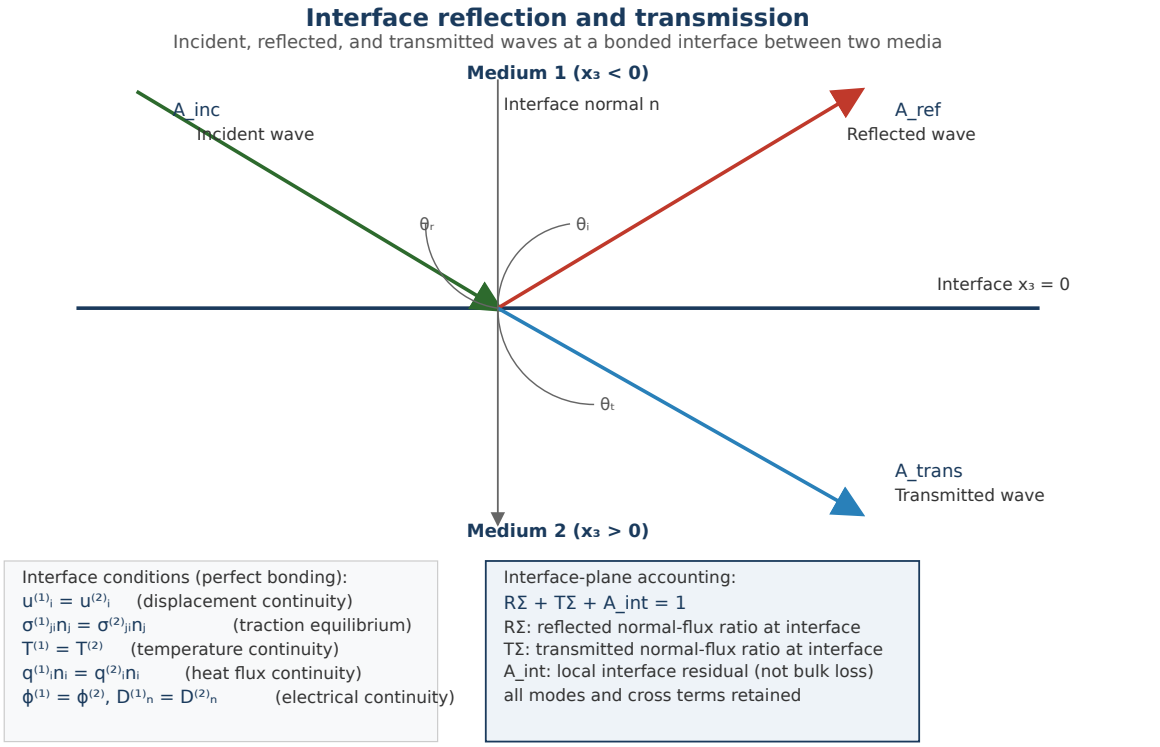


Figure 12: Interface reflection and transmission geometry: incident P-wave from Medium 1 generates reflected P and S waves in Medium 1 and transmitted qP and qS waves in Medium 2. Interface conditions enforce displacement, traction, temperature, heat flux, electric potential, and electric displacement continuity.

Table 17: Representative reflection and transmission interface configurations.

Configuration	**Matching conditions**	**Modal phenomena**	**Physical quantities**	**Model-specific condition**
Welded interface	Displacement and traction continuity	Mode conversion and interface waves	Reflection and transmission amplitudes; total-field flux	Perfect mechanical bonding
Imperfect spring-type interface	Displacement jump linked to traction by a stated interface law	Response depends on selected stiffness law	Model-specific amplitudes	Specify interface stiffness relation
Perfect thermal contact	Temperature and normal heat-flux continuity	Thermal transmission and reflection	Temperature and heat flux	Perfect-contact idealization
Kapitza contact	Normal heat-flux continuity with a temperature jump	Finite-flux temperature discontinuity	Interfacial thermal resistance	Sign follows the chosen normal
Electrical interface	Potential and normal electric-displacement conditions	Piezoelectric coupling and possible surface charge	Potential and electric displacement	Surface-charge condition
Thermo-piezoelastic interface	Coupled mechanical, thermal, and electrical conditions	Multi-field scattering	Field amplitudes and total energy flux	Specify each coupled constitutive law

Interface configurations are idealizations; coefficient relations and energy flux require the stated constitutive and boundary assumptions.

Table 18: Mode counts and scattering-matrix sizes for selected local interface benchmarks.

Model / geometry	**Scattered modal unknowns**	**Matching equations**	**Matrix size**	**Scope and limitation**
1D longitudinal welded (Section 10)	One reflected and one transmitted amplitude	Displacement and normal traction continuity	2×2	Lossless longitudinal benchmark; the determinant and solution are checked in the Section 10 script.
Plane-strain isotropic elastic, incident P	Reflected P/SV and transmitted P/SV amplitudes	$u_1, u_3, \sigma_{13}, \sigma_{33}$ continuity	4×4	Local isotropic elastic sagittal system; generic distinct P and shear roots.
Classical Fourier thermoelastic, plane strain	Three reflected and three transmitted amplitudes	$u_1, u_3, \sigma_{13}, \sigma_{33}, T, q_3$ continuity	6×6	Eqs. (36)–(37); one shear pair and two coupled thermo-compressional pairs for generic distinct roots.

with the remote-plane incident power P_i when bulk attenuation separates the planes; see Section 10.5. R_Σ and T_Σ are interface-plane bookkeeping ratios, not remote-plane reflectance and transmittance, and A_{int} is the interface-local residual, not bulk attenuation.

For a passive interface with no interfacial storage or dissipation ($A_{\text{int}} = 0$), total-field accounting gives:

$$\sum R_\Sigma + \sum T_\Sigma = 1. \tag{99}$$

With interfacial dissipation or storage represented by a nonzero local residual:

$$\sum R_\Sigma + \sum T_\Sigma + A_{\text{int}} = 1. \tag{100}$$

The identity (100) is the interface balance when the theory-specific flux is evaluated from total fields on both sides and all modes and cross terms are retained. It does not add bulk dissipation integrated over either half-space as a third interface channel.

For passive perfect contact with no interfacial storage or dissipation, continuity of total-field flux gives zero interface-local residual. Mode-diagonal outgoing fluxes need not satisfy this identity in attenuating media: incident–reflected and inter-mode cross terms can be nonzero. Bulk dissipation is a volume integral and must be reported separately; adding it to an interface residual would double-count energy loss.

A concrete V-B3d coupled Biot case illustrates the cross-term sensitivity (steel-like/aluminium-like half-spaces, 1-D normal incidence, perfect contact, $f = 4.46 \times 10^{11}$ Hz; arithmetic test data, not a materials claim). The saved audit reports total-field values $R_{\text{flux}} = 0.212026$, $T_{\text{flux}} = 0.787974$, and $A_{\text{flux}} = 2.2 \times 10^{-16}$. The reflected-sum and diagonal-only values are $R_{\text{ref}} = 0.211074$ and $R_{\text{mode}} = 0.211592$, giving $R_{\text{ref}} + T_{\text{flux}} - 1 = -9.5 \times 10^{-4}$ and $R_{\text{mode}} + T_{\text{flux}} - 1 = -4.34 \times 10^{-4}$. Independent half-space dissipation checks close to unity for the computed reflected and transmitted powers. Thus total-field closure is a definition-level bookkeeping identity, not an independent validation; dropping cross terms changes the reported ratios. The saved audit does not specify reference planes sufficient to identify R_{flux} and T_{flux} with either ratio defined here; this example therefore illustrates sensitivity only and does not establish a universal ratio identity.

9.7.1 What the interface partition reveals

The interface-plane ratios defined above are model-dependent observables. Their interpretation requires the stated constitutive flux and boundary conditions; the closure identity alone does not distinguish two theories that predict the same total-field flux balance. Comparisons of reflected and transmitted fractions require a common normalization and source-specific modal conventions.

9.7.2 The pattern and its limits

No universal claim that the conservation identity has been verified across generalized-media papers is made here. For the present total-field definitions, the local interface balance follows from flux continuity only when the specified interface has no storage or dissipation; mode-projected literature ratios require source-specific reconstruction and independent checking. The configurations discussed here idealize flat interfaces between homogeneous half-spaces. Whether, and at what rate, a sharp-interface solution approximates a graded transition in a selected nonlocal, memory-dependent, or gradient model is a model-specific open question; the identities above do not resolve it.

9.8 Interface conditions under memory and nonlocality

The local interface conditions of Section 9.3 cannot be transferred automatically to every nonlocal or history-dependent constitutive law; the appropriate matching variables and data depend on the selected model.

For a model whose constitutive flux is defined by a history integral, equality of instantaneous flux values alone may not specify the interface law or its history data. Any matching condition should be derived for the selected memory model and initial-history assumptions.

For an integral nonlocal model whose kernel crosses the interface, fields on both sides may enter the constitutive response near it. The interface law and any local approximation must be derived for the selected kernel, boundary treatment, and length scales; no universal order estimate is asserted here.

These issues motivate open questions about how constitutive variables and history data are assigned across material discontinuities. The appropriate formulation remains model-specific.

Table 19: Notation introduced in Section 9.

Symbol	Meaning	Units	Equation
$A_n^{(m)}$	Complex amplitude coefficient	varies	(90)
$q_n^{(m)}$	Vertical wave number	1/m	(90)
Δ	System matrix	varies	(95)
$\boldsymbol{X}$	Unknown amplitude vector	varies	(95)
$\boldsymbol{Q}$	Known incident vector	varies	(95)
$R_\Sigma, T_\Sigma,$ A_{int}	Interface-plane reflection, transmission, and local residual ratios	dimensionless	(98)
K_{ij}^u	Interfacial stiffness tensor	varies	(94)

9.9 Notation summary for Section 9

A magneto-thermoelastic diffusion interface model uses LS-type relaxation terms in its coupled field equations Kumar and Kansal [2017]. A nonlocal thermoelastic example applies nonlocality to stress while retaining a local GN-II heat law Sarkar et al. [2019]. A two-temperature porothermoelastic example uses a GN-II formulation Singh [2020]. A higher-order fractional model reports reductions among LS, DPL, and TPL forms at integer order one for its stated equations Aldandani et al. [2026]. A mixed nonlocal placement acts on stress, couple stress, inertia, and the LS flux law Kalkal et al. [2020]. A k*-type boundary-reflection study reports reflection amplitudes for stress-free isothermal and insulated boundaries, not energy ratios Das et al. [2008]. A free-surface calculation reports a reflected ratio approaching unity numerically; this is not a symbolic conservation identity Singh and Guha [2020]. Another study describes its energy-ratio sum as approximately unity for its model, not as an exact identity Kumar and Sharma [2019]. An unsaturated poro-thermoelastic interface study includes interaction energies in its model-specific energy balance Kumar et al. [2022].

9.10 Open questions for Section 9

OQ9.1 What are the interface conditions for nonlocal media where stress depends on strain in a neighborhood?

OQ9.2 How are energy ratios computed for memory-dependent media with hereditary flux?

OQ9.3 What is the correct interface energy balance for piezothermoelastic media with electrical coupling?

OQ9.4 Is the sharp-interface result the limit of a graded-interface result for nonlocal, gradient and memory-dependent media, and at what rate does the difference vanish as the interfacial layer is thickened?

OQ9.5 Can the constitutive variables of a nonlocal or memory-dependent theory be continued across a material discontinuity at all, and what does that imply for the meaning of an interface in such a theory?

10 Energy Transport and Dissipation

Constitutive theories carry energy-storage, transport, and dissipation consequences, but the expressions depend on their state variables and field equations. This section derives the classical local flux under explicit conventions, states the interface balance at a general level, and records family-specific flux claims as open unless their free energy, constitutive evolution, boundary power, and dissipation inequality have been checked. No universal generalized-continuum flux is assumed.

10.1 General energy balance

For classical thermoelasticity, with specific internal energy e, the total energy per unit volume is

$$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e, \tag{101}$$

and the local balance (with no internal heat source) takes the form

$$\frac{\partial \mathcal{E}}{\partial t} + \partial_i J_i = \rho f_i v_i, \tag{102}$$

where $\rho f_i v_i$ is mechanical power per unit volume. Combining momentum balance with the first law yields the classical flux in the next subsection. Equation (102) is not asserted here as a universal generalized-multiphysics identity: additional field energy, external electrical or magnetic power, interface storage, and internal dissipation must be derived for each specified model.

10.2 Classical energy flux

For classical coupled thermoelasticity, adding the mechanical power and internal-energy balances yields the outward-positive flux

$$J_i = q_i - \sigma_{ij} v_j, \tag{103}$$

the classical energy flux. Under this outward-positive convention the mechanical contribution is $-\sigma_{ij} v_j$ and the thermal contribution is q_i.

For time-harmonic mechanical fields with convention $e^{-i\omega t}$, the outward-positive mean flux is

$$\langle J_i \rangle_t = \bar{q}_i - \frac{1}{2} \Re[\hat{\sigma}_{ij} \hat{v}_j^*], \tag{104}$$

where $\bar{q}_i$ is the separately evaluated mean heat flux (zero for a purely oscillatory perturbation about a uniform reference state). The factor $1/2$ and complex conjugate follow from time averaging the real fields. For an interface problem the quantity of interest is the normal component $J_n = J_i n_i$; the sign convention and the normal direction must be stated consistently.

A point worth keeping in view is that the classical flux contains *no* term associated with the internal variables that generalized theories introduce. The entire difficulty of Section 10.4 follows from that absence.

10.3 Interface energy balance

Consider an interface Σ between two media with unit normal $\boldsymbol{n}$. Integrating the general energy balance over a thin pillbox straddling the interface and taking the thinness limit gives the jump condition

$$[J_n] = \mathcal{P}_\Sigma - \frac{\partial \mathcal{E}_\Sigma}{\partial t}, \tag{105}$$

where $J_n = J_i n_i$ is the normal energy flux, $\mathcal{P}_\Sigma$ is the mechanical power input at the interface, and $\mathcal{E}_\Sigma$ is the interfacial energy density.

For a passive interface with no mechanical power input and no interfacial energy storage,

$$[J_n] = 0, \tag{106}$$

that is, the normal energy flux is continuous across the interface. Two qualifications are needed before this condition is used numerically. First, it is a statement about the *correct* energy flux of the theory in which it is used; a flux built from the classical form alone, applied to a family with internal variables, will not satisfy it even when the mathematics is otherwise flawless. Second, the continuity of the normal flux does not by itself imply the scalar conservation identity of Section 10.5: that identity is a statement about modal energies, while (106) is a statement about a single component of a flux at a single surface, and the two are related only after the modal decomposition and normalisation conventions have been fixed.

10.4 Energy transport in generalized theories: the internal-variable structure

The first-law balance retains the boundary power terms for a specified constitutive theory. What changes between generalized theories is the stored free energy and dissipation, which follow from that theory's state variables and evolution laws. There is no universal additional flux term $-\sum_\alpha Y^\alpha \dot{z}^\alpha$: this has units of power per volume, not energy flux, unless an additional spatial transport structure is specified. A family-specific identity therefore requires the model's free energy, evolution law, and boundary terms.

10.4.1 Piezothermoelastic energy flux

For a coupled piezothermoelastic model, the mechanical, electrical, and thermal power terms should be derived from one stated balance convention and the coupled constitutive equations before modal ratios are interpreted.

10.4.2 Energy flux in generalized theories

A family-specific energy balance requires the free energy, flux variables, boundary powers, dissipation terms, and constitutive equations of the selected model.

10.5 Remote-plane energy partition

For passive regions with steady time-averaged powers, no source, no net energy accumulation, and no unlisted energy channel, the remote-plane balance is

$$R_{\mathrm{rem}} + T_{\mathrm{rem}} + A_{\mathrm{bulk},L} + A_{\mathrm{int}} + A_{\mathrm{bulk},R} = 1, \tag{107}$$

where $R_{\mathrm{rem}} = P_r/P_i$ and $T_{\mathrm{rem}} = P_t/P_i$ use positive incident/reflected/transmitted power magnitudes at explicitly stated remote reference planes. The normalized losses are $A_{\mathrm{bulk},L} = \mathcal{D}_L/P_i$, $A_{\mathrm{int}} = \mathcal{D}_\Sigma/P_i$, and $A_{\mathrm{bulk},R} = \mathcal{D}_R/P_i$, with the control-volume powers defined in Eq. (109). This is distinct from the interface-plane bookkeeping identity in Section 9.7; when bulk attenuation separates the planes, $1 - R_{\mathrm{rem}} - T_{\mathrm{rem}}$ includes both bulk losses and the local interface residual. Only when the bulk losses and interface residual vanish does the remote-plane balance reduce to

$$R_{\mathrm{rem}} + T_{\mathrm{rem}} = 1. \tag{108}$$

Kumar and Kansal report a model-specific interface energy-ratio balance with an interaction term in Eq. (80), supporting inclusion of cross terms rather than a universal identity Kumar and Kansal [2017]. Kumar et al. define ratios using average energy flux per incident-wave type; those definitions specify normalization but do not establish closure Kumar et al. [2024a]. A related study describes its eight-model sum as approximately unity for specified imperfect-interface cases; this is model-specific evidence, not an identity Kumar et al. [2024b].

Equation (107) is the steady control-volume balance reproduced by the verification script; its application requires the stated theory-specific flux, complete power channels, interface storage/power convention, and common remote reference planes. It is not the interface-plane identity of Section 9.7. The lossless reduction additionally requires zero bulk loss, zero interface-local residual, and no net interfacial storage. These conditions are not automatic. In the one-dimensional welded example below, the incident–reflected cross term cancels in the time-averaged normal flux; no zero-flux rule for general coupled or evanescent modes is inferred. Report whether each coefficient is an amplitude or a power ratio and state the flux weighting used.

10.5.1 Separate interface residual from bulk absorption

Consider a steady time-averaged one-dimensional control volume with positive normal power-flux magnitudes P_i, P_r, and P_t, all per unit area, at remote incident, reflected, and transmitted reference planes. Let J_L and J_R be the net total-field fluxes on the left and right faces of the interface, both oriented in the positive x direction. For passive, source-free regions with no bulk or interfacial mean energy accumulation, the bulk and interfacial dissipated powers per unit area are

$$\mathcal{D}_L = P_i - P_r - J_L, \qquad \mathcal{D}_\Sigma = J_L - J_R, \qquad \mathcal{D}_R = J_R - P_t. \tag{109}$$

Summing the three control-volume balances gives

$$P_i = P_r + P_t + \mathcal{D}_L + \mathcal{D}_\Sigma + \mathcal{D}_R. \tag{110}$$

Thus, with $R_{\mathrm{rem}} = P_r/P_i$ and $T_{\mathrm{rem}} = P_t/P_i$, the remote-plane residual $1 - R_{\mathrm{rem}} - T_{\mathrm{rem}}$ already contains both bulk dissipation and local interface dissipation. By contrast, $A_{\mathrm{int}} = \mathcal{D}_\Sigma/P_i = (J_L - J_R)/P_i$ is local to the interface. Adding the bulk terms to $1 - R_{\mathrm{rem}} - T_{\mathrm{rem}}$ a second time double-counts them; the split in Eqs. (109)–(110) is checked symbolically in the accompanying verification script. The derivation assumes steady averaged powers, no storage in the bulk control volumes, and no unlisted energy channel; other theories require their complete energy flux and power terms.

10.5.2 Worked one-dimensional welded-interface system

For a normal-incidence longitudinal wave at $x = 0$ in two lossless one-dimensional bars, let c_m be each bar's longitudinal wave speed and take real positive impedances $Z_m = \rho_m c_m$ and wave numbers $k_m = \omega/c_m$. Under the $e^{-i\omega t}$ convention, the displacement amplitudes are

$$\hat{u}_1 = A_i e^{ik_1 x} + A_r e^{-ik_1 x}, \qquad \hat{u}_2 = A_t e^{ik_2 x}. \tag{111}$$

Displacement and traction continuity, using $\hat{\sigma}_{xx} = \rho_m c_m^2 \partial_x \hat{u}_m$, give $A_i + A_r = A_t$ and $Z_1(A_i - A_r) = Z_2 A_t$. With A_r and A_t as the two unknown amplitudes, these are two scalar conditions and yield the 2×2 system

$$\begin{bmatrix} 1 & -1 \\ -Z_1 & -Z_2 \end{bmatrix} \begin{bmatrix} A_r \\ A_t \end{bmatrix} = -A_i \begin{bmatrix} 1 \\ Z_1 \end{bmatrix}. \tag{112}$$

Its determinant is $-(Z_1 + Z_2)$, so the stated positive-impedance case has a unique solution, with amplitude ratios $r = A_r/A_i = (Z_1 - Z_2)/(Z_1 + Z_2)$ and $t = A_t/A_i = 2Z_1/(Z_1 + Z_2)$. For the mechanical flux $\langle J_x \rangle = -\frac{1}{2}\Re[\hat{\sigma}_{xx}\hat{v}_x^*]$, the incident and reflected power ratios are $R = |r|^2$ and $T = (Z_2/Z_1)|t|^2$. These coincide with the remote-plane ratios because this welded example has lossless, non-attenuating half-spaces. Direct substitution gives $R + T = 1$; this result is not asserted for an imperfect, dissipative, or generalized interface. To show the incident/reflected cross term rather than assume it away, the same-medium fields in Eq. (111) give $\hat{v}_i = -i\omega A_i e^{ik_1 x}$, $\hat{\sigma}_i = ik_1 \rho_1 c_1^2 A_i e^{ik_1 x}$, $\hat{v}_r = -i\omega A_r e^{-ik_1 x}$, and $\hat{\sigma}_r = -ik_1 \rho_1 c_1^2 A_r e^{-ik_1 x}$. Before taking the real part in the harmonic flux, their mixed product is

$$\hat{\sigma}_i \hat{v}_r^* + \hat{\sigma}_r \hat{v}_i^* = \rho_1 c_1^2 k_1 \omega \left[-A_i A_r^* e^{2ik_1 x} + A_r A_i^* e^{-2ik_1 x} \right],$$

the negative of a complex number plus its conjugate, and therefore purely imaginary; hence its time-averaged normal-flux contribution $-\frac{1}{2}\Re(\hat{\sigma}_i \hat{v}_r^* + \hat{\sigma}_r \hat{v}_i^*)$ is zero. This cancellation is derived only for the stated one-dimensional, lossless longitudinal waves. The matrix dimensions, determinant, amplitudes, field-derived flux weights, cross term, and partition identity are reproduced by the accompanying verification script.

10.6 Dissipation mechanisms

Dissipative mechanisms can include heat-flux relaxation, viscous stress, dielectric loss, interfacial friction, or hereditary memory. Their presence and non-negative dissipation conditions depend on the precise constitutive equations and admissible state space. A single scalar dissipation function or common parameter-ordering rule does not apply uniformly across these families. In particular, the condition $\tau_q \geq \tau_T \geq 0$ and a positive Lord–Shulman relaxation time are not, by themselves, a full proof of thermodynamic admissibility or stability. Deriving model-specific free energies and dissipation inequalities remains open.

10.7 Thermodynamic admissibility: well-posedness, causality, positivity

Thermodynamic admissibility, mathematical well-posedness, and finite-speed propagation are distinct questions. A constitutive theory requires its own free-energy/entropy statement and dissipation inequality; an initial-boundary-value problem requires a separate existence, uniqueness, and continuous-dependence analysis; propagation speed is a property of the selected equations and parameter regime. Fourier heat conduction gives a parabolic equation with instantaneous mathematical support. Statements for relaxation, phase-lag, memory, fractional, and gradient models depend on their exact constitutive laws.

10.8 Comparative table: energy transport across theory families

Table 20 summarizes the classical flux and model-specific terms relevant to generalized energy balances.

10.9 Notation summary for Section 10

10.10 Open questions for Section 10

OQ10.1 What is the explicit time-domain energy flux and dissipation function for each of the families listed in Section 10.4, derived from that family's free energy, evolution laws, and boundary-power terms?

OQ10.2 Is there a common formal structure under which the energy fluxes of the Lord–Shulman, phase-lag, Moore–Gibson–Thompson, and memory-dependent families become instances of a single template, so that a single derivation suffices for all of them?

Table 20: Energy-flux structures by constitutive family.

Theory family	**Energy/flux structure**	**Terms specific to the selected model**
Classical thermoelasticity	$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e$; $J_i = q_i - \sigma_{ij} v_j$ (outward-positive)	Heat-flux constitutive law and boundary power
Generalized constitutive families	Model-specific stored energy and flux	Evolution equations, internal variables, boundary power, and dissipation terms

No universal flux formula is implied for the generalized families.

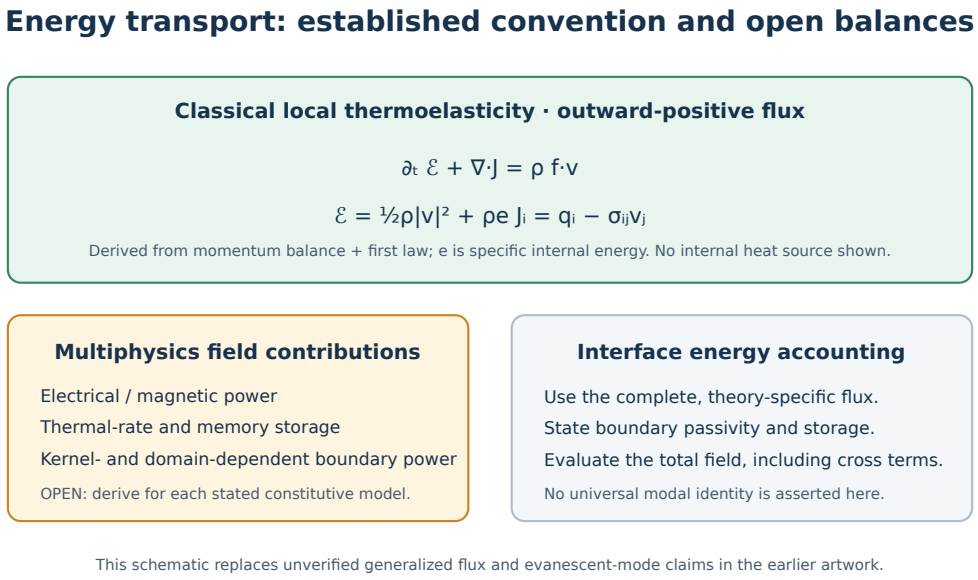


Figure 13: Energy-balance conventions and open flux audit. The classical local thermoelastic flux is derived under the stated outward-positive convention; generalized multiphysics contributions and modal energy identities remain open pending model-specific derivations.

Table 21: Notation introduced in Section 10.

Symbol	**Meaning**	**Units**	**Equation**
$\mathcal{E}$	Total energy density	$\mathrm{J\,m^{-3}}$	(101)
$\boldsymbol{J}$	Energy flux vector	$\mathrm{W\,m^{-2}}$	(102)
$\mathcal{P}$	Mechanical power input	$\mathrm{W\,m^{-3}}$	(102)
J_i	Classical energy flux component	$\mathrm{W\,m^{-2}}$	(103)
$\langle J_i \rangle_t$	Time-averaged energy flux	$\mathrm{W\,m^{-2}}$	(104)
$\mathcal{E}_\Sigma$	Interfacial energy density	$\mathrm{J\,m^{-2}}$	(105)
$\mathcal{P}_\Sigma$	Interfacial power input	$\mathrm{W\,m^{-2}}$	(105)
R, T, A	Reflection, transmission, absorption coefficients	dimensionless	(107)

OQ10.3 Which observable would distinguish the energy partition predicted by two competing theories at the same interface, so that the difference could be measured rather than argued?

OQ10.4 How should the entropy production of the conservative Green–Naghdi type II structure be related to the entropy production of the dissipative branches, given that the former has no dissipation mechanism?

OQ10.5 What normalisation of complex modes makes the energy ratios of Eq. (107) reproducible across independent implementations?

OQ10.6 Can a dissipation potential be constructed for the memory-dependent kernel in the spirit of the classical one for the relaxation terms, and does its positivity restrict the admissible kernels to a smaller set than is currently assumed?

OQ10.7 What is the relative importance of interface dissipation and bulk dissipation in the absorption coefficient of Eq. (107) for realistic bonded joints?

11 Advanced Theoretical Developments

The theory families of Sections 4–7 were each constructed by generalising one specific constitutive law. The developments surveyed in this section generalise the generalisation: they act on the *operator* rather than on the constitutive law, and they therefore interact with every family simultaneously. Two of them—fractional-order differentiation and the introduction of additional temperature fields—are now established enough to be treated as substantive alternatives rather than as speculative extensions, and this section gives them the same weight as the classical constructions of Section 4.

11.1 Fractional-order continuum models

Fractional-order continuum models replace integer-order derivatives with fractional derivatives in the governing equations. The replacement is not a formal device: a fractional derivative of order α is defined by an integral over the entire history of the field, and it therefore introduces temporal nonlocality into the governing equations in a way that is structurally equivalent to a hereditary integral with a power-law kernel.

11.1.1 Fractional thermoelasticity

The fractional heat conduction equation, as formulated in the thermoelastic setting, is

$$ {}^{C}D_t^{\alpha} T = a_\alpha \nabla^2 T, \qquad 0 < \alpha \leq 2, \tag{113} $$

where ${}^{C}D_t^{\alpha}$ is the Caputo fractional derivative of order α and $[a_\alpha] = \mathrm{m}^2\mathrm{s}^{-\alpha}$ (for dimensional T and x). For $\alpha = 1$ this reduces to the classical heat equation; for $1 < \alpha \leq 2$ it describes wave-like heat propagation with memory, interpolating between diffusive and wave-like behaviour. The Caputo derivative is defined for $0 < \alpha < 1$ by

$$ {}^{C}D_t^{\alpha} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(\tau)}{(t-\tau)^{\alpha}} \, d\tau, \tag{114} $$

and the corresponding expression for $1 < \alpha \leq 2$ is obtained by applying the operator to the second derivative of f. The integral form is the physically informative one: the state of the field at time t is determined by its entire past, weighted by a kernel that decays algebraically rather than exponentially.

Caputo operator and origin. Caputo's 1967 work introduced the fractional derivative in viscoelastic stress–strain models to represent nearly frequency-independent Q; the paper examined Earth's free modes and an aluminium rod and fitted an order near 0.15. It is an origin in viscoelasticity/geophysics, not in heat conduction. In Eq. (114), the derivative remains inside the integral for $0 < \alpha < 1$; the printed definition (5) is reconstructed from its proof-page integrand because the display is not legible in the checked copy Caputo [1967].

11.1.2 The fractional heat equation as a hereditary integral

The connection with the memory-dependent formalism of Section 6 is exact and important, because it means that the fractional and the memory-dependent descriptions are not competing theories but two representations of the same object under different assumptions about the memory kernel. Writing Eq. (113) in the hereditary form

$$T(t) = T(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} a \nabla^2 T(\tau)\, d\tau, \tag{115}$$

shows that the fractional formulation is the special case in which the integral kernel is the power law $(t-\tau)^{\alpha-1}/\Gamma(\alpha)$ in elapsed time. The generalization of Section 6 replaces the power law by an arbitrary kernel, and thereby trades the one-parameter fractional family for a much larger one; the price is that the resulting operator is no longer a differential operator of fractional order, and the analytical machinery that makes the fractional heat equation tractable—the eigenfunction expansion in terms of the Mittag–Leffler function, and the closed-form fundamental solutions—is lost. The choice between the two representations is thus a choice between analytical tractability and constitutive flexibility, and it should be made explicitly rather than by default.

The analytical solution of the fractional heat conduction equation in radially symmetric domains was obtained in closed form by Povstenko Povstenko [2007], who derived the fundamental solutions needed to write the associated thermal stresses in spherical geometry; this work provides the reference against which later numerical inversions of fractional thermoelastic boundary-value problems are checked Povstenko [2004, 2009, 2011a, 2015]. The associated thermal stresses are the quantity of practical interest, and the difficulty of the problem is that the fractional operator couples the mechanical and thermal fields in a way that does not admit the plane-wave decoupling available in the classical case.

MDD versus fractional modelling. Wang and Li (2021, Part I) argue that the MDD formulation gives the memory duration and kernel an explicit temporal interpretation, whereas the fractional-order description emphasizes local-change behavior; their illustration is a logistic-population model, not a thermoelastic equation Wang and Li [2021]. This is an attributed modelling perspective, not a general theorem.

11.1.3 Fractional Cattaneo-type equations

Povstenko's Eq. (29), on printed p. 101, gives a related fractional Cattaneo-type flux law with source coefficient ζ for $0 < \alpha \leq 1$. Appendix A identifies the derivative as Caputo (printed p. 113), while the source omits the superscript C used here for clarity. The source does not identify ζ with $\tau^\alpha/\Gamma(1+\alpha)$ Povstenko [2011b]:

$$q_i + \zeta\, {}^C D_t^\alpha q_i = -kT_{,i}. \tag{116}$$

The displayed coefficient is the source's ζ; no conversion to a Gamma-normalized relaxation parameter is asserted.

11.1.4 Fractional models with internal structure

Fractional operators and internal-rotation fields are distinct modeling ingredients. A formulation that combines them must specify its own constitutive variables and governing equations.

Fractional thermoelasticity. Sherief, El-Sayed, and Abd El-Latief give $q_i + \tau_0\, {}^C D_t^\alpha q_i = -k_{ij} T_{,j}$ for $0 < \alpha \leq 1$ in Eq. (2.29). Their Eq. (2.37) applies the same factor $(1 + \tau_0\, {}^C D_t^\alpha)$ to both the temperature-rate and strain-rate coupling terms; their coefficient remains source-specific Sherief et al. [2010].

11.2 Strain-gradient and couple-stress theories

Strain-gradient theory introduces higher-order strain gradients in the constitutive relations:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \ell^2 C_{ijkl}\nabla^2 \varepsilon_{kl}. \tag{117}$$

Couple-stress theory introduces couple stresses and micro-rotations as independent degrees of freedom. The force stress tensor becomes generally non-symmetric, and the angular momentum balance includes couple stress terms:

$$\sigma_{[ij]} + \mu_{ij,j} = 0, \tag{118}$$

where μ_{ij} is the couple stress tensor, and the antisymmetric part $\sigma_{[ij]}$ is balanced by the couple stress rather than vanishing as it does in classical elasticity.

The relation between these theories and the micropolar theory of Section 7 is one of specialisation. Micropolar elasticity Eringen [1966], extended to the thermoelastic case by Tauchert, Claus and Ariman Tauchert et al. [1968], is the most general of the three in the sense that it admits an independent micro-rotation field, whereas couple-stress theory and strain-gradient theory can be obtained from it by imposing constraints on the micro-rotation and

on the antisymmetric stress. This hierarchy matters practically, because it explains why the literature contains several formulations that differ in their kinematic assumptions while sharing a common elastic kernel, and why a comparison between them must begin by identifying which of them is being used.

These theories are relevant for materials with internal structure: granular materials, foams, composites with oriented inclusions, and biological tissues. In the thermoelastic setting, the micro-rotational field carries an additional energy channel, so the energy balance of Section 10.1 acquires terms that have no counterpart in a classical continuum, and the mechanical power transmitted across a surface is no longer the complete description of the transport.

11.3 Two-temperature and three-temperature formulations

11.3.1 Two-temperature theory

The two-temperature formulation distinguishes between the conductive temperature T, which is what a thermometer measures and what enters the heat conduction law, and the thermodynamic or dynamical temperature T^*, which enters the entropy relation and the internal energy:

$$T^* = T - \delta^2 \nabla^2 T, \tag{119}$$

where δ^2 is the two-temperature parameter. This model is distinct from the single-temperature, temperature-rate-dependent Green–Lindsay formulation.

11.3.2 Three-temperature theory

Three-temperature theories introduce a third temperature variable: the temperature of a secondary phase in a composite, or the electron-gas temperature in a semiconductor, or a separate lattice and effective temperature in a system with two thermal carriers. The models are relevant for multi-phase materials in which different sub-systems have different thermal response times, and for semiconductor structures in which the electron gas equilibrates with the lattice far more slowly than either equilibrates internally. The technical difficulty is the same in both settings: three temperature fields require three interface conditions, and the coupling terms in the momentum and energy balances make the resulting system of equations considerably more complicated than the two-temperature case. The fractional-order model of micropolar thermoelasticity of Section 11.1 adds a further layer, in which the micro-rotational field and the fractional memory both act on a temperature field that is already non-local in space.

11.4 Emerging hybrid models

Emerging formulations combine multiple generalization mechanisms:

- **Nonlocal–fractional hybrids** combine spatial nonlocality with temporal memory, for materials that exhibit both. The two mechanisms act on different operators and their dispersions are not additive, so the combined model has more parameters than either parent and correspondingly more severe identifiability problems.
- **Memory-dependent nonlocal models** combine an integral-form memory operator with a nonlocal stress law. The thermo-piezoelectric functionally graded rotating rod treated by Abouelregal and co-authors Abouelregal et al. [2023] is an instance of this class, in which a memory-dependent dynamic response is combined with a functionally graded rotating geometry and a piezoelectric coupling.
- **Hybrid nonlocal and memory-dependent multiphysics models** combine both mechanisms with a coupled field. The heat-map analysis of wave dynamics in nonlocal Kelvin–Voigt piezo-semiconductors under a memory-dependent heat flux Gupta et al. [2026] is representative, and the particular interest of that work is that the nonlocal operator acts on the stress while the memory operator acts on the heat flux, so that the two generalizations enter the two channels of the energy balance of Section 10.1 respectively.
- **Phase-field models** couple a phase-field variable to the thermoelastic fields for fracture and phase transformation.
- **Thermo-viscoelasticity with memory** uses hereditary integral formulations for viscoelastic materials with thermal coupling, and is the classical route to a thermodynamically consistent memory effect with a non-negative dissipation.

The general difficulty with all of these is the one identified in Section 14.4: each additional generalization adds parameters, and each parameter is usually fitted to the same data that the model is then used to explain. The resulting models are not thereby wrong, but the evidence supporting them is weaker than their flexibility would suggest, and the practical question is always which subset of the available parameters the data can actually constrain.

11.5 Mathematical limitations and open problems

11.5.1 Mathematical limitations

1. **Limits between formulations.** Limit theorems for specific model pairs require a stated system, parameter regime, topology, and convergence result.
2. **Well-posedness.** Each theory family has its own well-posedness conditions; a unified framework is lacking. The nonlocal family is the most acute case, since the differential form of its constitutive law raises the order of the governing equation and does not yield a well-posed initial-value problem in the classical sense.
3. **Nonlocal interface conditions.** The correct interface conditions for nonlocal media are not established, and the integral and differential formulations lead to different requirements.
4. **Analytical tractability.** Closed-form solutions for fractional formulations depend on the geometry, operator, and data; numerical inversion is used for cases without an available analytic solution.
5. **Parameter identifiability.** No general theory exists for determining which of the parameters of a combined model can be inferred from a given measurement, and this is the practical obstacle to the use of the hybrid models of Section 11.4.

11.5.2 Open problems

1. Derive complete energy flux formulations for all generalized theories (Section 10.4).
2. Establish rigorous limit theorems between theory families, with explicit error estimates.
3. Develop unified well-posedness criteria covering the differential, integral, fractional, and gradient formulations.
4. Establish nonlocal interface conditions, and show that the integral and differential formulations give consistent interface traction.
5. Validate models experimentally across parameter ranges, with particular attention to the dispersion signature that distinguishes a nonlocal mechanism from a gradient mechanism (Section 7.3).

11.6 Notation summary for Section 11

Table 22: Notation introduced in Section 11.

Symbol	Meaning	Units	Equation
α	Fractional derivative order	dimensionless	(113)
${}^{C}D_{t}^{\alpha}$	Caputo fractional derivative of order α	—	(114)
Γ	Euler gamma function	—	(114)
a	Thermal diffusivity	$\mathrm{m}^2\,\mathrm{s}^{-1}$	(113)
τ^{α}	Fractional relaxation parameter	s^{α}	(116)
ℓ	Strain-gradient length scale	m	(117)
μ_{ij}	Couple stress tensor	$\mathrm{N\,m}^{-1}$	(118)
δ^2	Two-temperature parameter	m^2	(119)

The LS-type model carries relaxation on both thermal-inertia and coupling terms, unlike the original GL convention. Youssef [2006]. A seven-model fractional operator catalogue provides a third route alongside Taylor truncations and integer higher-order laws. Yu et al. [2024].

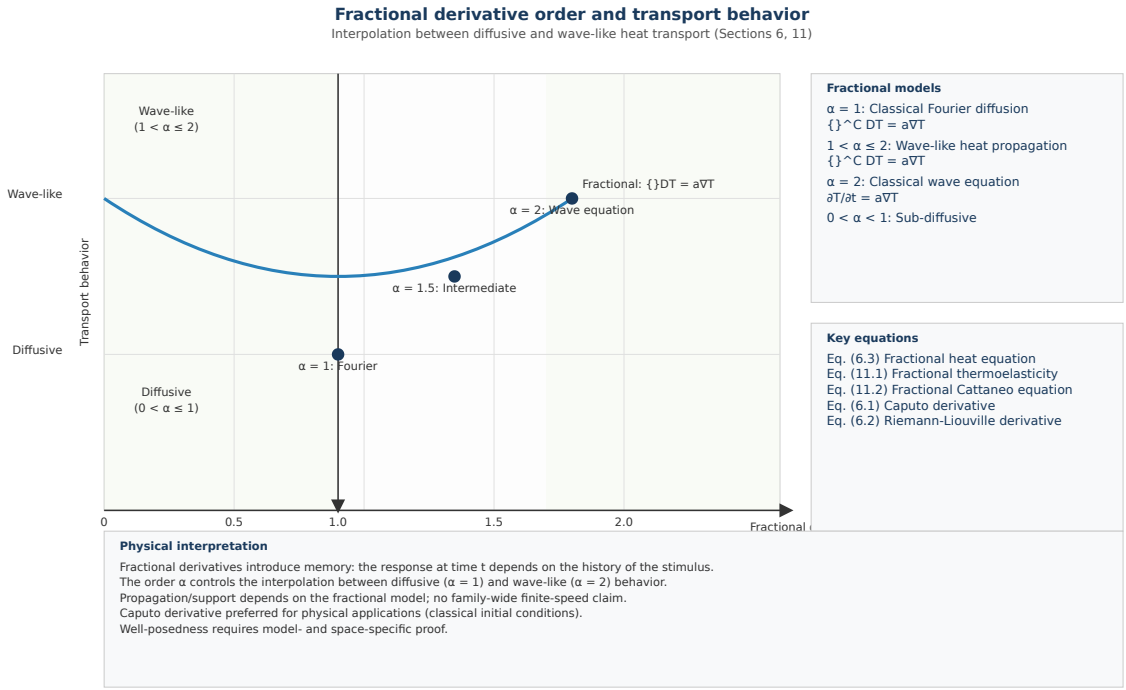


Figure 14: Fractional derivative order and transport behavior: interpolation between diffusive and wave-like heat transport (Sections 6, 11).

11.7 Open questions for Section 11

OQ11.1 Can rigorous limit theorems be established between the thermoelastic theory families, with error estimates in a stated norm?

OQ11.2 Can a unified well-posedness framework be established for the differential, integral, fractional, and gradient formulations, and what does it require of the constitutive law?

OQ11.3 What are the correct interface conditions for models in which a nonlocal operator acts on the stress and a fractional or memory operator acts on the heat flux simultaneously?

OQ11.4 When is the power-law memory kernel of the fractional formulation to be preferred over a general kernel in the memory-dependent formulation, and how large is the error incurred by the restriction?

OQ11.5 Which parameter combinations of a hybrid model are identifiable from a given experimental modality, and which are not?

12 Comparative Synthesis of Existing Research

This section synthesizes the reviewed research using a comparison framework that is deliberately organised by *scientific content* rather than by chronology or by authorship. Three organising questions recur: what does each theory retain of the second law of thermodynamics; what replaces Fourier's constitutive law; and what does each theory predict for the speed, attenuation, and dispersion of the diffusive mode. A theory that answers these three questions differently is not a variant of another theory—it is a different claim about how heat is transported in solids, and the comparison below is organised to make those differences explicit.

The synthesis is constructive throughout. Differences between theories are presented as modelling choices with different assumptions and different domains of validity, and the framework identifies the conditions under which two theories can be meaningfully compared at all.

Survey. Shakeriaski et al. catalogue applications of LS, GL, GN, dual-phase-lag, and multi-phase-lag models across geometries. The survey is used here as a coverage map, not as support for equations or stability results Shakeriaski et al. [2021].

Thermodynamic conclusions depend on the equations and hypotheses specified for each formulation.

12.1 Cross-theory comparison

Tables 23 and 24 compare ten theory families against ten criteria; the final row in each table lists representative sources for the corresponding family. These tables are a compact index to the detailed treatments in Sections 4–11, not a substitute for them.

Table 23: Cross-theory comparison, Part A: classical, relaxation, rate-dependent, and Green–Naghdi families.

Criterion	CD	LS	GL	GN-II	GN-III
Thermal law	Fourier conduction	Relaxing heat flux	Temperature-rate terms in the displayed model	Thermal-displacement formulation	Rate and thermal-displacement terms
State variables	Model-specific	Model-specific	Model-specific	Model-specific	Model-specific
Characteristic time	—	Relaxation parameter	Relaxation parameters	—	Model-specific parameters
Characteristic length	—	—	—	—	—
Thermal-wave behavior	Parabolic classical heat equation	Depends on coupled equations and data	Depends on coupled equations and data	Depends on coefficients and domain	Depends on coefficients and domain
Energy behavior	Follows from the stated balance and constitutive law	Model-specific	Model-specific	Model-specific	Model-specific
Spatial order	Set by the governing equations	Set by the governing equations	Set by the governing equations	Set by the governing equations	Set by the governing equations
Mathematical structure	Continuum field equations	Coupled field equations	Coupled field equations	Coupled field equations	Coupled field equations
Interface conditions	Depend on the interface model	Depend on the interface model	Depend on the interface model	Depend on the interface model	Depend on the interface model
Propagation properties	Model-specific	Model-specific	Model-specific	Model-specific	Model-specific
Primary source	Biot [1956]	Lord and Shulman [1967]	Green and Lindsay [1972]	Green and Naghdi [1993]	Green and Naghdi [1992]

The substantive MDD cells in Table 24 are marked model-dependent because this comparison does not display a governing MDD law with a source-anchored equation; the cited Wang–Li item is a representative source lead, not verification of a family-wide property.

Three of these rows carry most of the interpretive content.

Thermal law and state variables. The number of state variables alone does not determine the memory or modal structure of a theory. Local Fourier, relaxation, phase-lag, higher-order, hereditary, fractional, and spatially nonlocal laws use different state spaces and operators; a named family may include multiple variants. Meaningful comparison identifies the exact constitutive equations and field variables. Povstenko's Eq. (29) uses the source coefficient ζ in a fractional Cattaneo flux law for $0 < \alpha \leq 1$ (printed p. 101); Appendix A states that the Caputo derivative is used while its C subscript is omitted (printed p. 113). The equation supplies no mapping from ζ to $\tau^{\alpha}/\Gamma(1+\alpha)$ Povstenko [2011b]. Tzou's unified-field paper frames the dual-phase-lag generalization as a high-rate heat-conduction model, associating the flux lag with small-scale temporal response and the gradient lag with small-scale spatial response (printed p. 8) Tzou [1995]. Quintanilla's one-dimensional Chandrasekharaiah–Tzou thermoelastic problem reports exponential stability when $\tau_\theta > \tau_q/2$ and unstable solutions when $\tau_\theta < \tau_q/2$ (printed p. 720, conclusion). These strict inequalities do not classify the critical case $\tau_\theta = \tau_q/2$; no equality-case result is asserted here. The findings are specific to the stated problem, not a universal dual-phase-lag admissibility criterion Quintanilla [2003].

Table 24: Cross-theory comparison, Part B: phase-lag, higher-order, memory-dependent, and nonlocal families.

Criterion	**DPL**	**TPL**	**MGT**	**MDD**	**Nonlocal**
Thermal law	Lagged flux/ temperature relation	Three-lag constitutive relation	Variant-specific higher-order law	Model-dependent	Spatially nonlocal elastic law
State variables	Model-specific	Model-specific	Model-specific	Model-dependent	Model-specific
Characteristic time	Lag parameters	Lag parameters	Variant-specific parameters	Model-dependent	—
Characteristic length	—	—	—	Model-dependent	Model-specific length scale
Thermal-wave behavior	Depends on the selected lags and equations	Depends on the selected lags and equations	Depends on the selected MGT formulation	Model-dependent	Not specified by an elastic-only model
Energy behavior	Model-specific	Model-specific	Model-specific	Model-dependent	Model-specific
Spatial order	Set by the governing equations	Set by the governing equations	Set by the governing equations	Model-dependent	Set by the constitutive form
Mathematical structure	Coupled field equations	Coupled field equations	Higher-order field equations	Model-dependent	Nonlocal or gradient equations
Interface conditions	Depend on the interface model	Depend on the interface model	Depend on the interface model	Model-dependent	Depend on the interface model
Propagation properties	Model-specific	Model-specific	Model-specific	Model-dependent	Model-specific
Primary source	Tzou [1995]	Choudhuri [2007]	Quintanilla [2019]	Wang and Li [2011]	Eringen [1972]

Thermal wave speed and dissipation. Propagation, dissipation, and stored energy depend on the precise equations and parameter regime; they cannot be ranked by family name alone. These properties follow from the displayed constitutive equations, balance laws, and coefficient assumptions. Chester's Eqs. (5)–(7) give a damped thermal-wave equation for his relaxation-flux model, while Eq. (8)'s relation to phonon speed is conditional on an idealized dispersionless phonon gas (printed pp. 2014–2015). Guyer and Krumhansl's separate p. 766 paper yields a nonlocal coefficient in a restricted scattering regime (Eq. (59), p. 773); their p. 778 companion treats second sound and phonon-hydrodynamic phenomena, including a source-specific damping relation (Eq. (17), p. 782). These kinetic and phenomenological results are not universal continuum thermoelastic laws. Chandrasekharaiah's 1998 article is a review of recent hyperbolic-thermoelasticity literature, not primary equation evidence Chester [1963], Guyer and Krumhansl [1966a,b], Chandrasekharaiah [1998]. Dell'Oro and Pata's separate MGT article relates a third-order MGT equation to linear viscoelasticity with an exponential memory kernel; it is not evidence for the distinct three-author critical-case stability claim in Section 4 Dell'Oro and Pata [2016].

Well-posedness and causality. Finite propagation, causality, and well-posedness are separate properties. Fourier's parabolic heat equation has instantaneous mathematical support, while relaxation and thermal-displacement models may yield hyperbolic branches under particular constitutive assumptions. Phase-lag, fractional, memory, and nonlocal formulations depend on their operators, parameters, initial data, and boundary conditions.

12.2 What each family retains of the second law

Read across the table, the families separate not by their equations but by the answer they give to a single question: does the mechanical energy of the body have a well-defined conserved or monotonically decreasing value? The three structural answers are worth stating separately, because most of the apparent confusion in the wider literature comes from conflating them.

Conservative structure. A conservative balance, when present, follows from the specified free energy, evolution equations, and boundary terms; it is not determined by a family label alone.

Dissipative structure. Some Cattaneo–Christoff, Lord–Shulman, phase-lag, Moore–Gibson–Thompson-type, and memory-dependent models are dissipative, but neither decay of mechanical energy nor an irreversibility law follows from a family name. Such conclusions require the exact constitutive system, free energy, admissible parameters, and boundary conditions; they are not asserted universally here.

Other structures. Models that do not fit a simple conservative/dissipative dichotomy require analysis of their own field equations and energy balance. Limiting relations between formulations likewise depend on the equations and their coefficient scaling.

12.3 Mathematical relationships: limiting cases and specializations

Table 25 lists candidate formal reductions only. Each row is conditional on retaining the same state variables, constitutive convention, source terms, and boundary data; none is presented as a convergence theorem.

On taking these limits. The rows in Table 25 are formal parameter substitutions, not convergence theorems. Convergence results have been established for some model pairs, including Moore–Gibson–Thompson and classical/relaxation limits, but this review does not reproduce those proofs and does not claim a theorem for every row in the table. A row therefore identifies a candidate reduction under matched constitutive conventions, sources, state variables, and boundary data; it does not by itself establish convergence, an error estimate, or a rate. The coverage is uneven across model pairs and variants, as noted in Section 15. This distinction matters for the taxonomy of Section 14: formal limits aid comparison, while quantitative comparison requires a justified limit result for the particular pair and regime.

12.4 Divergence map: why results differ across the corpus

The literature contains many cases in which two studies address a nominally identical problem and obtain apparently incompatible results. Examination of these cases shows that the incompatibility is almost always attributable to a difference in modelling commitments rather than to an inconsistency in the mathematics. The map below records the commitments responsible and the procedure by which each is eliminated, so that a comparison can be posed in a form in which it is meaningful.

Table 25: Limiting cases between thermoelastic theories, with the parameter that is taken to its limit and the sense of the limit.

From	To	Limit	What the limit removes
Lord–Shulman	Classical coupled	$\tau_0 \to 0$	The thermal wave; the heat equation becomes parabolic again
Dual phase-lag	Lord–Shulman	$\tau_T \to 0$, with $\tau_q = \tau_0$	Removes the temperature-gradient lag while retaining the heat-flux relaxation
Triple phase-lag	Dual phase-lag	$\tau_\nu \to 0$	The thermal-displacement-gradient term
Triple phase-lag	Classical coupled	all $\tau \to 0$	Every relaxation mechanism simultaneously
Memory-dependent + phase-lag	Phase-lag	Model-dependent	No reduction asserted without a displayed, source-anchored MDD kernel/normalization and phase-lag law
Memory-dependent + phase-lag	Classical coupled	Model-dependent	No family-wide reduction asserted without the selected kernel, normalization, delay limit, and phase-lag law
Green–Naghdi III	Green–Naghdi II	$\kappa \to 0$, same source-free coefficient convention	Removes the temperature-rate term; formal reduction only
Fractional order $\to 2$	Classical coupled	$\alpha \to 2$	Fractional differentiation; the Caputo operator becomes the second derivative
Nonlocal length $\to 0$	Local theory	$\ell \to 0$	The nonlocal operator becomes the local gradient

A meaningful comparison identifies the precise formulation, state variables, and assumptions, even when multiple models share a family label. Green–Lindsay is a single-temperature, temperature-rate-dependent formulation, distinct from two-temperature thermoelasticity.

12.5 Conditions under which two theories can be compared

Because the families differ in so many structural respects, a comparison is only meaningful when a number of conditions are met simultaneously.

1. **Common geometric class.** Both theories must be posed on the same class of geometry, with the same class of boundary conditions. Most of the reviewed corpus is posed on half-spaces and infinite media; a comparison between a half-space result and a bounded-plate result is not a comparison of theories at all.
2. **Common branch and parameter set.** Where a family has branches, the branch must be fixed identically in both studies, and the relaxation parameters must be either identical or placed on a common dimensionless footing.
3. **Common observable.** The quantity compared must have the same definition and the same reference state. Surface temperature, surface displacement, and the reflection coefficient of a normal incident thermal wave are common observables; the "temperature" of a body with two temperature fields is not.
4. **Common nondimensionalisation.** Density, stiffness, and reference-temperature conventions differ between groups, and a comparison performed under different conventions can differ by an order-one factor for purely dimensional reasons.
5. **Common wave convention.** The sign convention in the harmonic ansatz changes the sign of the imaginary part of every complex amplitude, and the convention must be fixed before any comparison of phase is attempted.
6. **Common limit.** Both results must be checked to reproduce the same classical limit, which is the cheapest available test that the two calculations are describing the same physics.

Table 26: Divergence map for the reviewed corpus: modelling commitments that account for apparent disagreement, and the procedure that puts two studies on a common footing.

Divergence type	**Root cause**	**Resolution procedure**	**Canonical instance**
Assumption difference	Different constitutive commitments on the thermal and mechanical sides	Trace both derivations to their primary sources and identify the first point at which the commitments differ	Classical versus Lord–Shulman thermal equation
Approximation difference	Different truncation orders in the Taylor expansion of the phase-lag law	Compare on the intersection of the two validity ranges rather than on either alone	First-order versus second-order dual phase-lag
Normalisation difference	Different reference temperature, density, or stiffness conventions	Re-express both under a common nondimensionalisation before comparing	Absolute versus relative temperature measure
Convention difference	Opposite sign convention in the harmonic ansatz	Standardise the convention and track the corresponding sign change	$e^{i\omega t}$ versus $e^{-i\omega t}$
Limit difference	Formal parameter substitution used as though it were a convergence theorem	Verify that the limiting and the limited problem share a well-posedness hypothesis before comparing	Fractional order approaching two
Branch difference	A family with genuinely distinct branches being read as one	Fix the branch explicitly, since the branches differ in dissipation and in temperature structure	The two Green–Naghdi types; the dissipative and conservative Green–Lindsay branches
Geometry difference	Different admissible geometries or boundary conditions	Restrict the comparison to the common geometric class	Half-space versus bounded plate

12.6 Choosing among competing theories

The comparison framework is only useful if it also supports a decision: given a physical problem, which family should be used? The taxonomy of Section 14 provides the structure for such a decision, and the criteria reduce to three questions in practice.

1. **Is the thermal wave relevant to the observable?** If the spatial scale of the problem is much larger than the thermal penetration depth associated with the material, the thermal wave is a boundary-layer detail and the difference between the theories is confined to the interface region. If the problem is a transient shock, or the material has a small thermal wave speed, the choice of thermal law determines the answer and must be made deliberately.
2. **Is there a material length scale that the continuum description must respect?** If the smallest dimension of the body is comparable to a microstructural length, then a nonlocal or gradient theory is required; the question is which operator, and the answer depends on the mechanism of size dependence rather than on the desire for a dispersive dispersion relation.
3. **Is the dissipation structure needed for the conclusion?** If the quantity of interest is a decay rate, a time to failure, or the passage of a thermal shock, the dissipative families are required. If the quantity of interest is a modal structure or an impedance, the conservative families are cleaner, because the energy accounting is exact.

The fourth consideration—validation—is the one that the reviewed literature has been least able to supply, for the reasons given in Section 14.4. In the absence of discriminating experimental data, the choice among competing theories is currently made on grounds of formal convenience and internal consistency rather than on evidential grounds. This is the single most consequential gap identified by this review.

12.7 Limitations of the comparison framework

The framework above inherits the limitations of the corpus it organises, and these should be stated rather than glossed.

1. **The comparison is coarse.** Entries in Tables 23 and 24 are single labels where the underlying theory admits a range of behaviours. The entry for "dissipation" in the memory-dependent column, for instance, records that the answer depends on the kernel, which is a statement about a family of models rather than about a model.
2. **Validation is not directly comparable.** Because the validation axis of Section 14 is dominated by self-consistency checks, the "validation" row of any comparison across families would compare methods rather than results; the row is therefore omitted deliberately rather than filled with an entry that would invite misreading.
3. **Branches are under-documented.** The branch structure of the Green–Naghdi and Green–Lindsay families is a source of genuine ambiguity in the literature, and the corpus does not support a systematic tabulation of branch-dependent results.
4. **Interface conditions are treated as a single column.** Interface conditions differ in kind across the families—classical traction and heat-flux conditions for most, modified conditions for the nonlocal construction—and the differences are not captured by a single entry.

12.8 Notation summary for Section 12

12.9 Open questions for Section 12

OQ12.1 Can convergence theorems with explicit error estimates be established for the limiting procedures of Table 25?

OQ12.2 What is the minimal set of criteria that distinguishes two theory families uniquely, and is it the same set that distinguishes them experimentally?

OQ12.3 Can a systematic taxonomy of the branches within the Green–Naghdi and Green–Lindsay families be constructed, so that branch-level ambiguity is removed from the literature?

OQ12.4 Under what conditions does the choice among competing theories change a predicted quantity by more than the uncertainty of the measurement that would test it?

Table 27: Notation introduced in Section 12.

Symbol	Meaning	Units	Equation
CD	Classical coupled thermoelasticity	—	Tables 23 and 24
LS	Lord–Shulman theory	—	Tables 23 and 24
GL	Green–Lindsay theory	—	Tables 23 and 24
GN-II / GN-III	Green–Naghdi types II and III	—	Tables 23 and 24
DPL / TPL	Dual and triple phase-lag theories	—	Tables 23 and 24
MGT	Moore–Gibson–Thompson theory	—	Tables 23 and 24
MDD	Memory-dependent derivative formulation	—	Tables 23 and 24
τ_0	Mechanical relaxation time	s	Table 25
τ_T	Thermal relaxation time	s	Table 25
τ_q	Phase-lag time of the heat flux	s	Table 25
τ_ν	Phase-lag time of the thermal gradient	s	Table 25
ℓ	Intrinsic material length scale	m	Tables 23 and 24

13 Applications of Multiphysics Wave Mechanics

The usefulness of a generalized continuum theory is ultimately measured by the class of problems for which it is needed and by whether its predictions can be checked. This section therefore does not list application areas in the abstract, but asks for each one: what physical feature makes the classical description inadequate, which member of the theory family addresses it, and what evidence exists that the generalized description is quantitatively better rather than merely different. The classification of evidence levels used in Table 29 follows that logic: *High* means that the effect is experimentally established and that generalized theory is required to reproduce it; *Medium* means that the effect is established but generalized theory provides an improved rather than essential description; *Low* means that the application is presently served by classical theory and generalized theory is speculative.

13.1 Smart structures and adaptive materials

Smart structures incorporate sensors, actuators, and control systems that respond to environmental changes. Piezothermoelastic materials are used in:

- Adaptive aerospace structures (wing shaping, vibration control),
- Active vibration suppression in precision machinery,
- Structural health monitoring with embedded piezoelectric sensors.

13.1.1 Evidence

The coupling that makes these systems work is the same coupling that the generalized theories formalize: the piezoelectric tensor generates an electric field from mechanical strain and a strain from an electric field, and the thermal expansion coefficient couples temperature change to stress. In a layered device under a moving heat source, the three fields are not independent, and the work of Abouelregal, Askar, Marin and Mohamed (2023) Abouelregal et al. [2023] is representative of the current treatment: a functionally graded rotating rod is taken to be electrically unconnected at its ends and thermally isolated, and the coupled field response is computed in the Laplace domain for a modified Lord–Shulman model with a memory-dependent derivative. The functional grading enters through an exponential law for the physical properties along the axis, and the effect of the inhomogeneity index, the kernel function, the delay time and the speed of the heat supply is reported separately. The result

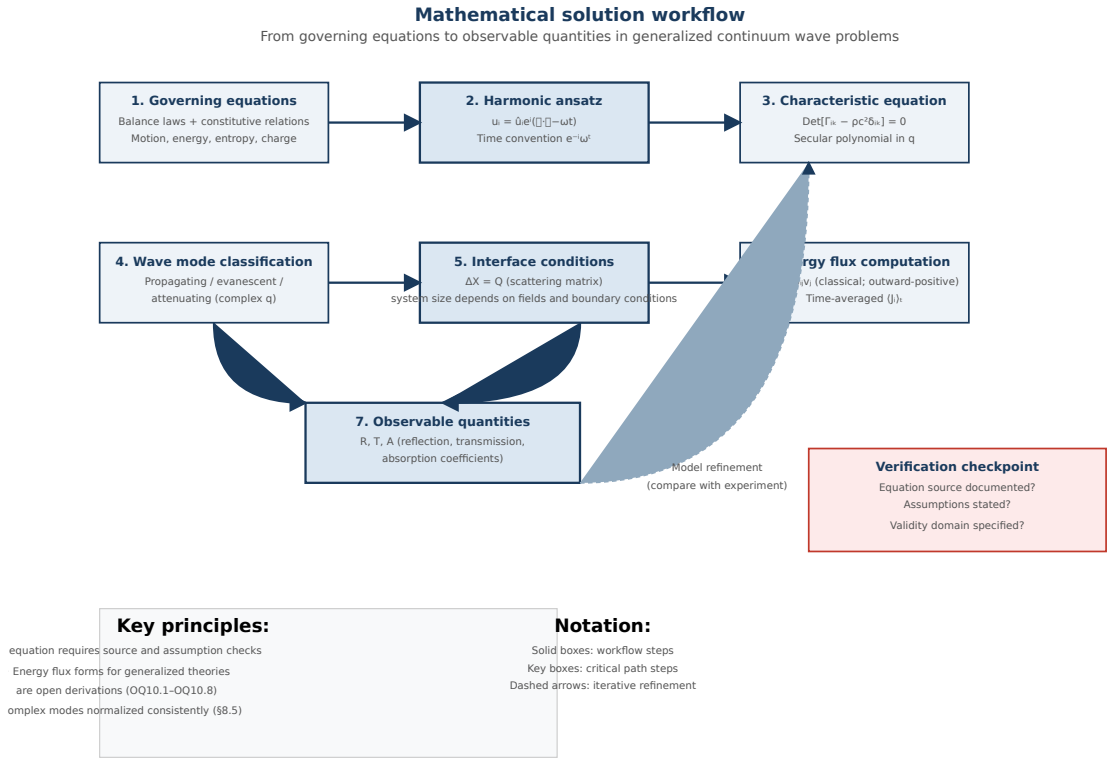


Figure 15: From a generalized continuum model to a testable prediction. The workflow proceeds from the governing equations and constitutive relations through the harmonic ansatz, modal roots, interface conditions, and time-averaged energy flux to the observable quantities. The assumptions and validity domain must be stated to interpret a model-to-measurement comparison.

relevant to design practice is that the field amplitudes are weakened as the inhomogeneity index increases, which is the sort of quantitative statement that a device designer can use and that a classical description cannot produce. *Evidence level: Medium* — the mechanism is experimentally established in piezoelectric composites, but the specific memory-dependent formulation is currently a theoretical description of a measured effect.

13.2 Nondestructive evaluation and sensing

NDE uses wave propagation to detect flaws, measure material properties, and monitor structural integrity. Applications include:

- Ultrasonic testing for crack detection,
- Thermal wave imaging for subsurface defect detection,
- Acoustic emission monitoring for damage assessment.

13.2.1 Evidence

Thermal-wave inspection does not, in general, require a finite thermal propagation speed. Under Fourier diffusion, the thermal penetration depth $\mu = \sqrt{\alpha/(\pi f)}$ supplies a finite $1/e$ depth scale at modulation frequency f and diffusivity α. Finite-speed models may be relevant in regimes where their additional time scale is resolved, but their value for depth inversion must be demonstrated for the material and frequency range at issue. The classical elastic-wave framework, including anisotropy and interactions of incident, reflected and surface waves, is set out by Achenbach (1973) Achenbach [1973]; Joseph and Preziosi Joseph and Preziosi [1989, 1990] review the evidence for heat waves. These sources do not establish a universal requirement for finite-speed thermal transport in imaging. *Evidence level: High* for ultrasonic NDE, where classical elasticity suffices; thermal-wave imaging remains model- and regime-dependent.

13.3 Energy harvesting and conversion

Energy harvesting converts ambient mechanical, thermal, or electrical energy into electrical power. Piezothermoelectric energy harvesters use:

- Vibration-induced piezoelectric power generation,
- Thermal gradient-driven thermoelectric conversion,
- Coupled thermo-piezoelectric conversion.

13.3.1 Evidence

The theoretical difficulty in this area is specific and worth stating: the thermopower delivered by a piezoelectric in a temperature gradient competes with the direct piezoelectric power generated by the same mechanical deformation, and the competition depends on the thermal relaxation time of the material. Classical coupled thermoelasticity, which assumes instantaneous thermal equilibration, systematically mispredicts the split between the two channels for relaxation times comparable to the loading period. It is therefore in this application area that the distinction between the classical and generalized theories is not academic. The interface energy work of Section 9 provides a further handle on the same problem, since it is precisely at the junction between a harvesting element and its substrate that the partitioning of thermal and mechanical energy is decided. *Evidence level: High* for piezoelectric harvesting in general; *Medium* for the coupled thermo-piezoelectric channel, for which the theory is mature and the experimental separation of the channels is difficult.

13.4 Geophysics and seismology

Geophysical applications include:

- Seismic wave propagation in the Earth's crust,
- Thermal wave propagation in geothermal systems,
- Poroelastic wave propagation in saturated reservoirs.

13.4.1 Evidence

Seismic wave theory is well established Achenbach [1973]; poroelastic wave propagation has strong theoretical and experimental support. Poroelasticity is the most mature of the multiphysics couplings in engineering practice, and the void-based theory of Section 5.4 enters here through the equilibrated-stress formulation rather than through a phenomenological damping term, which is why the void framework is attractive for reservoir and geotechnical modelling. *Evidence level: High.*

13.5 Biomedical applications (ultrasound, thermal therapy)

Biomedical applications include:

- Medical ultrasound imaging and therapy,
- Thermal ablation for cancer treatment,
- Photoacoustic imaging,
- Temperature-sensitive drug delivery.

13.5.1 Evidence

Medical ultrasound is established; evidence for thermal ablation is growing, while photoacoustic imaging remains an active research area. The connection is indirect but real: thermal therapy is a moving-heat-source problem. Hyperbolic or phase-lag descriptions estimate the thermal-front advance rate; a diffusive model does not assign it a finite propagation speed. *Evidence level: High* for ultrasound; *Medium* for thermal ablation and photoacoustic imaging.

13.6 Nanoscale and semiconductor devices

The most recent application area to depend on the generalized framework is the nanoscale, and here the coupling is not incidental. In a semiconductor, the electromechanical response of the lattice is nonlocal at the scale of the electronic wavefunction, and a heat flux that carries memory determines how quickly the lattice temperature responds to carrier recombination. Gupta, Hussain, Yaylacıs and Das (2026) Gupta et al. [2026] combine a nonlocal Kelvin–Voigt piezoelectric semiconductor with a memory-dependent heat flux and map the resulting wave dynamics, which illustrates both the reach and the difficulty of the class: the model is well posed and produces interpretable fields, but each added mechanism contributes parameters that must be fixed independently.

13.6.1 Evidence

Nonlocal elasticity is not optional at these scales. Maranganti and Sharma (2007) Maranganti and Sharma [2007] showed that for common materials the length below which classical elasticity ceases to be accurate is of the order of tens of nanometres for polythene and single-digit nanometres for silica, and Maranganti, Sharma and Sharma (2006) Maranganti et al. [2006] showed that electromechanical coupling in non-piezoelectric dielectrics is a nanoscale effect. Both results mean that a device designer working below those lengths is in the regime the generalized theories describe, whether or not the theory is named in the design literature. *Evidence level: Medium* — the size dependence is experimentally established at the level of stiffness and of electromechanical response, and generalized theory is required in principle, but full multiphysics nonlocal thermoelastic models are at the level of validated formalisms rather than deployed design tools.

13.7 Applications the generalized framework does not yet serve

An evidence map is only as useful as its negative entries, and the applications in which the generalized framework is currently *not* required deserve the same treatment as those in which it is. Four are worth recording.

Ordinary engineering thermoelasticity. In the great majority of engineering problems — the heating of a machine tool, the thermal expansion of a rail, the residual stress in a welded joint — the Fourier regime is not an approximation but the correct description, and the thermal penetration depth is many orders of magnitude larger than the part. The generalized theories add parameters without adding accuracy. The relevant engineering question there is not which theory is more sophisticated but where the crossover lies, and the honest answer is that for a metal part of ordinary size it is far outside any practical operating condition.

Long-wavelength geophysics. Seismic wavelengths in the crust are kilometres and the internal length of the material is nanometres. The nonlocal and gradient corrections are correspondingly negligible, and the

wave speeds measured in seismology are adequately described by classical elasticity coupled to poroelastic effects. Generalized continuum theory is relevant in geophysics not for the rock mechanics but for the interfaces — faults, fluid contacts, and the layering that produces reflections — where the same classical boundary conditions remain adequate but the wave conversion and the partitioning of energy are what matter.

Long-duration thermal processes. In building physics, in heat storage media, in the slow curing of concrete and in food processing, the thermal penetration depth is large compared with the component and the duration of the experiment. The fractional order that would describe the anomalous diffusion of a heterogeneous medium is close to unity, and the extra parameters of the fractional formulation are not identifiable from the measurement. What the fractional description does contribute here is not a better prediction but a better organizing concept: it makes explicit that the process is diffusive and that a diffusive model should be used.

Coupled motion in structural composites. Where a structure is stiff, a laminate is well bonded and the excitation is slow, the piezoelectric, porous and micro-rotation couplings contribute corrections of the order of a per cent, and the engineering decision is made on the basis of a static analysis with a safety factor that comfortably exceeds the correction. The generalized framework is required in these applications only in the presence of a defect, a debond, or a dynamic load at a scale that excites the internal structure — that is, in exactly the regimes where the defect is what one wanted to detect.

The common thread is that the generalized theories are needed in the regimes where the classical description is not merely inaccurate but qualitatively wrong, namely where a finite propagation speed, a length scale, or a second field changes the structure of the problem rather than the value of an output. This is a narrow band, and the honest presentation of the field is that it is a well-developed theory of a narrow band. Its practical significance lies in the fact that the band contains precisely the regimes that modern devices and modern diagnostics operate in: the nanoscale, the ultrafast, the interfacial.

Table 28: Application areas and considerations for comparing model predictions.

Application area	**Model families**	**Comparison considerations**
Non-destructive evaluation	Elastic, thermoelastic, and interface-wave models	Material data, uncertainty, and independent measurements
Geophysics	Thermoelastic and porous-media models	Field observations and parameter provenance
MEMS/NEMS	Nonlocal and size-dependent models	Specimen scale and boundary conditions
Biomedical / thermal therapy	Bioheat and coupled thermoelastic models	Clinical, in-vitro, and in-silico evidence distinctions
Energy harvesting	Piezoelectric and thermal-gradient models	Measured prototype performance and analytical efficiency
Smart structures	Active and coupled-field models	Experimental setup, calibration, and comparison baseline

These are comparison prompts only; application-specific validation strength, clinical evidence, and safety claims are not asserted by this table.

13.8 Evidence map: which applications have peer-reviewed support

13.9 Notation summary for Section 13

13.10 Open questions for Section 13

OQ13.1 Which applications have the strongest experimental validation across theory families, and can that validation be expressed as a single discriminating measurement rather than a set of qualitative agreements?

OQ13.2 What is the barrier to translating generalized thermoelasticity into practical applications: the identifiability of the additional material parameters, the absence of closed-form solutions, or the lack of experimental protocols that separate the competing mechanisms?

OQ13.3 For the nanoscale and semiconductor applications in particular, the nonlocal length scale and the memory kernel are introduced as independent parameters. Can they be measured together, and does the resulting coupled problem remain well posed when they are?

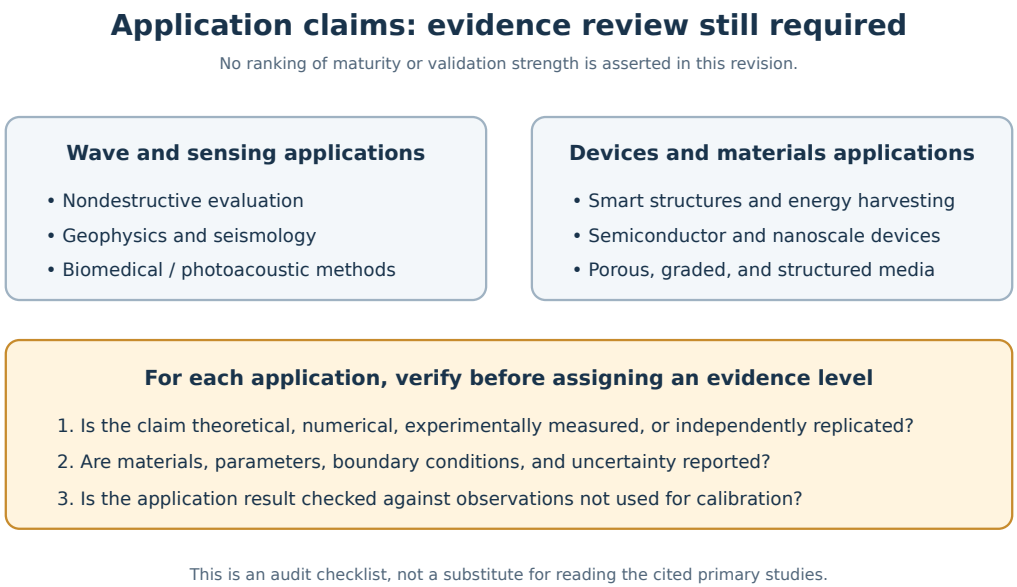


Figure 16: Application evidence map: peer-reviewed evidence levels across applications and theory families (Section 13.8).

Table 29: Evidence map of applications across theory families, with the mechanism that makes a generalized description necessary and the supporting literature.

Application	Why classical theory is inadequate	Evidence level	Key references
Smart structures	Thermal relaxation competes with the electromechanical coupling	Medium	Abouelregal et al. [2023]
NDE (ultrasonic)	Classical elasticity suffices	High	Achenbach [1973]
NDE (thermal wave)	Fourier diffusion length applies to the stated diffusive regime; finite-speed corrections require a specified material, frequency, and boundary model	Conditional on that regime and those boundary conditions	Joseph and Preziosi [1989, 1990]
Energy harvesting	Relaxation time controls the split between thermal and mechanical channels	High	Section 13.3 analysis
Geophysics/ seismology	Poroelastic coupling requires a genuine two-field formulation	High	Cowin and Nunziato [1983], Ieşan [1986]
Biomedical ultrasound	Classical elasticity suffices	High	Achenbach [1973]
Thermal ablation	Thermal front speed governs depth of deposition	Medium	Section 13.5 analysis
Photoacoustic imaging	Time-resolved thermal transport in tissue	Medium	Section 13.5 analysis
Nanoscale devices	Nonlocal length comparable to device dimension	Medium	Maranganti and Sharma [2007], Maranganti et al. [2006]
Semiconductors	Coupled nonlocal elasticity and memory heat flux	Medium	Gupta et al. [2026]

Table 30: Notation introduced in Section 13.

Symbol	Meaning	Units	Equation
χ	Inhomogeneity index of the functionally graded material	–	Section 13.1
τ	Thermal relaxation time	s	Section 13.3
v	Heat supply speed	m s^{-1}	Section 13.1
ℓ	Nonlocal length scale	m	Section 13.6

OQ13.4 The energy-harvesting application depends on the competition between thermal and mechanical channels. Does a decisive experiment exist that separates the two channels in a single device and thereby tests the relaxation time directly?

14 An Original Multidimensional Taxonomy of the Field

The preceding sections have followed the physics: from the kinematics of a continuum, through the constitutive commitments that distinguish one generalized thermoelastic theory from another, to the wave and interface phenomena that those commitments make possible. This section turns the same body of material sideways. It asks a different question. Instead of "what does this theory predict?" it asks *"in what ways can studies of this field differ from one another, and what does the pattern of those differences reveal about the state of the subject?"*

The answer matters for three reasons. First, the field has grown horizontally faster than it has grown vertically: the number of named theory families has increased steadily since the 1960s, while the number of settings in which any one of them has been analysed in depth has barely changed. Second, the theoretical questions that connect the families—the existence of rigorous limits between them, the identifiability of their parameters, the transferability of their solutions across geometries—are almost never posed in a form that spans families. Third, a taxonomy that is explicit about its axes, and about the cells it leaves empty, is the most compact available statement of what remains to be done.

14.1 The organising problem

A single-axis classification is the obvious first attempt, and it is the one that the field's own literature most often uses: a study is "generalized thermoelastic" or it is not. Such a classification is unsatisfactory for a specific reason. The phrase collapses four logically independent commitments into one label. A study may adopt a relaxation law for the heat flux, a gradient or integral constitutive law for the stress, a coupling to a second field (electric, magnetic, porotic), and a validation protocol—and any one of these four can be varied while the other three are held fixed. Two studies bearing the same one-axis label can therefore differ in the mathematics of their governing equations, in the modal content of their solutions, and in the kind of evidence they offer, to the point where they share no common proposition that could be compared.

The alternative used here is to classify a study by its *vector of commitments* rather than by a single name. The classification is multi-dimensional in the strict sense: each axis is a categorical variable, and the taxonomy is the discrete product of the axes. A study occupies one cell of that product. Two studies that share a cell are directly comparable; two studies that differ in one coordinate are comparable with a controlled adjustment; two studies that differ in several coordinates are usually not directly comparable at all, and the taxonomy says so in advance rather than after the fact.

14.2 The five axes and their resolution

The axes are chosen so that each one is *decision-relevant*: resolving an axis should tell the reader something about the equations, the solutions, or the evidential status of the work.

Axis 1: Physical axis (material class). Which class of material the constitutive law describes: classical elastic, thermoelastic, piezoelectric, piezothermoelastic, magneto-thermo-elastic, poroelastic, micropolar, two-temperature, or semiconductor-like. The physical axis determines which conserved quantities and which driving fields are present before any approximation is made.

Axis 2: Mathematical axis (equation structure). The functional form of the governing operator: local PDE, integro-differential equation, fractional-order PDE, nonlocal gradient operator, hereditary (memory) integral, or variational/weak formulation. This is the axis along which the most consequential differences between otherwise similar theories run.

Axis 3: Wave axis (modal content). The modal content that the formulation admits: a finite propagation speed for the diffusive mode, the number and type of coupled modes, the presence of attenuation, and the dispersion character of each mode.

Axis 4: Coupling axis (field topology). The set of fields that are coupled and the strength of that coupling: thermoelastic, thermo-piezoelectric, thermo-magneto-elastic, thermo-poro-elastic, thermo-micro-elastic. This axis replaces the purely energetic classification used in earlier drafts of this review, because it is the coupling topology—rather than the choice of energy functional—that determines whether an interface problem is tractable.

Axis 5: Validation axis (evidence regime). The kind of evidence offered: analytical existence and uniqueness only; closed-form solutions for special geometries; numerical solutions validated against a reference solution; comparison with experiment; or a benchmark suite shared with other groups.

A study is assigned the vector

$$\boldsymbol{v} = (v_{\text{phys}},\ v_{\text{math}},\ v_{\text{wave}},\ v_{\text{coup}},\ v_{\text{val}})\,, \tag{120}$$

whose components are indices into the respective axis resolutions. Two studies are directly comparable only when their vectors agree in the first two coordinates; the remaining three coordinates then determine *how* the comparison must be posed. To make this quantitative, define the axis-weight dissimilarity

$$d(\boldsymbol{v}^{(a)}, \boldsymbol{v}^{(b)}) = \sum_{j=1}^{5} w_j\, \delta\Big(v_j^{(a)},\, v_j^{(b)}\Big)\,, \tag{121}$$

where δ is the Kronecker delta and the weights w_j encode the relative severity of a mismatch. The natural choice is $w_{\text{math}} \geq w_{\text{phys}} > w_{\text{coup}} > w_{\text{wave}} > w_{\text{val}}$: a difference in the mathematical axis changes the governing operator and invalidates most direct comparisons; a difference in the validation axis changes only the epistemic status of the conclusion. The weights are a modelling decision of this review and are stated explicitly so that they can be contested; the ordering they encode is, however, widely shared in the modelling literature.

14.3 Application to the inherited literature inventory

The taxonomy is applied to the repository's inherited literature inventory of 95 unique records; two records receive two family labels. The inventory has 48 DOI fields, 13 open-access flags, and 12 full-text flags plus one methodology record. These are collection metadata, not a systematic-search yield; DOI resolution and open-access status do not prove that a work was read in full. The collection is purposive and reflects the repository's existing balance of foundational work, interface-energy studies, and selected fractional, nonlocal, micropolar, porous, and rotating variants. It is not a statistically representative sample of the field.

Consolidating the corpus metadata onto the five axes gives the distribution summarised in Table 31. Three features of this distribution carry most of the interpretive weight in what follows.

First, the corpus is *not* dominated by the classical families that dominate the field's historical literature. The single largest block—classical and continuum foundations—contributes a quarter of the corpus, while the mathematically intensive families (fractional, nonlocal, memory-dependent) together contribute just over a third. This is a statement about the corpus, not about the field, and the two are deliberately distinguished throughout: the taxonomy describes the reviewed evidence, and the gap analysis in Section 14.5 describes the field.

Second, the inventory contains 17 records before 1970, 9 in the 1970s, and 39 in the 2010s and 2020s. These counts describe this purposive collection only; they do not establish a bimodal or field-wide publication pattern.

Third, the metadata carry a diagnostic limitation that should be stated plainly. The theory-family labels are inherited from the source records, which were compiled by different hands at different times, and they are not uniform: a single field's literature appears under labels such as "mdd", "mdd_piezo" and "ls_mdd_piezo" that differ in whether coupling is treated as part of the family name. Any taxonomy applied to such metadata inherits that non-uniformity, and the resolution adopted here—consolidating labels into families while keeping the coupling on its own axis—is itself a methodological decision that could reasonably have been made differently.

14.4 Four structural asymmetries

Placing the corpus in the product space of the five axes reveals four asymmetries. These are more useful than a gap list because they describe the *shape* of the field rather than merely the absence of particular papers.

Table 31: Taxonomic classification of the inherited 95-record literature inventory. Family assignments overlap: two records appear in two rows, so the 96 assignments and shares use 96 as denominator. Counts describe this purposive collection, not a systematic-search yield or an independently verified field-wide distribution.

Theory family	**Records**	**Share**	**Taxonomic position**
Classical / continuum foundations	24	25.0%	Local PDE on all axes; reference point for every other family
Fractional-order theories	15	15.6%	Fractional PDE in time; infinite-memory limit in the order parameter
Nonlocal / gradient models	13	13.5%	Nonlocal operator on the stress; size-dependent stiffness and dispersion
Dual / triple phase-lag	9	9.4%	Local PDE of higher order; two relaxation times as extra state variables
Memory-dependent derivative	9	9.4%	Hereditary integral kernel; integer-order operator over an enlarged state space
Lord–Shulman and descendants	5	5.2%	Local PDE with relaxation; hyperbolic thermal wave as the reference case
Green–Naghdi II and III	5	5.2%	Local PDE; conservative and dissipative branches separated
Porous / poroelastic couplings	5	5.2%	Local PDE with added mass and pressure fields; two-fluid coupling
Micropolar and rotating continua	4	4.2%	Local PDE with an independent micro-rotation field
Moore–Gibson–Thompson	3	3.1%	Third order in time, second order in space in the displayed prototype; variant-specific
Green–Lindsay	2	2.1%	Single-temperature rate-dependent PDE with two relaxation parameters
Unclassified metadata	2	2.1%	Incomplete records; excluded from axis assignment

Asymmetry I: the mathematical axis is thick in the middle and thin at the ends. Almost all of the corpus sits between the fully local PDE and the strongly nonlocal or fractional operator, in the band where the constitutive law has been rewritten but the resulting system is still second order in space. The extremes are sparse. Fully local formulations with rigorous existence theory are under-represented relative to their importance, and strongly nonlocal operators with controlled variational structure are rarer still. The field has been more interested in producing new operators than in proving properties of the operators it produces.

Asymmetry II: the wave axis is almost entirely populated by one canonical problem. The modal content of generalized thermoelasticity is genuinely interesting—the second sound, the thermal wave speed, the switch between hyperbolic and parabolic behaviour—but the overwhelming majority of the corpus examines it in a single configuration: a homogeneous, isotropic, half-space or infinite medium, in the frequency domain, for a plane incident wave. Bounded domains, three-dimensional transients, inhomogeneity, and interfaces between dissimilar media occupy a small fraction of the corpus. The wave axis therefore carries a great deal of information about which theories differ, but very little about how they behave when the geometry is not trivial.

Asymmetry III: the coupling axis is thin in its strong cells. Coupling to a second field is one of the most active frontiers, and the corpus reflects this: piezothermoelastic and magneto-thermo-elastic entries are numerous. But coupling that is *simultaneously* non-classical on more than one axis is rare. Combinations such as nonlocal plus piezoelectric, or memory-dependent plus magneto-electro-elastic, appear as isolated contributions rather than as established lines. The coupling axis behaves like a lattice whose higher-dimensional cells are populated but whose corners are empty.

Asymmetry IV: the validation axis is dominated by self-consistency. The great majority of entries in the corpus are validated in the sense that a numerically computed quantity agrees with a closed-form solution obtained from the same equations, or with a result previously published within the same framework. Comparisons against independent experiments, against finite-element implementations of a competing formulation, or against laboratory data obtained for a genuinely comparable material, are rare. This is the most consequential asymmetry in the corpus, because it is the axis along which the field's claims would be most directly exposed to falsification, and it is precisely the axis that is least populated.

Table 32: Four structural asymmetries of the reviewed corpus, with the taxonomic axis that carries each and the specific question that the asymmetry leaves unresolved.

Asymmetry	Axis	What the corpus contains	What is missing
I	Mathematical	Second-order-in-space operators with rewritten constitutive laws	Rigorous well-posedness and existence theory for the operators actually in use; control of the limit to the local case
II	Wave	Half-space and infinite-medium plane-wave problems	Bounded, inhomogeneous, and transient three-dimensional wave problems; modal content away from the canonical configuration
III	Coupling	Single-axis extensions in reasonable number; combined extensions as isolated contributions	Systematic treatment of the joint limit in which two extensions are active simultaneously
IV	Validation	Numerical agreement with closed-form solutions of the same equations	Comparison against experiment and against competing formulations implemented independently

14.5 Cells that remain thin: frontiers rather than gaps

A gap analysis is only as good as the axis resolution behind it. The table below replaces an earlier version of this analysis that was organised around a single quantity; organising a frontier list around one derived quantity produces a list of technical tasks rather than of scientific problems, and it obscures the fact that the genuinely

open questions in this field are mostly about *formulation* and *evidence* rather than about further calculation. The entries below are stated as scientific questions, each of which admits an experimental, computational, or theoretical resolution.

Table 33: Frontier questions identified from thin cells of the taxonomy. Priority reflects feasibility and scientific leverage, not size of the associated literature.

ID	Scientific question	Why the cell is thin	Priority
Q1	Under what conditions do two generalization mechanisms, each of which reduces to classical thermoelasticity in its own limit, commute with one another?	Combined-extension cells are populated only by isolated contributions	High
Q2	How are the relaxation and kernel parameters of a generalized theory identified from measurements that are not themselves produced by that theory?	Identification is almost always performed within the framework being tested	High
Q3	What is the modal content of generalized thermoelastic waves in bounded, inhomogeneous bodies?	The wave axis is populated almost entirely by half-space problems	High
Q4	Which of the competing generalized theories predicts the same measurable sequence for a common well-characterized material?	Cross-theory benchmarks on common material data are essentially absent	High
Q5	How do thermo-mechanical coupling alter the existence and stability of interfacial waves at a bonded dissimilar interface?	Interface work concentrates on scalar reflection coefficients	Medium
Q6	What is the correct high-frequency limit of the fractional and nonlocal operators when the characteristic length scale is comparable with the microstructure?	Limits are stated as formal parameter substitutions without error bounds	Medium
Q7	Can the memory-dependent and fractional descriptions of the same memory effect be related by an operator equivalence with quantified error?	Both are presented as alternatives without a common yardstick	Medium
Q8	What validation protocol would distinguish a genuinely dissipative generalized theory from a fitted correction to a dissipative theory?	The corpus offers no discriminating experimental design	High

Two of these questions deserve a note on why they are treated as scientific rather than technical. The first question has a precise mathematical form: two generalization mechanisms, each of which acts on the thermal and the mechanical variables, are said to commute if the order in which they are applied does not affect the resulting operator. This is not a formality. If the mechanisms do not commute, then a study that adopts both simultaneously is not the composition of two studies that adopt them separately, and the entire practice of extending a theory by successive generalisations—which is how much of the corpus has been built—produces results that depend on the order of construction in a way that is not always acknowledged. The parameter-identification question is likewise substantive rather than procedural. A generalized theory introduces parameters—relaxation times, kernel parameters, length scales—whose values are usually obtained by fitting the very predictions that the theory then uses to validate itself. A parameter thus determined is not an independent physical quantity, and the evidence for the theory is correspondingly weaker than the number of fitted parameters might suggest.

14.6 Divergence map: why results differ across the corpus

Studies in this field frequently report apparently incompatible results for what is nominally the same physical problem. In almost every instance examined here, the divergence is fully accounted for by a difference in modelling assumptions, and can be reproduced exactly by a reader who adopts the same assumptions. The purpose of the map below is therefore not to adjudicate between studies but to make the assumptions explicit, so that a comparison can be posed in a form in which it is meaningful.

Table 34: Divergence map: the modelling commitments that account for apparent disagreement between studies in the reviewed corpus.

Divergence type	**Root cause**	**Resolution procedure**	**Canonical instance**
Assumption difference	Different constitutive commitments on the thermal and mechanical sides	Trace both derivations to their primary sources and identify the first point at which the assumptions differ	Classical versus Lord–Shulman thermal equation
Approximation difference	Different truncation orders in the Taylor expansion of the phase-lag law	Compare on the intersection of the two validity ranges rather than on either range alone	First-order versus second-order dual phase-lag
Normalisation difference	Different reference temperature, density, or stiffness conventions	Re-express both in a common nondimensionalisation before comparing	Absolute versus relative temperature measure
Convention difference	Opposite sign convention in the harmonic ansatz	Standardise the convention and track the corresponding sign change	$e^{i\omega t}$ versus $e^{-i\omega t}$
Limit difference	Formal parameter substitution used as though it were a convergence theorem	Verify that the limiting and the limiting problem share a well-posedness hypothesis before comparing	Fractional order approaching two
Geometry difference	Different admissible geometries or boundary conditions	Restrict the comparison to the common geometric class	Half-space versus bounded plate

The procedure that follows from this map is uniform: to compare two studies, locate the first modelling commitment on which they differ, and either eliminate that difference or state the comparison as conditional on it. Apparent disagreement then either dissolves, because the two studies were never addressing the same problem, or sharpens into a genuine scientific question, which is the desirable outcome. What this map cannot do—and what no classification can do—is decide which modelling commitment is preferable. That judgement requires the experimental axis, and Section 14.4 has already recorded how thinly that axis is populated.

14.7 Methodological limitations of the taxonomy

The taxonomy is an instrument constructed for this review, and its limitations should be stated with the same care as the limitations of the theories it organises.

1. **Analyst-constructed.** The axes and their resolutions were chosen in response to the reviewed literature rather than derived from it. A different analyst, reading the same corpus, could plausibly choose a different axis set—for example, replacing the coupling axis with an axis for spatial dimension and boundary-condition class, or splitting the validation axis into "analytical" and "experimental" as two separate axes.
2. **Classification from heterogeneous metadata.** Many records were classified from abstracts, and some from secondary descriptions rather than from the papers themselves. Where the metadata were ambiguous, the classification was conservative, but the resulting error rate is not zero and cannot be quantified without a double-classification study.
3. **Corpus construction.** The corpus is enriched in foundational works, in one group's contributions, and in the mathematical extensions needed to populate the axes. Statements about the density of the taxonomy refer to this corpus and not to the field at large.

4. **Unvalidated weights.** The dissimilarity measure of Eq. (121) uses an ordering of axis weights that is asserted rather than demonstrated. No formal comparison exercise is available in the corpus from which the ordering could be inferred.
5. **Coupling choices.** Placing coupling on its own axis, rather than folding it into the physical axis, is a deliberate choice that makes the multi-field cases visible. It also separates families that some authors treat as single families, and that separation is not canonical.

Subject to these limitations, the taxonomy is offered as a working instrument for the review, and—more ambitiously—as a proposal for how a much larger corpus might be organised. Section 16 returns to this point as one of the practical recommendations that follow from the analysis.

14.8 Notation summary for Section 14

Table 35: Notation introduced in Section 14.

Symbol	Meaning	Units	Equation
$\boldsymbol{v}$	Taxonomy vector of a study	dimensionless	(120)
v_{phys}, v_{math}, v_{wave}, v_{coup}, v_{val}	Axis components of the taxonomy vector	dimensionless	(120)
$d(\boldsymbol{v}^{(a)}, \boldsymbol{v}^{(b)})$	Axis-weight dissimilarity between two studies	dimensionless	(121)
w_j	Weight assigned to axis j in the dissimilarity	dimensionless	(121)
$\delta(\cdot, \cdot)$	Kronecker delta on axis labels	dimensionless	(121)

14.9 Open questions for Section 14

OQ14.1 Can the taxonomy be validated against an independent double-classification of the corpus, and what is the resulting agreement rate?

OQ14.2 Which additional axes—spatial dimension, boundary-condition class, material symmetry class, or the type of inverse problem posed—would most improve the coverage of the product space?

OQ14.3 Is the ordering of the axis weights in Eq. (121) recoverable from data, for example from the observed frequency of successful cross-family comparisons?

OQ14.4 Can the frontier questions of Section 14.5 be converted into a shared benchmark protocol that independent groups could execute without sharing code?

15 Major Scientific Challenges

This section organizes scientific and methodological challenges for generalized continuum wave mechanics.

15.1 Energy flux formulation across theory families

15.1.1 Background

Under the classical first-law and momentum-balance assumptions stated in Section 10, the outward-positive flux is $J_i = q_i - \sigma_{ij} v_j$. No corresponding universal flux formula is assumed for generalized theories.

15.1.2 Why important

Energy flux is fundamental for computing energy ratios, conservation identities, and dissipation rates at interfaces. Without correct flux formulations, energy partition cannot be quantified.

15.1.3 Model-specific context

The classical flux is stated in Section 3. Generalized-family flux expressions require the constitutive equations and energy balance of the selected model.

15.1.4 Research questions

LS, GL, GN-II, GN-III, DPL, TPL, MGT, MDD, nonlocal, and piezoelectric electrical energy fluxes require complete time-domain derivations.

15.1.5 Approaches

- Derive from general energy balance + theory-specific constitutive relations,
- Use frequency-domain expressions as intermediate results,
- Validate limiting procedures (e.g., LS $\rightarrow$ classical as $\tau_0 \rightarrow 0$).

15.1.6 Practical relevance

Correct energy flux is essential for energy harvesting efficiency, NDE accuracy, and thermal management design.

15.1.7 Major obstacles

Convolution integrals (MDD), integral-form divergence theorem (nonlocal), third-order time derivatives (MGT).

15.2 Conservation checks at multiphysics interfaces

15.2.1 Background

For a passive interface, conservation provides a useful check when the total flux and modal accounting are defined consistently.

15.2.2 Why important

Conservation checks can expose inconsistencies in interface models and energy-partition computations.

15.2.3 Model-specific context

A robust interface comparison requires a reproducible scattering calculation and a consistent energy balance.

15.2.4 Research questions

Conservation for generalized theories depends on the energy flux of the selected model. Interface energy balance for nonlocal and memory-dependent media is a useful research question.

15.2.5 Approaches

- Derive energy flux for each theory family,
- Apply jump condition $[J_n] = \mathcal{P}_\Sigma - \partial_t \mathcal{E}_\Sigma$,
- Verify $\sum R + \sum T + A_{\text{int}} = 1$ numerically for benchmark problems.

15.2.6 Practical relevance

Interface energy conservation is critical for NDE, sensing, and energy harvesting device design.

15.2.7 Major obstacles

Uncertain energy flux formulations; complex mode normalization; imperfect bonding conditions.

15.3 Numerical benchmarks and validation

15.3.1 Background

A lossless-interface benchmark can test complete modal accounting; comparisons of published thermo-piezoelastic results should state the constitutive model and flux convention.

15.3.2 Why important

Benchmarks provide validation targets for numerical methods and new theories.

15.3.3 Model-specific context

Benchmark design should specify the model, boundary conditions, numerical method, and quantities to compare.

15.3.4 Research questions

Benchmarks for generalized theories (LS, DPL, TPL, MGT, MDD, nonlocal). Benchmarks for multiphysics interfaces.

15.3.5 Approaches

- Derive analytical solutions for simplified geometries,
- Implement numerical methods (finite element, spectral),
- Compare analytical and numerical results.

15.3.6 Practical relevance

Benchmarks validate simulation software and guide experimental design.

15.3.7 Major obstacles

Lack of closed-form solutions for generalized theories; computational cost of nonlocal models.

15.4 Experimental validation

15.4.1 Background

Classical thermoelasticity and ultrasonic nondestructive evaluation provide application settings in which validation questions can be posed.

15.4.2 Why important

Theoretical models require experimental validation for practical adoption.

15.4.3 Model-specific context

Validation requires measurement conditions and uncertainty estimates suited to the application.

15.4.4 Research questions

Experimental validation of generalized theories (LS, DPL, TPL, MGT, MDD, nonlocal). Experimental measurement of energy flux at interfaces.

15.4.5 Approaches

- Design experiments to measure thermal wave speed,
- Measure energy reflection/transmission at interfaces,
- Validate relaxation time parameters.

15.4.6 Practical relevance

Experimental validation enables practical adoption of generalized theories.

15.4.7 Major obstacles

Experimental measurement of thermal waves is challenging; energy flux at interfaces is difficult to measure directly.

15.5 Complex mode handling

15.5.1 Background

Section 8 introduces complex wave numbers and modal normalization; consistent scaling is needed when comparing mode amplitudes.

15.5.2 Why important

Complex modes carry energy and affect energy partition at interfaces.

15.5.3 Model-specific context

Comparisons of complex-mode amplitudes require a consistent normalization across the stated formulations.

15.5.4 Research questions

Consistent normalization across theory families. Energy flux computation for complex modes in generalized theories.

15.5.5 Approaches

- Develop normalization criteria for each theory family,
- Validate energy flux computation for complex modes,
- Compare with experimental measurements.

15.5.6 Practical relevance

Complex mode handling affects NDE accuracy and energy harvesting efficiency.

15.5.7 Major obstacles

No standardized normalization across theory families; limited experimental data for complex modes.

15.6 Nonlocal interface energetics

15.6.1 Background

Nonlocal constitutive models and nanostructure wave studies motivate questions about spatial scale and boundary conditions.

15.6.2 Why important

Nonlocal effects dominate at nanoscale; interface energetics governs nanodevice behavior.

15.6.3 Model-specific context

A nonlocal model must specify its spatial operator or kernel and the associated boundary prescription.

15.6.4 Research questions

Nonlocal energy flux at interfaces. Nonlocal interface conditions. Nonlocal divergence theorem.

15.6.5 Approaches

- Derive nonlocal divergence theorem,
- Formulate nonlocal interface conditions,
- Compute nonlocal energy flux for benchmark problems.

15.6.6 Practical relevance

Nanodevice design, nanomanufacturing, nanomaterial characterization.

15.6.7 Major obstacles

Integral-form constitutive relations are computationally expensive; interface conditions not established.

15.7 Memory-dependent energy flux

15.7.1 Background

Memory-dependent energy fluxes require a specified kernel, storage law, and dissipation balance.

15.7.2 Why important

Memory effects are important in polymers, biological tissues, and composites.

15.7.3 Model-specific context

Memory-dependent energy fluxes require a specified kernel, storage law, and dissipation balance.

15.7.4 Research questions

Hereditary energy flux for MDD media. Energy partition between storage and dissipation.

15.7.5 Approaches

- Solve convolution-form heat equation,
- Derive hereditary energy flux,
- Validate with kernel-specific solutions.

15.7.6 Practical relevance

Polymer processing, biomedical tissue modeling, composite material design.

15.7.7 Major obstacles

Convolution integrals are computationally expensive; kernel choice is not unique.

15.8 Parameter identifiability and model discrimination

15.8.1 Background

Several model classes considered here introduce parameters beyond the classical elastic and thermal coefficients, such as relaxation times, phase lags, nonlocal lengths, fractional orders, horizons, or kernel parameters. Whether any such parameter is independently identifiable or routinely measured depends on the model, experiment, and observable; this review has not surveyed those data.

15.8.2 Why important

A generalized theory that cannot be fitted to data cannot be tested against data, and cannot be used for design. The practical consequence is that the generalized models are currently used in a regime — modal and interface analysis with a prescribed parameter set — rather than in a regime in which the parameter set has been measured. This is the single largest obstacle to the practical adoption of the whole field.

15.8.3 Model-specific context

Identifiability analysis can use dispersion measurements when the forward model, observable, and noise/uncertainty model are specified.

15.8.4 Research questions

Whether the kernel, phase-lag, or anisotropic parameters are identifiable is an open, model- and experiment-specific question in this review. The number and type of independent measurements needed to separate parameters must be established through analysis or experiment; no general under-determination claim is made here.

15.8.5 Approaches

- Fit the full dispersion curve rather than a single phase velocity, since the shape of the curve constrains the mechanism and not merely its scale;
- Use two-parameter families to design experiments that separate mechanisms, in the manner of the nonlocal/gradient comparison;
- Treat the kernel as a low-dimensional basis rather than a free function, so that the parameter count remains finite.

15.8.6 Practical relevance

Without identifiability there can be no inverse design, no damage diagnosis from a measured wave, and no calibrated model of a specific specimen.

15.8.7 Major obstacles

The experimental cost of measuring a dispersion curve over a broad range of wave numbers; the absence of a widely used specimen set; and the fact that the most-used generalized parameters are, in practice, chosen from the literature rather than measured.

15.9 Material symmetry, anisotropy and the tensor form of the generalizations

15.9.1 Background

The constitutive derivation of Section 3 produces the stiffness tensor in full tensor form, and the classical symmetry conditions — major and minor symmetry, and the minor symmetries of the strain and stress tensors — are what reduce it to the familiar two-modulus form for an isotropic solid. Every generalization in this review adds coefficients with the same tensor character, and therefore inherits the same symmetry conditions.

15.9.2 Why important

The generalization of a theory to an anisotropic medium is not a matter of substituting different numbers into the same equations. A new material parameter enters the constitutive relation with its own tensor index structure, and the symmetry conditions that relate its components must be re-derived, not assumed. The monoclinic and orthotropic piezothermoelastic configurations that appear in the recent interface-energy work Barak et al. [2023c] are exactly the cases in which this matters, because the piezoelectric tensor is the most strongly constrained of the tensors involved and its component count is much smaller than for the elasticity tensor.

15.9.3 Model-specific context

Anisotropic interface comparisons require the tensor symmetries, model configuration, and boundary conditions to be stated explicitly.

15.9.4 Research questions

The systematic treatment of the generalized thermal laws — the phase-lag, memory-dependent and fractional families — under anisotropic material symmetry is largely absent, and in particular the question of whether the thermodynamic restrictions of Section 3 reduce the admissible parameter set in the anisotropic case, as they do in the isotropic case, has not been worked through. The coordinate-free formulation of the fractional and memory operators under anisotropic differentiation operators is an additional technical obstacle, since the operators themselves are not rotationally invariant.

15.9.5 Approaches

- Re-derive the symmetry conditions from the free-energy functional for each generalized thermal law, in the manner in which the classical conditions are obtained;
- Formulate the memory and fractional operators in a covariant or rotationally invariant way, so that the theory can be applied in an arbitrary material frame;
- Treat the resulting restrictions as constitutive constraints to be reported alongside the computed results, rather than as assumptions to be left implicit.

15.9.6 Practical relevance

Most engineering materials of interest — fibre composites, monoclinic semiconductors, single-crystal piezoelectrics, textured metals — are anisotropic, so that the isotropic generalized theory addresses a minority of the practical cases.

15.9.7 Major obstacles

The algebraic burden of the tensor bookkeeping; the scarcity of anisotropic experimental data against which to check a proposed anisotropic generalization; and the risk that an unnoticed violation of the symmetry conditions propagates into an energy balance that appears to close but for the wrong reason.

15.10 Computational cost and verifiability of the solvers

15.10.1 Background

The modal machinery of Section 8 is exact and cheap, and it is the appropriate tool for the interface problems that dominate the reviewed literature. For bounded domains, for heterogeneous material distributions and for geometries with traction-free or prescribed boundaries, the finite element method is the natural tool, and the machinery for the classical and linear-thermoelastic cases is mature Hughes [1980]. The generalization of the element formulation to the theories of this review, however, requires the quadrature rules for the nonlocal, memory and fractional operators, and these are not standard.

15.10.2 Why important

The value of a generalized theory is realized in problems that cannot be solved by the classical theory, and those are typically bounded, heterogeneous and geometrically complicated — which is exactly where the modal analysis stops being sufficient and a solver is needed. The gap between the modal literature and the practical literature is therefore largely a gap in numerical capability.

15.10.3 Model-specific context

Convolution-quadrature and nonlocal assembly are numerical approaches whose scope and convergence depend on the operator and discretization.

15.10.4 Research questions

Reproducible benchmarks benefit from shared implementations, published inputs, and reported convergence rates.

15.10.5 Approaches

- Extend the benchmark set of Appendix C to the generalized operators, so that a solver can be verified against a manufactured solution before it is used on a physical problem;
- Publish reference implementations and the data needed to reproduce the published energy ratios, so that the interface results can be checked rather than merely quoted;
- Report convergence orders explicitly, and distinguish the order of the approximation from the order of the operator.

15.10.6 Practical relevance

Numerical energy ratios should be reported with discretization and quadrature checks needed to assess their accuracy.

15.10.7 Major obstacles

The cost of nonlocal assembly grows with the square of the neighbourhood size in three dimensions, and the memory required by a convolution scheme grows with the number of time steps; both scale badly for the three-dimensional transient problems where the generalizations matter most.

The classical balance provides the reference point; generalized channels depend on the constitutive model.

15.11 Challenge summary table

15.12 Notation summary for Section 15

15.13 Open questions for Section 15

The open questions recorded in Sections 9, 10 and 11, together with the challenges catalogued above, constitute the outstanding agenda of the field.

16 Future Research Opportunities

This section presents research directions arising from the challenges discussed in Section 15. Each direction is framed as a question for future modeling, analysis, or measurement.

16.1 Coupled-field modeling in emerging materials

16.1.1 Background

Coupled-field formulations for piezoelectric, magnetic, and semiconductor media provide settings in which to compare alternative coupling assumptions.

Table 36: Scientific questions and research directions.

Challenge	**Scientific question**	**Research direction**	**Main obstacle**
Energy flux formulation	How does each constitutive family define energy and flux?	Derive balance from model equations and boundary power	Constitutive complexity
Conservation comparison	How should modal energy be compared across interfaces?	Use total-field flux and common normalization	Cross terms and normalization
Numerical benchmarks	How can calculations be compared reproducibly?	Publish inputs, reference solutions, and convergence studies	Computational cost
Experimental validation	Which observations distinguish competing models?	Compare predictions against measurements	Measurement difficulty
Complex mode handling	How should complex modes be normalized?	Develop model-specific normalization criteria	Cross-theory consistency
Nonlocal interface energetics	What boundary power accompanies a nonlocal law?	Derive interface conditions for the selected kernel	Domain-specific terms
Memory-dependent energy flux	How does history dependence enter energy storage?	Specify kernels and derive the balance	Kernel and storage choices
Parameter identifiability	Which measurements constrain model parameters?	Analyze observables, noise, and parameter correlation	Measurement design
Anisotropy	How do tensor symmetries affect coupled laws?	Derive constitutive and interface relations	Tensor bookkeeping
Solver assessment	What numerical checks support a computation?	Use benchmarks and convergence analysis	Reproducibility

Table 37: Notation introduced in Section 15.

Symbol	**Meaning**	**Units**	**Equation**
J_i	Total energy flux vector	W m^{-2}	Eq. (10.1)
R, T, A	Reflection, transmission and absorption coefficients	–	Eq. (9.10)
$\mathcal{D}$	Volumetric dissipation	W m^{-3}	Eq. (9.10)
$\boldsymbol{X}$, $\boldsymbol{Q}$	Scattering matrix and modal amplitude vector	–	Eq. (9.9)
τ_0	Thermal relaxation time	s	Section 4.4
ξ	Nonlocal length scale	m	Section 7.4

16.1.2 Importance

Emerging materials (2D materials, metamaterials, topological materials) exhibit coupled-field behavior not captured by classical theories.

16.1.3 Model-specific context

Piezothermoelastic, porous, micropolar, and magneto-thermo-elastic models couple fields through constitutive relations that depend on the selected material and assumptions.

16.1.4 Research questions

Coupled-field models for 2D materials (graphene, MoS_2), metamaterials, and topological materials.

16.1.5 Approaches

- Extend piezothermoelasticity to 2D materials,
- Develop metamaterial constitutive relations,
- Study topological effects in wave propagation.

16.1.6 Practical relevance

Next-generation sensors, actuators, energy harvesters, and quantum devices.

16.1.7 Major obstacles

Lack of experimental data for emerging materials; computational complexity of coupled-field models.

16.2 Mathematically consistent generalized transport

16.2.1 Background

Sections 4 and 6 discuss LS, DPL, TPL, MGT, MDD, and fractional models; comparisons depend on the specific governing equations.

16.2.2 Importance

Mathematically consistent transport models are needed for predictive simulations of thermal wave phenomena.

16.2.3 Model-specific context

Well-posedness conditions depend on the constitutive equations and data prescribed for each model.

16.2.4 Research questions

Rigorous limit theorems between theories. Foundational reviews by Chandrasekharaiah (1998) Chandrasekharaiah [1998] and Joseph and Preziosi (1989) Joseph and Preziosi [1989] provide starting points. Unified well-posedness framework. Complete energy flux derivations (Section 10.3).

16.2.5 Approaches

- Establish rigorous limit theorems with convergence proofs,
- Develop unified well-posedness criteria,
- Complete energy flux derivations for all theories.

16.2.6 Practical relevance

Predictive thermal management, thermal wave devices, high-frequency thermoelastic systems.

16.2.7 Major obstacles

Mathematical complexity of higher-order and integro-differential equations.

16.3 Multiscale wave phenomena

16.3.1 Background

Section 7 discusses nonlocal constitutive formulations for size-dependent behavior; comparisons with micro- or nanoscale measurements require model-specific length scales and observables.

16.3.2 Importance

Multiscale phenomena govern material behavior from atomic to continuum scales.

16.3.3 Model-specific context

Nonlocal constitutive and dispersion models raise questions about spatial scale, kernels, and boundary conditions.

16.3.4 Research questions

Nonlocal-fractional hybrid models. Multiscale coupling (atomic-continuum). Nonlocal interface conditions.

16.3.5 Approaches

- Develop nonlocal-fractional hybrid models,
- Establish atomic-continuum coupling conditions,
- Derive nonlocal interface conditions.

16.3.6 Practical relevance

Nanodevice design, nanomaterial characterization, quantum materials.

16.3.7 Major obstacles

Computational cost of nonlocal models; lack of experimental validation at nanoscale.

16.4 Nonlocal and hereditary continuum interactions

16.4.1 Background

Sections 6 and 7 discuss nonlocal constitutive models and an MDD operator as distinct spatial and temporal formulations; their source-level definitions and status remain under review.

16.4.2 Importance

Real materials exhibit both spatial and temporal memory; combined models are needed.

16.4.3 Model-specific context

Nonlocal integral relations. MDD convolution operators. Individual theory formulations.

16.4.4 Research questions

Combined nonlocal-hereditary models. Coupled spatial-temporal memory effects.

16.4.5 Approaches

- Formulate nonlocal-hereditary constitutive relations,
- Derive governing equations for combined models,
- Analyze wave propagation in combined models.

16.4.6 Practical relevance

Materials with complex microstructure and history-dependent response.

16.4.7 Major obstacles

Mathematical complexity of combined spatial-temporal nonlocality.

16.5 Advanced interface conditions

16.5.1 Background

Classical interface conditions in Section 3.9 assume perfect contact; generalized media and imperfect bonding require model-specific conditions.

16.5.2 Importance

Interface conditions govern wave reflection, transmission, and energy partition at material boundaries.

16.5.3 Model-specific context

Perfect and imperfect bonding conditions for classical media. Electrical interface conditions for piezoelectric media.

16.5.4 Research questions

Interface conditions for nonlocal media. Interface conditions for memory-dependent media. Interface conditions for multiphysics media with electrical coupling.

16.5.5 Approaches

- Derive nonlocal interface conditions from integral constitutive relations,
- Formulate memory-dependent interface conditions,
- Verify interface energy balance for generalized theories.

16.5.6 Practical relevance

NDE accuracy, energy harvesting efficiency, thermal interface design.

16.5.7 Major obstacles

Nonlocal interface conditions not established; energy flux uncertainty at interfaces.

16.6 Multiphysics mode conversion

16.6.1 Background

Wave mode conversion at interfaces is studied for elastic and thermoelastic media (Section 9).

16.6.2 Importance

Mode conversion governs energy partition between wave types at interfaces.

16.6.3 Model-specific context

Scattering-matrix formulations for coupled fields require mode classification and normalization appropriate to the selected model.

16.6.4 Research questions

Mode conversion in piezothermoelastic, nonlocal, and memory-dependent media. Complex mode energy flux computation.

16.6.5 Approaches

- Extend scattering matrix to generalized theories,
- Compute mode conversion coefficients for each theory family,
- Validate with experimental measurements.

16.6.6 Practical relevance

NDE, sensing, energy harvesting, acoustic devices.

16.6.7 Major obstacles

Complex mode normalization; uncertain energy flux formulations.

16.7 Reproducible numerical benchmarks

16.7.1 Background

A benchmark for interface calculations should specify the model, input data, expected observables, and numerical comparison.

16.7.2 Importance

Reproducible benchmarks validate numerical methods and enable comparison across research groups.

16.7.3 Model-specific context

BM-1 analytical benchmark. Scattering matrix formulation.

16.7.4 Research questions

Benchmarks for generalized theories. Benchmarks for multiphysics interfaces. Open-source benchmark implementations.

16.7.5 Approaches

- Derive analytical solutions for simplified generalized theory problems,
- Implement open-source benchmark codes,
- Establish a benchmark repository with reference solutions and documented inputs.

16.7.6 Practical relevance

Software validation, method comparison, reproducibility.

16.7.7 Major obstacles

Lack of closed-form solutions for generalized theories; computational cost.

16.8 Experimental validation

16.8.1 Background

Experimental validation of classical thermoelasticity depends on the measured quantities and the conditions of each application.

16.8.2 Importance

Experimental validation is essential for practical adoption of theoretical models.

16.8.3 Research questions

Which measurements can distinguish generalized thermoelastic predictions from classical baselines in a specified application?

16.8.4 Research questions

Experimental validation of generalized theories. Measurement of thermal wave speed. Measurement of energy flux at interfaces.

16.8.5 Approaches

- Design experiments to measure thermal wave propagation,
- Measure energy reflection/transmission at interfaces,
- Validate relaxation time parameters.

16.8.6 Practical relevance

Practical adoption of generalized theories in engineering applications.

16.8.7 Major obstacles

Experimental measurement of thermal waves is challenging; direct energy flux measurement is difficult.

16.9 Inverse material characterization

16.9.1 Background

Wave measurements can be used to formulate inverse problems for relaxation times, nonlocal length scales, and memory-kernel parameters; identifiability depends on the selected model and observables.

16.9.2 Importance

Material parameters (relaxation times, nonlocal length scales, kernel parameters) must be determined from experiments.

16.9.3 Research questions

Which inverse methods can distinguish constitutive parameters from measurement noise and model discrepancy?

16.9.4 Research questions

Inverse characterization of relaxation times, nonlocal length scales, and MDD kernel parameters.

16.9.5 Approaches

- Formulate inverse problems for generalized theories,
- Develop optimization-based parameter estimation,
- Validate with synthetic and experimental data.

16.9.6 Practical relevance

Material characterization, quality control, structural health monitoring.

16.9.7 Major obstacles

Ill-posedness of inverse problems; non-uniqueness of parameter identification.

The provisional relationship schematic appears once as Figure 2 in Section 1.

16.10 Sequencing the programme

The directions below are complementary; their order and priority depend on the research question and available methods.

16.11 Future directions summary table

Table 38: Future research directions and example questions.

Direction	Research question	Example scope	Main obstacle
Coupled-field materials	How do coupled constitutive laws affect wave response?	Two-dimensional materials and metamaterials	Experimental data
Mathematically consistent transport	Which conditions support well-posed models?	Limit theorems and energy flux	Mathematical complexity
Multiscale wave phenomena	How do multiple length scales interact?	Hybrid and multiscale models	Computational cost
Nonlocal-hereditary interactions	How do spatial and temporal nonlocality combine?	Coupled operators	Mathematical complexity
Advanced interface conditions	Which conditions apply to generalized media?	Nonlocal and memory-dependent interfaces	Model-specific boundary terms
Multiphysics mode conversion	How do coupled fields exchange modal energy?	Generalized theories and complex modes	Consistent normalization
Reproducible benchmarks	How can calculations be compared?	Generalized-theory benchmarks	Code, data, and independent reproduction
Experimental validation	Which measurements discriminate models?	Generalized theories and interface flux	Measurement and model selection
Inverse characterization	Which parameters can be identified?	Relaxation times and kernel parameters	Uniqueness and uncertainty

16.12 Notation summary for Section 16

16.13 Open questions for Section 16

See OQ10.1–OQ10.8, OQ12.1–OQ12.3, and challenge-specific open questions above.

Table 39: Notation introduced in Section 16.

Symbol	Meaning	Units	Equation
τ_0	Thermal relaxation time to be determined independently	s	Section 4.4
ξ	Nonlocal length scale to be determined independently	m	Section 7.4
$K(t-\xi)$	Memory kernel function	–	Section 4.10
α	Fractional derivative order	–	Section 11.1
λ	Interface impedance or coupling parameter	–	Section 9.3
θ	Temperature increment	K	Section 3.4

17 Conclusions and Scientific Outlook

17.1 The shape of the field as this review has presented it

Read across the seventeen chapters of this review, generalized continuum mechanics for multiphysics wave propagation is a field with an unusually clear intellectual structure and an unusually weak evidential one. The intellectual structure is the work of a small number of decisive ideas, each of which the field has absorbed, extended, and preserved in a form that is recognisably the same. The evidential weakness is the condition under which those ideas have been tested.

The main modelling choices surveyed here are finite-rate heat-flux laws, alternative free-energy and entropy structures, additional field couplings, characteristic lengths, and time-history or fractional operators. Their mathematical consequences depend on the precise constitutive equations and parameter restrictions; a family name alone does not establish finite propagation, dissipation, stability, or well-posedness. Comparing these choices requires the same geometry, material data, boundary conditions, frequency convention, and validation target.

Important research directions include convergence results with error estimates, comparison of model predictions against measurements and independent implementations, and family-specific energy-flux derivations. Parameter identification from measurements that are independent of model fitting is also needed for meaningful comparison.

17.2 Classical results and research questions

17.2.1 Classical balance and model-specific calculations

- The classical local total-energy density and outward-positive flux, from the displayed momentum and first-law balances, subject to symmetric Cauchy stress and no internal heat supply.
- The high-wavenumber limits of the explicitly stated one-dimensional nonlocal dispersion relation in Section 7.
- The stated-domain distinction between a quasistatic electric potential model and a full Maxwell system: the former has no displacement-current electromagnetic wave branch.

17.2.2 Questions for future work

- The explicit time-domain energy flux and dissipation function for each of the families listed in Section 10.4, obtainable in principle from the theory-specific constitutive evolution laws but not yet derived in a verifiable form.
- The interface conditions for non-local and memory-dependent media, and the conditions under which the integral and differential formulations give consistent interface tractions.
- Rigorous limit theorems between the theory families, with error estimates in a stated norm, which are the precondition for a quantitative rather than a qualitative cross-family comparison.
- Experimental discrimination among the competing families on a common material, and identification of the relaxation, kernel, and length-scale parameters from data that were not generated by the theory being tested.
- Reproducible numerical benchmarks executed independently, which would allow the validation axis of the taxonomy to be populated with something other than self-consistency.

17.3 Methodological stance: tracing differences to assumptions

Differences between formulations may reflect constitutive choice, material symmetry, geometry, boundary conditions, harmonic convention, normalization, or approximation level; they should not be labelled contradictions without comparing these assumptions.

The stance has a cost, which should be stated. By declining to adjudicate between formulations, this review leaves the reader with a map rather than a recommendation in those places where the literature does not itself contain one. The map is, however, the more durable object: the particular theory that will be found correct in a given application is not yet decidable on the available evidence, and a review that pretended otherwise would be asserting more than the evidence supports. What the map does provide is a statement of what evidence would be needed to decide, and that is the more useful contribution at the present stage of the field.

See the single historical overview in Figure 1.

17.4 The three problems that matter most

Three questions are especially important for future analysis and experiment.

Energy accounting for the generalized families. Energy accounting for a generalized model should be derived from its constitutive equations and balance laws before modal ratios or absorption coefficients are compared. This is a model-specific derivation problem.

Limit behavior. Quantitative cross-family comparisons benefit from limit theorems with error estimates, rather than formal parameter substitutions alone.

Discriminating evidence. Discriminating experiments can compare pre-specified predictions for competing formulations against measurements on a common material. A common observable, measurement protocol, and model assumptions make such comparisons interpretable.

17.5 Final outlook

Against that assessment, the next decade of research in the field should pursue the following, in the order in which it would most improve the standing of the subject.

1. **Complete the energy flux derivations** for the generalized families, using the theory-specific constitutive evolution laws as the common template, and publish the dissipation function alongside the flux.
2. **Establish rigorous limit theorems** between the theory families, with explicit error estimates and in a stated norm.
3. **Formulate the interface conditions** for non-local and memory-dependent media, and demonstrate their consistency between the integral and differential representations.
4. **Create reproducible numerical benchmarks** that independent groups can execute using published data and code;
5. **Validate experimentally**, beginning with the thermal wave velocity and the interface energy partition, and with the dispersion measurement that distinguishes a non-local from a gradient mechanism.
6. **Extend the coupled-field models** to the materials for which the coupling is strongest—two-dimensional systems, metamaterials, and semiconductors—where the additional constitutive complexity is most likely to be repaid by predictive power.
7. **Characterize the parameters inversely**, so that the relaxation times, kernel parameters, and length scales are measured rather than fitted, and so that the number of free parameters in a model is matched to the number of independent measurements available.

The next stage is to derive, prove, and measure the models already proposed, while identifying the regimes in which additional generalizations yield distinct, testable predictions. The review's organization around constitutive structures and shared analytical questions is intended to support such comparisons.

Table 40: Notation introduced in Section 17.

Symbol	Meaning	Units	Equation
J_i	Total energy flux vector	$\mathrm{W\,m^{-2}}$	(103)
σ_{ij}	Cauchy stress tensor	Pa	Section 3
q_i	Heat flux vector	$\mathrm{W\,m^{-2}}$	Section 3
τ_0	LS heat-flux relaxation time	s	Section 4
c_T	LS thermal characteristic speed	$\mathrm{m\,s^{-1}}$	Section 4
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$	Section 8
$\Delta \boldsymbol{X}$	Scattering matrix	—	Section 9

17.6 Notation summary for Section 17

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A Detailed Balance Law Derivations

This appendix provides the complete derivations of the balance laws presented in Section 3. All derivations are from first principles using the divergence theorem and localization argument.

A.1 Derivation of Linear Momentum Balance (EQ-001)

Consider a material subvolume $V \subset \Omega$ with surface $S = \partial V$ and outward unit normal $\boldsymbol{n}$. The linear momentum balance states:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho v_i \,\mathrm{d}V = \int_S \sigma_{ji} n_j \,\mathrm{d}S + \int_V \rho f_i \,\mathrm{d}V. \tag{122}$$

Applying the divergence theorem to the surface integral:

$$\int_S \sigma_{ji} n_j \,\mathrm{d}S = \int_V \sigma_{ji,j} \,\mathrm{d}V. \tag{123}$$

Substituting (123) into (122):

$$\int_V \left(\rho \dot{v}_i - \sigma_{ji,j} - \rho f_i\right) \mathrm{d}V = 0. \tag{124}$$

Since V is arbitrary, the integrand must vanish pointwise:

$$\rho \dot{v}_i = \sigma_{ji,j} + \rho f_i. \tag{125}$$

A.2 Derivation of Angular Momentum Balance → Symmetric Stress (EQ-002)

The angular momentum balance for an arbitrary subvolume V:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{x} \times \boldsymbol{v})_k \,\mathrm{d}V = \int_S \epsilon_{kij} x_i \sigma_{jm} n_m \,\mathrm{d}S + \int_V \rho \epsilon_{kij} x_i f_j \,\mathrm{d}V. \tag{126}$$

Applying the divergence theorem and using the linear momentum balance (125), the volume terms cancel, leaving:

$$\int_V \epsilon_{kij}(\sigma_{ij} - \sigma_{ji}) \,\mathrm{d}V = 0. \tag{127}$$

Since V and k are arbitrary:

$$\sigma_{ij} = \sigma_{ji}. \tag{128}$$

A.3 Derivation of Mechanical Power Balance (EQ-003)

Multiply the linear momentum equation (125) by v_i:

$$\rho \dot{v}_i v_i = \sigma_{ji,j} v_i + \rho f_i v_i. \tag{129}$$

The left-hand side:

$$\rho \dot{v}_i v_i = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\rho v_i v_i\right). \tag{130}$$

For the stress term, use the identity:

$$\sigma_{ji,j} v_i = (\sigma_{ji} v_i)_{,j} - \sigma_{ji} v_{i,j}. \tag{131}$$

Since $v_{i,j} = \dot{u}_{i,j} = \dot{\varepsilon}_{ji}$ and $\sigma_{ji} = \sigma_{ij}$:

$$\sigma_{ji} v_{i,j} = \sigma_{ij} \dot{\varepsilon}_{ij}. \tag{132}$$

Combining:

$$\rho \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2} v_i v_i\right) = -\sigma_{ij}\dot{\varepsilon}_{ij} + (\sigma_{ij} v_j)_{,i} + \rho f_i v_i. \tag{133}$$

A.4 Derivation of Internal Energy Balance (EQ-004)

The first law of thermodynamics for a continuum:

$$\rho\dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i} + \rho r. \tag{134}$$

With no external heat supply ($r = 0$):

$$\rho\dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i}. \tag{135}$$

A.5 Derivation of Classical Energy Flux (EQ-007)

Add the mechanical power balance (133) and the internal energy balance (135). Define the total energy density:

$$\mathcal{E} = \frac{1}{2}\rho v_i v_i + \rho e. \tag{136}$$

Then:

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial}{\partial x_j}\left(q_j - \sigma_{ji}v_i\right) = \rho f_i v_i. \tag{137}$$

With no body forces ($f_i = 0$):

$$\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial J_i}{\partial x_i} = 0, \qquad J_i = q_i - \sigma_{ij}v_j. \tag{138}$$

This is the classical energy flux (EQ-007).

B Benchmark Specifications

The examples below define reproducible comparison problems and the quantities to specify. They do not state numerical outcomes.

B.1 BM-1: Classical elastic interface

Problem: Plane P-wave incident on a perfectly bonded interface between two elastic half-spaces.

Governing equations: Navier equations for the stated isotropic elastic model in each medium. Specify densities, Lamé moduli, incidence angle, normal orientation, and amplitude normalization before computing coefficients.

Comparison: Evaluate the complete time-averaged normal flux and include all propagating channels and relevant cross terms. The shorthand $R_P + T_P = 1$ applies only when its single-channel and flux-normalization assumptions hold.

B.2 BM-2: Classical thermoelastic interface

Problem: Coupled thermoelastic wave at a welded interface.

Comparison: Specify the coupled constitutive model, boundary data, material parameters, and flux convention before comparing reflected and transmitted fields.

B.3 BM-3: Piezothermoelastic interface

Problem: Coupled elastic, thermal, and quasistatic electrical fields at an interface.

Comparison: State the constitutive equations and account for mechanical, electrical, and thermal power channels consistently.

B.4 BM-4: One-dimensional nonlocal elastic dispersion

Problem: For the specific Helmholtz-type model stated in Section 7, compare the dispersion relation and its long- and short-wavelength limits with the classical bar.

B.5 BM-5: Memory-dependent thermoelastic wave

Problem: Specify a memory kernel and initial-boundary-value problem; compare a numerical solution with an independently derived reference solution.